

Expert Report / Eksperto ataskaita

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1 Introduction / Įvadas

This document is my expert report in the matter of UAB Teso L.T. [Claimant] versus Luminati Networks, LTD [Respondent].

UAB Teso L.T., is hereafter referred to as “Teso” and Luminati Networks, LTD is hereafter referred to as “Luminati”.

On September 3, 2019, Attorney Rapolas Kasparavičius with the law firm Ellex Valiunas retained my services. Attorney Rapolas represents Luminati in this legal case.

Teso claims that Luminati has stolen Teso’s intellectual property and incorporated it into Luminati’s product offerings.

Šis dokumentas yra mano eksperto ataskaita UAB „Teso L.T.“ byloje. [Ieškovas], palyginti su Luminati Networks, LTD. UAB „Teso L.T.“, toliau vadinama „Teso“, o „Luminati Networks, LTD“, toliau vadinama „Luminati“.

2019 m. Rugsėjo 3 d. Advokatas Rapolas Kasparavičius su advokatų kontora „Ellex Valiunas“ išlaikė mano paslaugas. Advokatas Rapolas atstovauja „Luminati“ šioje teisinėje byloje.

„Teso“ tvirtina, kad „Luminati“ pavogė „Teso“ intelektinę nuosavybę ir įtraukė ją į „Luminati“ siūlomus produktus.

2 Scope of Report / Ataskaitos apimtis

I was asked to provide expert opinions on:

- The report of Jonas Butenis, an expert retained by Teso. Mr. Butenis identifies 13 areas of commonality between the features of the specific product offerings of Teso and Luminati.

Manęs buvo paprašyta pateikti ekspertų nuomones dėl:

- „Teso“ išlaikytas eksperto Jono Butenio pranešimas. Ponas Butenis išskiria 13 bendrų „Teso“ ir „Luminati“ gaminių pasiūlos ypatumų.

- Comparison of the Teso and Luminati products, including both similarities and differences. For similarities between the Teso and Luminati products, evaluate possible ways in which such similarity might arise, and whether there is evidence that this similarity is due to use of Teso's source code, either through reverse engineering or unauthorized access to Teso's source code.

In preparing my report, I have adopted the same terminology used by Mr. Butenis in his report regarding the specific products that I have evaluated. Thus, I refer to "OX" when describing the "Oxylabs" product offered by Teso and to "LU" when describing the "data center proxy" product offered by Luminati.

3 Qualifications / Kvalifikacija

I received my undergraduate degree in 1987 from the University of Chicago in mathematics with a computer science concentration. I earned my Master of Science degree in 2017 from the Georgia Institute of Technology in Computer Science.

From 1987 to 1989, I was a research staff member at Stanford University, where my research primarily revolved around operating systems, including the development of kernel

- „Teso“ ir „Luminati“ produktų palyginimas, apimantis panašumus ir skirtumus. Norėdami įvertinti „Teso“ ir „Luminati“ produktų panašumus, įvertinkite galimus tokio panašumo atsiradimo būdus ir ar nėra įrodymų, kad tokį panašumą lemia „Teso“ šaltinio kodo panaudojimas, naudojant atvirkštinę inžineriją ar nesankcionuotą prieigą prie „Teso“ šaltinio kodo.

Rengdamas savo pranešimą aš pasirinkau tą pačią terminiją, kuri savo pranešime vartojo ponas Butenis, kalbėdamas apie mano įvertintus produktus. Taigi, aš apibūdinu „Teso“ siūlomą produktą „Oxylabs“ ir „LU“, kai apibūdinu „Luminati“ siūlomą „duomenų centro tarpinį“ produktą, ir „LU“.

Bakalauro laipsnį gavau 1987 m. Čikagos universitete matematikos srityje su a informatikos koncentracija. 2017 metais iš Gruzijos gavau mokslo magistro laipsnį Kompiuterių technologijos institutas.

Nuo 1987 m. Iki 1989 m. Dirbau Stanfordo universiteto mokslo darbuotoju, kurio pagrindiniai tyrimai buvo susiję su operacinėmis sistemomis, įskaitant

mode debuggers, network device drivers, and computer network protocol drivers for multiple operating systems. From 1989 through 1992 I worked for Transarc Corporation, where I was part of a team that designed and developed computer software for sharing information over computer networks; this work involved implementation, analysis, and debugging of network protocols, including DCE RPC, TCP/IP, UDP/IP, DNS, and Kerberos. From 1992 through 1993 I worked at FORE Systems, Inc. where I led the engineering team responsible for enabling asynchronous transfer mode (ATM) networking hardware to work with multiple computer operating systems. I also represented FORE Systems at the ATM Forum and was an active member in the local area network (LAN) Emulation working group that developed the first Ethernet over ATM standard to enable transparent interoperability between existing Ethernet networks and new ATM networks. From 1993 to 2016 I was a Consulting Partner for OSR Open Systems Resources, Inc. In that role I was the key architect and technical leader for leading development of a broad range of operating systems software projects, including extensive work with computer network communications protocols. I taught professional seminars on developing operating systems software, including software related to computer networking to

branduolio režimo derinimo priemonių, tinklo įrenginių tvarkykles ir kompiuterio tinklo protokolų tvarkykles, skirtas kelioms operacinėms sistemoms. Nuo 1989 m. Iki 1992 m. Dirbau „Transarc Corporation“, kur buvau komandos, kuri suprojektavo ir sukūrė kompiuterinę programinę įrangą, skirtą keistis informacija kompiuterių tinklais, dalis; šis darbas apėmė tinklo protokolų, įskaitant DCE RPC, TCP / IP, UDP / IP, DNS ir Kerberos, įgyvendinimą, analizę ir derinimo veiksmus. Nuo 1992 m. Iki 1993 m. Dirbau FORE Systems, Inc., kur vadovavau inžinierių komandai, atsakingai už asinchroninio perdavimo režimo (ATM) tinklo aparatūros įjungimą darbui su keliomis kompiuterio operacinėmis sistemomis. Aš taip pat atstovavau „FORE Systems“ ATM forume ir buvau aktyvus vietinio tinklo (LAN) emuliacijos darbo grupės narys, kuris sukūrė pirmąjį „Ethernet over ATM“ standartą, kad užtikrintų skaidrų esamų Ethernet tinklų ir naujų ATM tinklų suderinamumą. Nuo 1993 iki 2016 m. Buvau „OSR Open Systems Resources, Inc.“ konsultacinis partneris. Atlikdamas šias pareigas buvau pagrindinis architektas ir techninis vadovas vadovaujant plataus spektro operacinių sistemų programinės įrangos projektų plėtrai, įskaitant platų darbą su kompiuterių tinklo komunikacijos protokolais. Keliems tūkstančiams kūrėjų mokiau profesionalių seminarų apie

several thousand developers. Since 2015 I have been an Instructional Associate for the Georgia Institute of Technology (Georgia Tech) assisting in the delivery of an online course in computer operating systems. Since 2016 I have been an independent consultant working in the field of computer technology, as I work to complete my PhD in Computer Science at the University of British Columbia.

I am the co-author of Windows NT Device Driver Development (with Peter G. Viscarola) (1998) and lex & yacc (with Doug Brown and John Levine) (1st ed. 1990 and 2nd ed. 1992). I am also the author or co-author on three peer reviewed papers, over fifty articles relating to computer systems architecture, and am the first named inventor of 11 patents in the field. I am currently a member of the Association for Computing Machinery, the Institute for Electrical and Electronic Engineers, and the USENIX Association.

My area of expertise includes operating systems software design, debugging, and development for multiple operating systems including Windows, Linux, and UNIX. This work includes design, debugging, and development of operating systems software specific to computer networking. The course I assist delivering at Georgia Tech includes

operacinių sistemų programinės įrangos kūrimą, įskaitant programinę įrangą, susijusią su kompiuterių tinklais. Nuo 2015 m. Buvau Džordžijos technologijos instituto („Georgia Tech“) instruktorius, padedantis organizuoti internetinius kompiuterių operacinių sistemų kursus. Nuo 2016 m. Buvau nepriklausomas konsultantas, dirbantis kompiuterinių technologijų srityje, nes dirbu baigdamas informatikos mokslų daktaro laipsnį Britų Kolumbijos universitete.

Aš esu „Windows NT įrenginio tvarkyklės tobulinimo“ (kartu su Peteriu G. Viscarola) (1998 m.) Ir „lex & yacc“ (kartu su Dougu Brownu ir Johnu Levine'u) (pirmasis leidimas 1990 m. Ir antrasis leidimas 1992 m.) Bendraautoris. Taip pat esu trijų recenzuotų straipsnių, daugiau nei penkiasdešimt straipsnių, susijusių su kompiuterinių sistemų architektūra, autorė ar bendraautorė ir esu pirmoji įvardijama 11 šios srities patentų išradėja. Šiuo metu esu skaičiavimo mašinų asociacijos, Elektros ir elektronikos inžinierių instituto ir USENIX asociacijos narė.

Mano kompetencija apima operacinių sistemų programinės įrangos kūrimą, derinimo ir daugelio skirtingų operacinių sistemų, įskaitant „Windows“, „Linux“ ir „UNIX“, kūrimą. Šis darbas apima kompiuterinių tinklų operacinės sistemos programinės įrangos projektavimą, derinimą ir plėtrą. Kursas, kurį padedu pristatyti

teaching students about the development of an HTTP-like protocol and a proxy server for that HTTP-like protocol. In preparing my opinion I have reviewed the literature relating to HTTP and HTTPS proxy servers, as well as protocols commonly used in the development of such proxy servers. I am familiar with these protocols, the concepts and implementation of internet protocols and proxy servers. My *curriculum vitae* is attached.

I have served as an expert in multiple United States District Court cases, as well as for actions brought to the United States Patent and Trademark Office, for Inter-Partes Review, for cases involving Internet technologies, computer networking, distributed systems, storage systems, and computer operating systems.

4 Scope of my investigation / Mano tyrimo sritis

On September 3, 2019, Attorney Rapolas Kasparavičius with the law firm Ellex Valiunas commissioned me to review the English language translation of a report by Jonas Butenis comparing features of two specific products offered by Teso and Luminati. I reviewed Mr. Butenis' report and share my analysis of it. I also researched the history of Teso and Luminati, including their websites and technology, with a focus

„Georgia Tech“, apima pagalbą studentams kuriant į HTTP panašų protokolą ir tarpinį serverį tam HTTP protokolui. Rengdamas savo nuomonę apžvelgiau literatūrą, susijusią su HTTP ir HTTPS tarpiniais serveriais, taip pat protokolus, dažniausiai naudojamus kuriant tokius tarpinius serverius. Aš esu susipažinęs su šiais protokolais, interneto protokolų ir tarpinių serverių koncepcijomis ir jų įgyvendinimu. Mano gyvenimo aprašymas pridedamas.

Aš dirbau kaip ekspertas keliose Jungtinių Valstijų apygardos teismo bylose, taip pat dėl ieškinių, pareikštų Jungtinių Valstijų patentų ir prekių ženklų biurui, dėl „Inter-Partes Review“, bylų, susijusių su interneto technologijomis, kompiuterių tinklais, paskirstytomis sistemomis, saugojimo sistemomis, ir kompiuterių operacinės sistemos.

2019 m. Rugsėjo 3 d. Advokatas Rapolas Kasparavičius kartu su advokatų kontora „Ellex Valiunas“ man pavedė peržiūrėti Jono Butenio pranešimo vertimą į anglų kalbą, lyginant dviejų specifinių „Teso“ ir „Luminati“ siūlomų produktų ypatybes. Aš peržiūriu pono Butenio ataskaitą ir pasidalinu jos analize. Aš taip pat tyrinėju „Teso“ ir „Luminati“ istoriją, įskaitant jų svetaines ir technologijas, daugiausia dėmesio

on evaluating the features identified by Mr. Butenis in his report and in the Statement of Claims.

5 Opinions / Nuomonės

The basis and reasons for my opinions come from a combination of:

- My education, training, and experience in the field of Computer Science, including networking, network protocols, network software implementation, proxy servers, operating systems, and distributed systems.
- My experience developing networking software, performing forensic analysis of network protocol behaviors, teaching courses in which forensic analysis of network protocol behaviors is used for analysis, developing and teaching about the development of proxy servers in a network environment.

skirdamas p. Butenio pranešime ir ieškinių pareiškime nurodytoms savybėms įvertinti.

Mano nuomonės pagrindas ir priežastys yra derinamos:

- Mano išsilavinimas, mokymas ir patirtis kompiuterių mokslo srityje, įskaitant tinklų kūrimą, tinklo protokolus, tinklo programinės įrangos diegimą, tarpinius serverius, operacines ir paskirstytas sistemas.
- Mano patirtis kuriant tinklo programinę įrangą, atliekant kriminalistinę tinklo protokolų elgsenos analizę, rengiant mokymo kursus, kuriuose naudojama tinklo protokolų elgsenos kriminalistinė analizė analizuojant, kuriant ir mokant apie tarpinių serverių plėtrą tinklo aplinkoje.

- My knowledge of the evolution of computer networking, computer network protocols, internet protocol standards processes, and common methods of implementation for software related to computer networking and computer network protocols.

- My knowledge and experience evaluating plagiarism, including my published, peer reviewed literature based upon my experience in detecting, evaluating, and substantiating plagiarism in computer software.

- The background material in §7

- The materials reviewed, which are described in §6 of this report.

I state the following opinions to a reasonable degree of certainty:

- The OX and LU products, while similar, are not identical. See §5.1.

- Mano žinios apie kompiuterinių tinklų, kompiuterių tinklo protokolų, interneto protokolų standartų procesų ir įprastų programinės įrangos, susijusios su kompiuterių tinklais ir kompiuterių tinklo protokolais, diegimo metodus.

- Mano žinios ir patirtis vertinant plagiatą, įskaitant publikuotą, recenzuojamą literatūrą, paremta mano patirtimi aptikti, įvertinti ir pagrįsti kompiuterinės programinės įrangos plagiatą.

- pagrindinę medžiagą 7 skyriuje.

- Peržiūrėta medžiaga, kuri aprašyta šios ataskaitos 6 skyriuje.

Pareiškiu šias nuomones pakankamai užtikrintai:

- Nors OX ir LU produktai yra panašūs, jie nėra tapatūs. Žr. §5.1.

- The features described by Mr. Butenis are general in nature and the techniques required to implement them would not normally require use of a third-party's confidential technology. See §5.2.

- Pono Butenio aprašytos savybės yra bendro pobūdžio, o joms įgyvendinti reikalingi metodai paprastai nereikalauja trečiosios šalies konfidencialios technologijos. Žr. §5.2.

- The features described by Mr. Butenis are general in nature and the techniques required to implement them would not normally require use of a third-party's confidential technology. See §5.2.

- Mr. Butenis' report indicates that he accessed nine servers during his analysis: five Teso managed servers, and four others. Only one of the five Teso managed servers is in Lithuania (80% are outside Lithuania). Only one of the nine servers used by Mr. Butenis in his report is located in Lithuania. See §5.4. Of the nine servers Mr. Butenis referenced in his report, seven are in the United States, one in Germany, and one in Lithuania.

- Pono Butenio ataskaitoje nurodoma, kad analizuodamas jis prisijungė prie devynių serverių: penkių „Teso“ valdomų serverių ir dar keturių. Tik vienas iš penkių „Teso“ valdomų serverių yra Lietuvoje (80% yra ne Lietuvoje). Tik vienas iš devynių serverių, kuriuos ponas Butenis naudojo savo pranešime, yra Lietuvoje. Žr. §5.4 skyrių. Iš devynių serverių, kuriuos ponas Butenis minėjo savo ataskaitoje, septyni yra JAV, vienas - Vokietijoje, kitas - Lietuvoje.

- Neither Mr. Butenis’ report nor Teso’s Statement of Claims provide any specific analysis of the code that implements the LU and OX products. The features described as being the same are well-known in the field. In most cases there are many compatible third-party implementations, including some that are freely available. Teso clearly establishes in their Statement of Claims that their source code is well-secured and thus would not have been available to Luminati. Thus, it is my opinion that none of the information provided supports a claim that Luminati’s products incorporate Teso’s source code or proprietary knowledge.

I reserve the right to amend this report if other evidence or different information becomes available. I may have additional opinions and/or reasoning depending on what is asked of me at deposition and/or trial.

5.1 Product Similarity / Produkto panašumas

In his report, Mr. Butenis identifies and compares 13 features of the OX and LU products that he claims are identical. Mr. Butenis’ report does not state that it is his opinion that the presence of any individual or even the collective presence of all these

- Nei p. Butenio ataskaitoje, nei „Teso“ ieškinių pareiškimė nepateikta jokios konkrečios kodų, įgyvendinančių „LU“ ir „OX“ produktus, analizės. Savybės, apibūdinamos kaip vienodos, yra gerai žinomos šioje srityje. Daugeliu atvejų yra daug suderinamų trečiųjų šalių įdiegimų, įskaitant tuos, kurie yra laisvai prieinami. „Teso“ ieškinių pareiškimė aiškiai nustato, kad jų šaltinio kodas yra gerai apsaugotas, todėl Luminati jų neturėjo. Taigi, mano nuomone, nė viena iš pateiktos informacijos nepatvirtina teiginio, kad „Luminati“ produktai turi „Teso“ kodą ar patentuotas žinias.

Pasilieku teisę taisyti šią ataskaitą, jei yra kitų įrodymų ar kitokios informacijos tampa prieinamas. Aš galiu turėti papildomų nuomonių ir (arba) argumentų, atsižvelgiant į ko iš manęs klausiama deponavimo ir (arba) teismo metu.

Savo pranešime ponas Butenis nustato ir lygina 13 OX ir LU produktų savybių, kurios, jo teigimu, yra tapačios. Pono Butenio pranešime nesakoma, kad, jo nuomone, bet kuris asmuo ar net kolektyvinis visų šių bruožų buvimas rodo, kad jie turi

features indicates they share a common code base. The statement of claims states that the presence of these features supports Teso's claim that their technology was misappropriated by Luminati.

It is my opinion that not all of these features are present in the LU product. It is also my opinion that each of these features can be realized by well-known implementation techniques of these common features. It is also my opinion that the presence of these features in the LU product are not persuasive evidence of piracy or plagiarism.

In the subsequent subsections, I will explain the basis of my opinions by discussing each of the points from Mr. Butenis' report.

5.2 Product Features / Prekės savybės

In his report, Mr. Butenis reviewed a list of 13 features that are common to the two products. In this section, I review those results. I note where I confirm his results as well as where my observations and/or analysis differs from his. In cases where my results differ, when possible I have provided an explanation for the difference. There may be other differences beyond those I identified.

Mr. Butenis provided a list of standard tools that he used for his analysis. I used those tools for my analysis as well, though I worked from both an Ubuntu 18.04 Linux

bendrą kodo bazę. Ieškinių pareiškime teigiama, kad šių funkcijų buvimas patvirtina „Teso“ teiginį, kad „Luminati“ pasisavino jų technologiją.

Manau, kad ne visos šios savybės yra „LU“ produkte. Taip pat manau, kad kiekviena iš šių savybių gali būti įgyvendinta gerai žinomomis šių bendrų bruožų įgyvendinimo technikomis. Taip pat manau, kad šių produktų buvimas LU produkte nėra įtikinantys piratavimo ar plagiato įrodymai.

Tolesniuose poskirsiuose paaiškinsiu savo nuomonės pagrindą aptardamas kiekvieną punktą iš p. Butenio pranešimo.

Savo pranešime p. Butenis apžvelgė 13 bruožų, būdingų abiemis gaminiams, sąrašą. Šiame skyriuje apžvelgiu tuos rezultatus. Aš atkreipiu dėmesį į tai, kur patvirtinu jo rezultatus, taip pat kur mano pastebėjimai ir (arba) analizė skiriasi nuo jo. Tais atvejais, kai mano rezultatai skiriasi, kai įmanoma, pateikiau skirtumo paaiškinimą. Gali būti ir kitų skirtumų, išskyrus tuos, kuriuos aš identifikuojau.

Ponas Butenis pateikė standartinių priemonių, kurias naudojo analizei, sąrašą. Aš tuos įrankius naudojausi ir savo analizei, nors dirbau ir iš „Ubuntu 18.04 Linux“

system and a Microsoft Windows 10 systems; the tools used by Mr. Butenis are available on both systems.

In addition to the tools that Mr. Butenis used, I also used the **nslookup** utility, which is used for interrogating Domain Name System (DNS) servers (see §7.2.4). This is important, because the domain name to IP address mapping is defined by the DNS server. Thus, using different DNS servers can yield different results.

5.2.1 Shared IP Addresses / Bendri IP adresai

I agree with the analysis in Mr. Butenis' report that both OX and LU offer shared IP addresses. This is not unique to these two companies, as this is a common feature of other proxy service company offerings.

sistemos, ir iš „Microsoft Windows 10“ sistemų; p. Butenio naudojami įrankiai yra prieinami abiejose sistemose.

Be įrankių, kuriais naudojosi ponas Butenis, aš taip pat naudojau įrankį `textbf nslookup`, kuris naudojamas užklausti domenų vardų sistemos (DNS) serverius (žr. 7.2.4 skyrių). Tai svarbu, nes domeno pavadinimą į IP adresą priskiria DNS serveris. Taigi, naudojant skirtingus DNS serverius, rezultatai gali būti skirtingi.

Sutinku su p. Butenio pranešime pateikta analize, kad tiek OX, tiek LU siūlo bendrus IP adresus. Šioms dviem įmonėms tai nėra būdinga, nes tai yra bendras kitų tarpinių paslaugų įmonių pasiūlymų bruožas.

Sign in

Oxylabs Learning HUB > FAQ > Data Center Proxies

Q Search

Which protocols are supported by Oxylabs?

Which locations are available in Oxylabs?

How many subnets do I get if I order 1000 proxies?

Can you whitelist my data center IPs?

I've whitelisted my IP, how long should I wait before its enabled?

Do you offer shared proxies?

No, we don't.



Was this article helpful?

0 out of 0 found this helpful

<https://learn.oxylabs.io/hc/en-us/articles/360024815772-Do-you-offer-shared-proxies->

September 13, 2019

Even though both Mr. Butenis and I observed that multiple clients may use the same proxy server, the `oxylabs.io` website states that they do not offer shared proxies. This information is contradictory with these findings.

The OX product website explains why this is a popular feature: it is lower in cost: “Shared proxies are IP addresses that are used by multiple users at the same time. Shared proxies are considered to be inferior in comparison to private proxies. However, shared proxies can be suitable for most tasks online. Shared proxies tend to be the cheapest type of proxies.”

Numerous other proxy services offer shared IP addresses. “Most of the proxies offered by proxy services or those you may found [sic] on proxy lists are shared as they are cheaper and easier to maintain. They come with unlimited bandwidth, multiple locations, multiple subnets, instant activation, money-back guarantee [1].”

Thus, it is my opinion that both services do offer shared IP addresses.

Nepaisant to, kad ir ponas Butenis, ir aš pastebėjome, kad keli klientai gali naudoti tą patį tarpinį serverį, svetainė `oxylabs.io` teigia, kad jie nesiūlo bendrų tarpinių serverių. Ši informacija prieštarauja šioms išvadoms.

„OX“ produkto svetainė paaiškina, kodėl tai yra populiari savybė: ji yra mažesnė kaina: “Bendri tarpiniai serveriai yra IP adresai, kuriuos vienu metu naudoja keli vartotojai. Laikoma, kad bendri įgaliotiniai yra žemesni, palyginti su asmeniniais įgaliotiniais. Tačiau bendri tarpiniai serveriai gali būti tinkami atlikti daugumą užduočių internetu. Bendri įgaliotiniai dažniausiai būna pigiausi įgaliotinių tipai.”

Daugybė kitų tarpinių paslaugų siūlo bendrus IP adresus. “Dauguma tarpinių serverių, kuriuos siūlo tarpinės tarnybos, arba tie, kuriuos galite rasti tarpinių serverių sąrašuose, yra bendrinami, nes jie yra pigesni ir lengviau prižiūrimi. Jie aprūpinti neribotu pralaidumu, keliomis vietomis, keliais potinkliais, greitu aktyvinimu, pinigų grąžinimo garantija [1].”

Taigi, mano nuomone, abi paslaugos siūlo bendrus IP adresus.

```
(base) tony@wam-desktop:~$ ifconfig
docker0: flags=4099<UP,BROADCAST,MULTICAST> mtu 1500
    inet 172.17.0.1 netmask 255.255.0.0 broadcast 172.17.255.255
    inet6 fe80::42:24ff:feb9:6a76 prefixlen 64 scopeid 0x20<link>
    ether 02:42:b9:6a:76 txqueuelen 0 (Ethernet)
    RX packets 0 bytes 0 (0.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 122 bytes 18714 (18.7 KB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 192.168.111.114 netmask 255.255.255.0 broadcast 192.168.111.255
    inet6 fe80::215:50ff:fe6f:9101 prefixlen 64 scopeid 0x20<link>
    ether 00:15:5d:6f:91:01 txqueuelen 1000 (Ethernet)
    RX packets 6012020 bytes 3684334854 (3.6 GB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 2389666 bytes 2535616165 (2.5 GB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0x10<host>
    loop txqueuelen 1000 (Local Loopback)
    RX packets 114426 bytes 2539960268 (2.5 GB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 114426 bytes 2539960268 (2.5 GB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

vmet1: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 172.16.160.1 netmask 255.255.255.0 broadcast 172.16.160.255
    inet6 fe80::250:56ff:fec0:1 prefixlen 64 scopeid 0x20<link>
    ether 00:50:56:c0:00:01 txqueuelen 1000 (Ethernet)
    RX packets 0 bytes 0 (0.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 39093 bytes 0 (0.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

vmet8: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 172.16.158.1 netmask 255.255.255.0 broadcast 172.16.158.255
    inet6 fe80::250:56ff:fec0:8 prefixlen 64 scopeid 0x20<link>
    ether 00:50:56:c0:00:08 txqueuelen 1000 (Ethernet)
    RX packets 0 bytes 0 (0.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 39092 bytes 0 (0.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

(base) tony@wam-desktop:~$ date; curl http://icanhazip.com
Sun Sep  8 17:28:40 PDT 2019
108.172.232.33
(base) tony@wam-desktop:~$
```

```
tony@wam-desktop:~$ ifconfig
eth1: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 192.168.40.241 netmask 255.255.255.240 broadcast 192.168.40.255
    inet6 fe80::45a3:dbf6:2809:a84e prefixlen 64 scopeid 0xfd<compat,link,site,host>
    ether 00:15:5d:5f:a2:7e (Ethernet)
    RX packets 0 bytes 0 (0.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 0 bytes 0 (0.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

eth2: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 169.254.90.67 netmask 255.255.0.0 broadcast 169.254.255.255
    inet6 fe80::618d:c007:c275:5a43 prefixlen 64 scopeid 0xfd<compat,link,site,host>
    ether 00:15:5d:6f:91:00 (Ethernet)
    RX packets 0 bytes 0 (0.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 0 bytes 0 (0.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

eth3: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 192.168.111.148 netmask 255.255.255.0 broadcast 192.168.111.255
    inet6 fe80::ec38:2656:9340:607 prefixlen 64 scopeid 0xfd<compat,link,site,host>
    ether 38:05:47:02:62:e4 (Ethernet)
    RX packets 0 bytes 0 (0.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 0 bytes 0 (0.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

eth4: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 10.0.75.1 netmask 255.255.255.240 broadcast 10.0.75.15
    ether 00:15:5d:6f:91:05 (Ethernet)
    RX packets 0 bytes 0 (0.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 0 bytes 0 (0.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0xfe<compat,link,site,host>
    loop txqueuelen 1000 (Local Loopback)
    RX packets 0 bytes 0 (0.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 0 bytes 0 (0.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

tony@wam-desktop:~$ date; curl http://icanhazip.com
Sun Sep  8 17:30:15 PDT 2019
108.172.232.33
tony@wam-desktop:~$
```

However, there is no useful significance to this similarity. I conducted an experiment, using Mr. Butenis' protocol from my home computer network using two different computers. From each computer I executed the same commands, except that I did not specify any credentials. Thus, the ability to map distinct clients to a common IP address is functionality that exists within the firewall, gateway, or router in a normal residential setting. I discuss this in greater detail in §7.1.3. It is also normal practice to use routers, gateways, and/or firewalls to protect data center environments. It is my opinion that the ability to map two or more distinct computers or users to a single IP address is neither novel nor interesting, within the context of evaluating the OX and LU products.

Tačiau šis panašumas neturi jokių reikšmingos reikšmės. Aš vedžiau eksperimentą, naudodamas pono Butenio protokolą iš savo namų kompiuterių tinklo, naudodamas du skirtingus kompiuterius. Iš kiekvieno kompiuterio vykdžiau tas pačias komandas, išskyrus tai, kad nenurodžiau jokių prisijungimo duomenų. Taigi galimybė priskirti atskirus klientus bendruoju IP adresu yra funkcija, egzistuojanti ugniasienėje, šliuze ar maršrutizatoriuje įprastoje gyvenamojoje aplinkoje. Aš tai išsamiau aptariu S ref poskirsnyje: proxy. Taip pat įprasta naudoti maršrutizatorius, šliuzus ir (arba) ugniasienes, kad būtų apsaugota duomenų centro aplinka. Mano nuomone, vertinant OX ir LU produktus, galimybė susieti du ar daugiau skirtingų kompiuterių ar vartotojų vienu IP adresu nėra nei nauja, nei įdomu.

“When an IP address is used by only one user, it is commonly known as a dedicated IP address. VPNs normally used IP addresses that are shared by many users, making it easier to keep each user anonymous.

Although a dedicated IP address is not as secure when it comes to anonymity, they do have their own advantages.

“Shared IP addresses are often blacklisted, and although this could also happen with a dedicated IP address, this only happens rarely if the user of the IP does not perform spamming actions. Media services, including Hulu, BBC, and Netflix, are very good at identifying shared IP addresses and blacklisting them. This will cause the public proxy to be blocked. With a dedicated IP address, it is easier to stay hidden and reduce the chance of being blacklisted [2].”

In §7.6 I have provided a list of data center proxy providers that were active prior to the date that `oxylabs.io` became available (§8.1.2). I have confirmed that the following providers were offering shared IP address proxies prior to the existence of `oxylabs.io`: Squid Proxies, Buy Proxies, SSL Private Proxy, Proxy-N-VPN, and Mexela.

Thus, it is my opinion that shared proxies were generally available prior to their support in OX. Their presence in the OX and LU products is inclusion of a common feature present in pre-existing competing products.

“Kai IP adresą naudoja tik vienas vartotojas, jis paprastai žinomas kaip skirtas IP adresas. VPT paprastai naudojo IP adresus, kuriais naudojasi daugelis vartotojų, todėl kiekvienam vartotojui lengviau išlaikyti anonimą. Nors skirtas IP adresas nėra toks saugus, kai kalbama apie anonimiškumą, jie turi savo privalumų.

”Bendrinami IP adresai dažnai įtraukiami į juodąjį sąrašą, ir nors tai gali nutikti ir su tam skirtu IP adresu, taip nutinka tik retai, jei IP vartotojas neatlieka šlamšto platinimo veiksmų. Žiniasklaidos paslaugos, įskaitant „Hulu“, BBC ir „Netflix“, labai gerai padeda nustatyti bendrus IP adresus ir juos įtraukti į juodąjį sąrašą. Tai užblokuos viešąjį tarpinį serverį. Turėdami tam skirtą IP adresą, lengviau paslėpti ir sumažinti tikimybę patekti į juodąjį sąrašą [2].”

7.6 skyriuje pateikiau duomenų centro tarpinių paslaugų teikėjų, kurie buvo aktyvūs iki datos, kai atsirado `oxylabs.io`, sąrašą (8.1.2 skyrius). Patvirtinau, kad šie paslaugų teikėjai siuntė bendrus IP adresų įgaliojinius iki egzistavimo `oxylabs.io`: „Squid Proxies“, „Buy Proxies“, „SSL Private Proxy“, „Prxy-N-VPN“ ir „Mexela“.

Taigi, mano nuomone, bendri įgaliojiniai buvo prieinami prieš jiems palaikant OX. Jų buvimas OX ir LU produktuose yra bendro ankstesnių konkuruojančių gaminių bruožo įtraukimas.

5.2.2 Dedicated Data Center IP Addresses / Specialūs IP adresai

I agree with the analysis in Mr. Butenis' report that both OX and LU offer dedicated data center IP addresses. This is not unique to these two companies, as this is a common feature of other proxy service company offerings.

"When an IP address is used by only one user, it is commonly known as a dedicated IP address. VPNs normally used IP addresses that are shared by many users, making it easier to keep each user anonymous.

Although a dedicated IP address is not as secure when it comes to anonymity, they do have their own advantages.

"Shared IP addresses are often blacklisted, and although this could also happen with a dedicated IP address, this only happens rarely if the user of the IP does not perform spamming actions. Media services, including Hulu, BBC, and Netflix, are very good at identifying shared IP addresses and blacklisting them. This will cause the public proxy to be blocked. With a dedicated IP address, it is easier to stay hidden and reduce the chance of being blacklisted [2]."

There are a lot of excellent dedicated proxy providers (unlimited bandwidth, money-back guarantee, HTTP proxy, SOCKS proxy). Take a look to my [proxy tools](#) page, or take Google search on "dedicated proxy," "dedicated IP proxies," "HTTP dedicated proxy," or "socks private proxy."

<https://hide-ip-proxy.com/what-is-a-dedicated-proxy/>

"There are a lot of excellent dedicated proxy

Aš sutinku su p. Butenio pranešime pateikta analize, kad tiek OX, tiek LU siūlo specialius IP adresus. Šioms dviem įmonėms tai nėra būdinga, nes tai yra bendras kitų tarpinių paslaugų įmonių pasiūlymų bruožas.

"Kai IP adresą naudoja tik vienas vartotojas, jis paprastai žinomas kaip skirtas IP adresas. VPT paprastai naudojo IP adresus, kuriais naudojasi daugelis vartotojų, todėl kiekvienam vartotojui lengviau išlaikyti anonimą. Nors skirtas IP adresas nėra toks saugus, kai kalbama apie anonimiškumą, jie turi savo privalumų.

"Bendri IP adresai dažnai įtraukiami į juodąjį sąrašą, ir nors tai gali atsitikti ir turint tam skirtą IP adresą, taip nutinka tik retai, jei IP vartotojas neatlieka šlamšto platinimo veiksmų. Žiniasklaidos paslaugos, įskaitant „Hulu“, BBC ir „Netflix“, labai gerai padeda nustatyti bendrus IP adresus ir juos įtraukti į juodąjį sąrašą. Tai užblokuos viešąjį tarpinį serverį. Turint tam skirtą IP adresą, lengviau likti paslėptam ir sumažinti tikimybę patekti į juodąjį sąrašą [2]. "

"Yra daugybė puikių tarpinių paslaugų

providers (unlimited bandwidth, money-back guarantee, HTTP proxy, SOCKS proxy). Take a look to my proxy tools page, or take Google search on ‘dedicated proxy,’ ‘dedicated IP proxies,’ ‘HTTP dedicated proxy,’ or ‘socks private proxy.’[2]”

The functionality of offering a dedicated data center IP address to a client machine is not unique and is frequently found in data center proxy services. It is my opinion that the ability to provide a dedicated data center IP address to a client is neither novel nor interesting, within the context of evaluating the OX and LU products.

5.2.3 Load Balancing/Rotation / Apkrovos balansavimas / pasukimas

I agree with the analysis in Mr. Butenis’ report that both OX and LU offer load balancing and rotation across servers. This is a common approach in the computer network business for load balancing.

teikėjų (neribotas pralaidumas, pinigų grąžinimo garantija, HTTP tarpinis serveris, SOCKS tarpinis serveris). Pažvelkite į mano įgaliojotojo serverio įrankių puslapį arba „Google“ paiešką naudodamiesi „dedikuotu tarpiniu serveriu“, „dedikuotais IP tarpiniais serveriais“, „HTTP dedikuotu proxy“ arba „kojinių privačiu tarpiniu serveriu.“[2]”

Siūlomo IP adreso kliento mašinai funkcionalumas nėra unikalus ir dažnai sutinkamas duomenų centro tarpinių serverių paslaugose. Mano nuomone, vertinant OX ir LU produktus, galimybė suteikti klientui skirtą IP adresą nėra nei nauja, nei įdomi.

Sutinku su p. Butenio pranešime pateikta analize, kad tiek OX, tiek LU siūlo apkrovos balansavimą ir sukimą tarp serverių. Tai yra įprastas požiūris į kompiuterių tinklo verslą apkrovos balansavimui.

Load balancing across multiple application instances is a commonly used technique for optimizing resource utilization, maximizing throughput, reducing latency, and ensuring fault-tolerant configurations.

It is possible to use nginx as a very efficient HTTP load balancer to distribute traffic to several application servers and to improve performance, scalability and reliability of web applications with nginx.

https://web.archive.org/web/20141227033342/https://nginx.org/en/docs/http/load_balancing.html

December 27, 2014.

“Load balancing across multiple application instances is a commonly used technique for

“Apkrovos balansavimas keliuose taikymo egzemplioriuose yra dažniausiai naudojamas

optimizing resource utilization, maximizing throughput, reducing latency, and ensuring fault-tolerant configurations.”

“It is possible to use nginx as a very efficient HTTP load balancer to distribute traffic to several application servers and to improve performance, scalability and reliability of web applications with nginx.”

In §5.2.9 I discuss the concept of vertical scaling. The idea of using load balancing was introduced in order to provide the type of scaling described there.

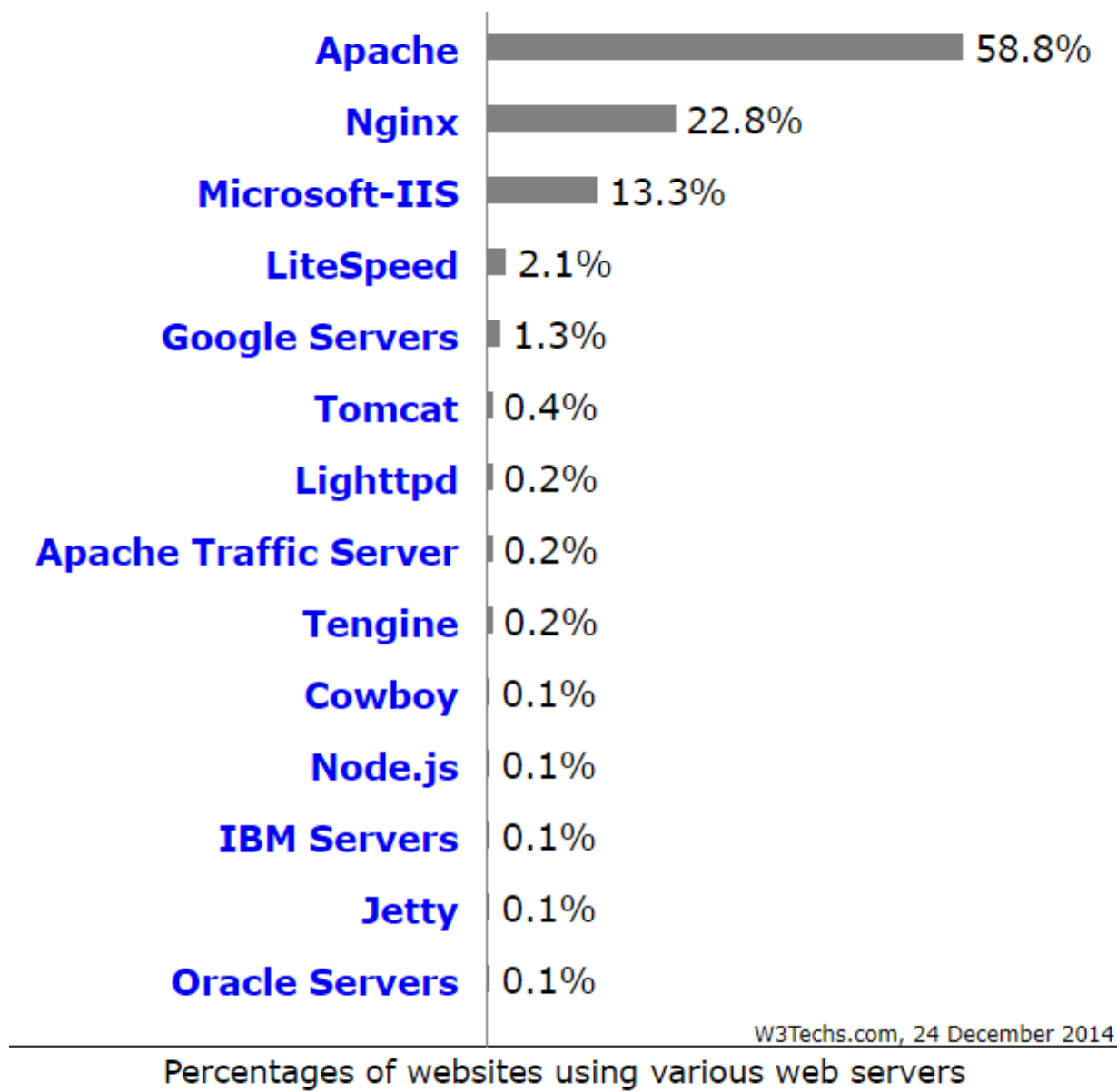
In §5.2.7 I discuss the concept of a *blacklist*. Websites may block a specific web client based upon any criteria it wishes to apply. For example, a company could block IP addresses associated with competing companies by placing them on a blacklist. Thus, one purpose of IP address rotation is to avoid such blacklists. This is described on one or more websites of the data center proxy providers described in §7.6.

būdas optimizuoti išteklių naudojimą, maksimaliai padidinti pralaidumą, sumažinti vėlavimą ir užtikrinti gedimams atsparias konfigūracijas.

“ Galima naudoti „nginx“ kaip labai efektyvų HTTP apkrovos balansavimo įrenginį srautui paskirstyti į kelis programų serverius ir pagerinti interneto programų su „nginx“ našumą, mastelį ir patikimumą. ”

5.2.9 skyriuje aptariu vertikalaus mastelio sąvoką. Buvo pristatyta apkrovos balansavimo idėja, kad būtų galima ten aprašyti mastelio tipą.

5.2.7 skyriuje aptariu juodojo sąrašo sąvoką. Svetainės gali blokuoti konkretų interneto klientą, remdamosi bet kokiais kriterijais, kuriuos ji nori taikyti. Pvz., Įmonė galėtų blokuoti konkuruojančių įmonių registruotus IP adresus įtraukdama juos į juodąjį sąrašą. Taigi vienas IP adresų sukimosi tikslas yra išvengti tokių juodųjų sąrašų. Tai aprašyta viename ar daugiau duomenų centro tarpinių paslaugų teikėjų, aprašytų 7.6 skyriuje.



https://web.archive.org/web/20141224093940/https://w3techs.com/technologies/overview/web_server/all
December 24, 2014.

In December 2014, Nginx was the second most popular web server on the internet. Nginx described its use for load balancing. Thus, load balancing was a common feature in internet web services, including proxy servers, prior to the existence of oxylabs.io (§8.1.2).

In §7.6 I have provided a list of data center

2014 m. Gruodžio mėn. „Nginx“ buvo antras populiariausias interneto serveris. „Nginx“ aprašė jo naudojimą apkrovai subalansuoti. Taigi, iki oxylabs.io egzistavimo (8.1.2 skyrius), apkrovos balansavimas buvo dažna interneto žiniatinklio paslaugų, įskaitant tarpinius serverius, ypatybė.

2014 m. Gruodžio mėn. „Nginx“ buvo antras

proxy providers that were active prior to the date that `oxylabs.io` became available (§8.1.2). I have confirmed that the following providers were clearly offering load balancing or rotation prior to the existence of `oxylabs.io`: Your Private Proxy, Squid Proxies, and Mexela.

In §5.2.10 I describe the “IBM WebSphere proxy server,” which includes the ability to route proxy clients to specific servers, or to a group of servers. Thus, this also demonstrates another data center focused technology that provided comparable functionality in 2012, which is prior to the date that `oxylabs.io` became available (§8.1.2).

In §5.2.6 I describe the `seo-proxies.com` website’s publication of their API. They provide code for using their API to do proxy server rotation. Changing proxy servers will change the IP address being used because each proxy server has its own unique IP address. This information was available at least by December 24, 2014, which is prior to the date that `oxylabs.io` became available (§8.1.2).

The functionality of load balancing and rotation is standard in the computer networking field, including the field of proxy servers. It is also clearly present in many product offerings that existed prior to the OX and LU products. It is my opinion that

populiariausias interneto serveris. „Nginix“ aprašė jo naudojimą apkrovai subalansuoti. Taigi, iki `oxylabs.io` egzistavimo, apkrovos balansavimas buvo dažna interneto žiniatinklio paslaugų, įskaitant tarpinius serverius, ypatybė.

5.2.10 skyriuje aprašau „IBM WebSphere proxy serverį“, kuris apima galimybę nukreipti tarpinius serverius į konkrečius serverius arba į serverių grupę. Taigi tai taip pat parodo kitą duomenų center nukreiptą technologiją, kuri 2012 m. Suteikė palyginamą funkcionalumą, ty iki datos, kai buvo paskelbtas `oxylabs.io` (8.1.2 skyrius).

5.2.6 skyriuje apibūdinu svetainės URL skelbimą `seo-proxies.com`. Jie pateikia kodą, kaip naudoti savo API, kad būtų galima atlikti tarpinio serverio sukimaši. Pakeitus tarpinius serverius, bus pakeistas naudojamas IP adresas, nes kiekvienas tarpinis serveris turi savo unikalų IP adresą. Ši informacija buvo prieinama bent iki 2014 m. Gruodžio 24 d., Ty iki datos, kai `oxylabs.io` tapo prieinama (§8.1.2 skyrius).

Apkrovos balansavimo ir pasukimo funkcijos yra standartinės kompiuterių tinklų srityje, įskaitant tarpinius serverius. Tai aiškiai matyti ir daugelyje produktų, egzistavusių prieš OX ir LU gaminius. Mano nuomone, vertinant OX ir LU produktus, galimybė

the ability to provide load balancing and rotation of IP addresses to a client is neither novel nor interesting, within the context of evaluating the OX and LU products.

5.2.4 SOCKS5 Protocol Support / SOCKS protokolo palaikymas

Mr. Butenis' report indicates that he found support for SOCKS5 in both OX and LU.

In §7.6 I have provided a list of data center proxy providers that were active prior to the date that `oxylabs.io` became available (§8.1.2). I have confirmed that at least Your Private Proxy was clearly offering SOCKS5 support prior to the existence of `oxylabs.io`.

In addition, I describe in §7.3 the lengthy history of the SOCKS5 protocol as well as the existence of numerous SOCKS5 implementations.

The concept of supporting SOCKS5 in a data center server proxy product is not new. SOCKS5 protocol support is clearly present in at least one product offering that existed prior to the OX and LU products.

The SOCKS5 protocol is well-understood after two decades of use and there are numerous implementations. It is also clearly present in a number of product offerings that existed prior to the OX and LU products. Thus, based upon the fact this protocol is

klientui pateikti apkrovos balansavimą ir IP adresų kaitaliojimą nėra nei nauja, nei įdomi.

Pono Butenio ataskaitoje nurodoma, kad jis atrado SOCKS5 tiek OX, tiek LU.

§7.6 skyriuje pateikiau duomenų centro tarpinių paslaugų teikėjų, kurie buvo aktyvūs iki datos, kai atsirado `oxylabs.io`, sąrašą (§8.1.2 skyrius). Patvirtinau, kad bent jau jūsų privatus tarpinis serveris aiškiai siūlė SOCKS5 palaikymą iki `oxylabs.io` egzistavimo.

Be to, 7.3 skyriuje ilga SOCKS5 protokolo istorija, taip pat daugybė SOCKS5 diegimų.

„SOCKS5“ palaikymo duomenų centro serverio įgaliojoje programoje koncepcija nėra nauja. „SOCKS5“ protokolo palaikymas aiškiai egzistuoja bent viename produkte, kurie egzistavo prieš OX ir LU produktus, pasiūlyme.

SOCKS5 protokolas yra gerai suprantamas po dviejų dešimtmečių naudojimo ir yra daugybė įdiegimų. Tai taip pat akivaizdžiai matoma daugelyje siūlomų produktų, egzistavusių prieš OX ir LU produktus. Taigi, atsižvelgiant į tai, kad šis protokolas

present in many products and is two decades old, it is my opinion that the ability to provide SOCKS5 support to a client is neither novel nor interesting, within the context of evaluating the OX and LU products.

yra daugelyje produktų ir yra dviejų dešimtmečių senumo, manau, kad vertinant OX ir LU produktus, galimybė teikti SOCKS5 palaikymą klientui nėra nei nauja, nei įdomi.

5.2.5 HTTP Protocol Support / HTTP protokolo palaikymas

I agree with the analysis in Mr. Butenis' report that both OX and LU support the HTTP protocol. I did not find this surprising, since using the HTTP protocol has been a fundamental part of web proxy servers for more than 20 years. See also §7.2.5.

Sutinku su p. Butenio pranešime pateikta analize, kad tiek OX, tiek LU palaiko HTTP protokolą. Manęs tai nestebino, nes HTTP protokolo naudojimas daugiau nei 20 metų buvo pagrindinė žiniatinklio tarpinių serverių dalis. Taip pat žiūrėkite 7.2.5 skyrių.



Introduction

[About Squid](#)

[Why Squid?](#)

[Squid Developers](#)

[How to Donate](#)

[How to Help Out](#)

[Getting Squid](#)

[Squid Source Packages](#)

[Squid Deployment Case-Studies](#)

[Squid Software Foundation](#)

Documentation

Configuration:

[Reference](#)

[Examples](#)

[FAQ and Wiki](#)

Guide Books:

[Beginners](#)

[Definitive](#)

[Non-English](#)

[More...](#)

Support

What is Squid?

Squid is a fully-featured HTTP/1.0 proxy which is almost (but not quite - we're getting there!) a fully-featured HTTP/1.1 proxy. Squid offers a rich access control, authorization and logging environment to develop web proxy and content serving applications. Squid offers a rich set of traffic optimization options, most of which are enabled by default for simpler installation and high performance.

Where did Squid come from?

Squid is based on the Harvest Cache Daemon developed in the early 1990's. It was one of two forks from the codebase after the Harvest project ran to completion. (The other fork being what became Netapp's Netcache.)

The Squid project was funded by an NSF grant (NCR-9796082) which covered research into caching technologies. The [ircache](#) funding ran out a few years later and the Squid project continued through volunteer donations and the occasional commercial investment.

Squid is currently being developed by a handful of individuals donating their time and effort to building current and next generation content caching and delivery technologies. An ever-growing number of companies use Squid to save on their internet web traffic, improve performance, deliver faster browsing to their end-clients and provide static, dynamic and streaming content to millions of internet users worldwide.

Who uses Squid today?

A good question! Many of you are using Squid without even knowing it! Some companies have embedded Squid in their home or office firewall devices, others use Squid in large-scale web proxy installations to speed up broadband and dialup internet access. Squid is being increasingly used in content delivery architectures to deliver static and streaming video/audio to internet users worldwide.

<http://www.squid-cache.org/Intro>

Squid is an early HTTP proxy server, written in the 1990s. It supports the HTTP protocol. Indeed, it would be difficult for me to imagine a web proxy server that did not support the HTTP protocol, because HTTP is the protocol used to communicate between a browser and web server.

In §7.6 I have provided a list of data center proxy providers that were active prior to the

„Squid“ yra ankstyvasis HTTP tarpinis serveris, parašytas 1990 m. Tai palaiko HTTP protokolą. Iš tikrųjų man būtų sunku įsivaizduoti žiniatinklio tarpinį serverį, kuris nepalaiko HTTP protokolo, nes HTTP yra protokolas, naudojamas ryšiui tarp naršyklės ir žiniatinklio serverio.

7.6 skyriuje pateikiau duomenų centro tarpinių paslaugų teikėjų, kurie buvo aktyvūs

date that `oxylabs.io` became available (§8.1.2). I have confirmed that all the listed providers (Your Private Proxy, Anonymous Proxies, Squid Proxies, Buy Proxies, My Private Proxy, SSL Private Proxy, Instant Proxies, Proxy-N-VPN, and Mexela) supported HTTP prior to the existence of `oxylabs.io`.

In §8.2.2 I describe my findings that Luminati was advertising HTTP support on their website, `luminati.io`, prior to the date that `oxylabs.io` became available (§8.1.2).

The support of HTTP by the OX and LU products is not unique; indeed, if either of them did not support the HTTP protocol, they would not have a viable web proxy server product. Thus, it is my opinion that the support for HTTP is neither novel nor interesting, within the context of evaluating the OX and LU products.

5.2.6 API IP Address List / API IP adresų sąrašas

I agree with the analysis in Mr. Butenis' report that both OX and LU provide an API for interacting with their respective websites.

However, this is a standard way that websites are used. There are standardized methods for constructing an API for *any* website. For example, Wrap API (<https://wrapapi.com>)

iki datos, kai atsirado `oxylabs.io`, sąrašą (8.1.2 skyrius). Patvirtinau, kad visi išvardyti paslaugų teikėjai (jūsų privatus tarpinis serveris, anoniminiai tarpiniai serveriai, kalmarų tarpiniai serveriai, pirkimo tarpiniai serveriai, mano privatus tarpinis serveris, SSL privatus tarpinis serveris, momentiniai tarpiniai serveriai, tarpinis serveris-N-VPN ir „Mexela“) palaikė HTTP iki `oxylabs.io`.

8.2.2 skyriuje aprašau savo išvadas, kad „Luminati“ reklamavo HTTP palaikymą jų svetainėje `luminati.io` iki datos, kai `oxylabs.io` tapo prieinama (8.1.2 skyrius).

HTTP palaikymas naudojant OX ir LU produktus nėra unikalus; Iš tikrųjų, jei kuris nors iš jų nepalaikytų HTTP protokolo, jie neturėtų perspektyvaus žiniatinklio tarpinio serverio produkto. Taigi, mano nuomone, vertinant OX ir LU produktus, palaikymas HTTP nėra nei naujas, nei įdomus.

Aš sutinku su p. Butenio pranešime pateikta analize, kad tiek OX, tiek LU teikia API sąveikai su atitinkamomis jų svetainėmis.

Tačiau tai yra standartinis internetinių svetainių naudojimo būdas. Yra standartizuoti bet kurios svetainės API kūrimo metodai. Pavyzdžiui, „Wrap API“

is a website that facilitates the automation of this task. In my personal experience, I have worked to develop and use numerous APIs for interacting with websites.

In 2000, Dr. Roy Fielding, published his doctoral dissertation describing a general framework for constructing web APIs [3]. This became the basis of what we now refer to as REST APIs. Today, essentially all companies providing web-based services use REST APIs for interacting with their web service. Thus, the feature of providing API access to web sites is now more than two decades old.

In §7.6 I have provided a list of data center proxy providers that were active prior to the date that `oxylabs.io` became available (§8.1.2). I have confirmed that the Mexela provider explicitly states that they provide API support in 2014. The absence of this statement on other websites does not necessarily indicate they did not support the use of APIs, as such a feature is quite common, as I previously noted, and might not be considered important enough to list. The Mexela service provided API support prior to the existence of `oxylabs.io`.

In §8.2.2 I describe my findings that Luminati was demonstrating the presence of API support on their website, `luminati.io`, prior to the date that `oxylabs.io` became

(<https://wrapapi.com>) yra svetainė, palengvinanti šios užduoties automatizavimą. Savo asmenine patirtimi dirbau kurdamas ir naudodamas daugybę API sąveikai su svetainėmis.

2000 m. Dr. Roy Fielding paskelbė daktaro disertaciją, kurioje aprašė bendrą interneto API kūrimo sistemą [3]. Tai tapo pagrindu to, ką mes dabar vadiname REST API. Šiandien iš esmės visos internetinės paslaugas teikiančios įmonės sąveikai su interneto paslauga naudoja REST API. Taigi API prieigos prie interneto svetainių teikimo funkcija yra senesnė nei du dešimtmečiai.

7.6 skyriuje pateikiau duomenų centro tarpinių paslaugų teikėjų, kurie buvo aktyvūs iki datos, kai atsirado `oxylabs.io`, sąrašą (8.1.2 skyrius). Patvirtinau, kad „Mexela“ teikėjas aiškiai nurodo, kad jie teikia API palaikymą 2014 m. Šis teiginys kitose interneto svetainėse nebūtinai reiškia, kad jie nepalaikė API naudojimo, nes tokia funkcija yra gana įprasta, kaip aš anksčiau pažymėjau, ir gali būti nelaikomas pakankamai svarbiu, kad galėtumėte išvardyti. „Mexela“ paslauga teikė API palaikymą iki `xyllabs.io` egzistavimo.

8.2.2 skyriuje aprašau savo išvadas, kad „Luminati“ demonstravo API palaikymą jų svetainėje `luminati.io` iki datos, kai `xyllabs.io` tapo prieinama (8.1.2 skyrius

available (§8.1.2).

skyrius). Aš taip pat aprašau savo išvadas, kad jie pasirodė turintys informacijos apie savo API savo svetainėje praėjus trimis dienoms nuo dienos, kai „oxylabs.io“ tapo prieinama.

SEO-Proxies.com is an advanced high quality private proxy solution. [Buy Private Proxies](#) here.

We deliver [highest quality IPs without abusive history](#), combined in a package that enables you to use them easily for your project, through your browser or as VPN.

All our IP addresses are clean (no gray/blacklists) and are true dedicated [private proxies](#). No one shares them with you, guaranteed !

Choose your IPs from 7 different US locations. We provide service in Denver (CO), SF-Bay-Area (CA), Los Angeles (CA), Seattle (WA), Dallas (TX), New Jersey (NJ), Atlanta (GA) and New York City (NY).

We issue only [US IP addresses](#), some other countries are possible on request.

Unlike various other services SEO-Proxies.com is not just an IP-reseller. We focus on highest reliability and the best possible IP quality.

Why we are The Most Advanced Proxy Provider

Order [Private Proxies](#) and gain access to our unique Webtool! Manage your proxies by clicking buttons only. It has never been that easy to change your IP or rotate through multiple IPs!

Automatic Browser Configuration allows you to [rotate your IP on-the-fly](#) without any configuration work! Watch the videos to see the magic work.

Switch the proxy type from [HTTP\(s\)](#) to [SOCKS](#) to [VPN](#) in a matter of seconds through your profile, you can immediately start using your chosen proxy protocol.

Integrate our IP solution into your scripts and tools in minutes ! The [API](#) is simple to use, free source code and expert support is available!

Give us a chance. [Order Exclusive Proxies](#) secure through Paypal and get started.

[SEO-Proxies.com](#) will be your reliable and professional partner, expert level customer service included.

<http://web.archive.org/web/20141218080300/http://www.seo-proxies.com/>

December 24, 2014

According to [archive.org](#) the website [seo-proxies.com](#) clearly indicated that they have an API for controlling the use of their proxy service. I have highlighted the section of their home page that indicates their API support.

Anot [archive.org](#), svetainė [seo-proxies.com](#) aiškiai nurodė, kad turi API, leidžiančią kontroliuoti savo tarpinio serverio paslaugų naudojimą. Aš išryškinau jų pagrindinio puslapio skiltį, kurioje nurodoma jų API palaikymas.

The seo-proxies.com API documentation, free tools and source code

- API documentation
 - [Introduction](#)
 - [General information](#)
 - [Using the API](#)
- [Free tools and source code](#)

The API documentation

Our API can be used to integrate automated proxy handling into tools, macros and scripts.
You could even create a bookmark in your browser to change your external IP on the fly!
Disclaimer: you may not use our API for illegal activities and you may only use our API if you own an active license
Hint: When configuring a tool, browser or script make sure you add our domain (www.seo-proxies.com) as proxy exception

Our API is designed to integrate our IP services (IP rotation) into custom tools and scripts without much effort, however it can still be quite a task to develop your data analysis or mining tools.
If you require help from experienced people please contact our customer support, we can recommend experienced developers.

Through the API you can do these actions

- * take IPs from the IP pool
- * start a proxy
- * stop a proxy
- * rotate your proxy to a random IP (IPs that were not used for a longer time are favored against IPs you used more recently)
- * rotate your proxy to a specific IP
- * verify your external IP

General

The API

The API is a HTTP protocol, that means you can access it through a browser, a TCP session, CURL or similar.
All API calls require: your userid, your encrypted password, a command name and optional parameters
Your API password: d24479f7966bdb4e57451b8e35bb16d
Your API userid : -1
Your first Proxy-Port : 7999

<http://web.archive.org/web/20141224192446/http://www.seo-proxies.com/?tab=api>

December 24, 2014

According to [archive.org](#) the website [seo-proxies.com](#) publishes details about their API and how a customer can use it. I have reproduced a portion of the relevant page above. The [archive.org](#) website provides a complete listing, including an example of how to rotate proxies (§5.2.3).

Thus, it is my opinion that the fact OX and LU products both support APIs is neither novel nor interesting, within the context of evaluating the OX and LU products.

Anot [archive.org](#), svetainė

[seo-proxies.com](#) skelbia informaciją apie jų API ir tai, kaip klientas gali ja naudotis. Atgaminu atitinkamo puslapio dalį aukščiau. Svetainėje [archive.org](#) pateikiamas visas sąrašas, įskaitant tarpinių serverių pasukimo pavyzdį (5.2.3 skyrius).

Taigi, mano nuomone, faktas, kad tiek OX, tiek LU produktai palaiko API, nėra nei naujas, nei įdomus vertinant OX ir LU produktus.

5.2.7 Whitelisting / Baltojo sąrašo sudarymas

The concept of whitelisting — or it's complement, blacklisting — is one that has been in use for decades. For example, in RFC 5782, a formal scheme was proposed for using the Domain Name System (§7.2.4) to create an Internet whitelist and blacklist, by

Baltojo sąrašo - arba jo papildymo, juodojo sąrašo - sąvoka vartojama dešimtmečius. Pavyzdžiui, RFC 5782 buvo pasiūlyta oficiali schema, kaip naudoti domenų vardų sistemą (7.2.4 skyrius), norint sukurti interneto baltąjį sąrašą ir juodąjį sąrašą, saugant IP

storing IP addresses within the DNS itself. There are numerous articles on whitelisting prior to 2015 [4]–[6]. Further, the LU product's use of them is consistent with the use of IP whitelisting as a security measure, which is a common use for IP whitelisting.

In §7.6 I have provided a list of data center proxy providers that were active prior to the date that `oxylabs.io` became available (§8.1.2). I have confirmed that the Squid Proxies service explicitly lists their support for whitelisting on their web page from 2010. The absence of this statement on other websites does not necessarily indicate they did not support the use of whitelisting, as such a feature is quite common, as I previously noted, and might not be considered important enough to list. The Squid Proxies service provided whitelisting prior to the existence of `oxylabs.io`.

Thus, it is my opinion that the fact the OX and LU products both support whitelisting is neither novel nor interesting, within the context of evaluating the OX and LU products.

5.2.8 Rate Limiting / Normos ribojimas

I disagree with the analysis in Mr. Butenis' report that both OX and LU offer rate limiting in their data center proxy products.

adresus pačiame DNS. Yra daugybė straipsnių apie baltojo sąrašo įtraukimą iki 2015 m. [4]–[6]. Be to, „LU“ produktas juos naudoja kaip IP priemonės įtraukimo į baltąjį sąrašą įtraukimą kaip saugumo priemonę, kuri yra įprasta naudoti IP baltąjį sąrašą.

7.6 skyriuje pateikiau duomenų centro tarpinių paslaugų teikėjų, kurie buvo aktyvūs iki datos, kai atsirado `oxylabs.io`, sąrašą (8.1.2 skyrius). Patvirtinau, kad „Squid Proxies“ tarnyba nuo 2010 m. Aiškiai nurodo jų palaikymą baltojo sąrašo sąraše. Šis teiginys, esantis kitose svetainėse, nebūtinai rodo, kad jie nepalaikė baltojo sąrašo naudojimo, nes tokia funkcija yra gana įprasta, kaip jau minėjau anksčiau ir gali būti nelaikomas pakankamai svarbiu sąrašą. Prieš pateikiant `oxylabs.io`, „Squid Proxies“ teikė baltųjų sąrašų sąrašą.

Taigi, mano nuomone, faktas, kad tiek OX, tiek LU produktai palaiko baltojo sąrašo sudarymą, nėra nei naujas, nei įdomus vertinant OX ir LU produktus.

Aš nesutinku su p. Butenio pranešime pateikta analize, kad tiek OX, tiek LU siūlo duomenų centro tarpinių produktų kainos ribojimą.

While Mr. Butenis does reference Luminati's website explaining how to configure rate limiting, he fails to observe that the instructions are how to configure the Luminati Proxy Manager, a program run on a computer the user controls, to support rate limiting; the data center proxy service does not support rate limiting, only the local proxy manager client — you can see the term "Luminati proxy manager" on the graphic that Mr. Butenis included in his report; this is the name of the local proxy manager, not a control interface for the data center service. This is different than the OX product features, where Teso implements rate limiting in their data center proxy server, as stated in Mr. Butenis' report. Thus, the support for rate limiting is **not** in LU, which is different than the OX product implementation of this feature.

According to Mr. Butenis' report, the OX product implements proxy server-side rate limiting, while according to my analysis the LU product does not. What Mr. Butenis demonstrates is using a client-side product, the Luminati Proxy Manager (§8.2.1) to perform rate limiting. This is distinct from a server-side rate limiting implementation, such as is provided by OX. Thus, it is my opinion that the OX and LU products differ in their support for rate limiting in the data center.

Nors p. Butenis daro nuorodą į „Luminati“ svetainę, paaiškinančią, kaip sukonfigūruoti greičio ribojimą, jis nepastebi, kad instrukcijos yra, kaip sukonfigūruoti „Luminati“ tarpinio serverio tvarkyklę, programą, vykdomą vartotojo valdomu kompiuteriu, kad palaikytų greičio ribojimą; duomenų centro tarpinio serverio paslauga nepalaiko greičio ribojimo, tik vietinis tarpinio serverio valdytojo klientas — grafike, kurį ponas Butenis įtraukė į savo pranešimą, galite pamatyti terminą „Luminati tarpinio serverio valdytojas“; tai yra vietinio tarpinio serverio valdytojo vardas, o ne duomenų centro paslaugos valdymo sąsaja. Tai skiriasi nuo OX produkto savybių, kai „Teso“ duomenų centro tarpiniame serveryje įgyvendina greičio ribojimą, kaip teigiama p. Butenio pranešime. Taigi greičio ribojimo palaikymas nėra LU, kuris skiriasi nuo šios funkcijos OX produkto įdiegimo.

Remiantis p. Butenio ataskaita, OX produktas įgyvendina tarpinio serverio greičio ribojimą, o mano analizei LU produktas to nedaro. Tai, ką demonstruoja ponas Butenis, naudoja normą ribojantį kliento produktą - „Luminati Proxy Manager“ (8.2.1 skyrius). Tai skiriasi nuo serverio greičio, ribojančio įgyvendinimą. Taigi, mano nuomone, „OX“ ir „LU“ produktai palaiko greičio ribojimą duomenų center.

So, there you have it. A nice little example of a multi-threaded, rate-limiting HTTP proxy in Go. Stay tuned to the next article, where I explain why `http.Get` and `http.Post` are not really what you want in an HTTP proxy, and how to get what you **DO** want from the `http` package.

Posted on [February 7, 2011](#) [February 7, 2011](#) by [jra](#)

<https://blog.nella.org/a-rate-limiting-http-proxy-in-go/>

February 7, 2011

John Regan posted an example proxy server, which included rate limiting, written in the Go computer programming language on his blog in February, 2011. This is prior to the date that `oxylabs.io` became available (§8.1.2).

John Regan savo tinklaraštyje 2011 m. Vasario mėn. Paskelbė įgaliootojo serverio, kuriame buvo nurodytas tarifų ribojimas, pavyzdį, parašytą kompiuterių programavimo kalba „Go“. Tai yra iki datos, kai `oxylabs.io` tapo prieinamas (8.1.2 skyrius).

Module ngx_http_limit_req_module

[Example Configuration](#)

[Directives](#)

[limit_req](#)

[limit_req_log_level](#)

[limit_req_zone](#)

The module ngx_http_limit_req_module (0.7.21) allow to limit the number of requests per defined key, in particular, the number of requests from a single IP address. The limitation is done using the “leaky bucket” method.

Example Configuration

```
http {
    limit_req_zone $binary_remote_addr zone=one:10m rate=1r/s;
    ...
    server {
        ...
        location /search/ {
            limit_req zone=one burst=5;
        }
    }
}
```

Directives

```
syntax: limit_req zone=name [burst=number] [nodelay] ;
default: —
context: http, server, location
```

Sets a zone and the maximum allowed burst of requests. If the rate of requests exceeds the rate configured for a zone, request processing is delayed such as that they are processed at a defined rate. Excessive requests are delayed until their number exceeds the defined number of bursts. In this case, the request is terminated with an error 503 (Service Temporarily Unavailable). By default, the number of bursts is equal to zero. For example, the directives

```
limit_req_zone $binary_remote_addr zone=one:10m rate=1r/s;

server {
    location /search/ {
        limit_req zone=one burst=5;
    }
}
```

allow not more than 1 request per second at an average, with bursts not exceeding 5 requests.

https://web.archive.org/web/20111116153759/https://nginx.org/en/docs/http/ngx_http_limit_req_module.html

November 26, 2011

The nginx web server, which can be used as a proxy server (§5.2.3) provided rate limiting at least by November 16, 2011. This is prior to the date that oxylabs.io became available (§8.1.2).

„Nginx“ žiniatinklio serveris, kurį galima naudoti kaip tarpinį serverį (§5.2.3 skyrius), teikė tarifą ribojantį bent jau iki 2011 m. Lapkričio 16 d. Tai yra iki datos, kai oxylabs.io tapo prieinama (§8.1.2 skyrius).

Proxy files for DotNet, Java and PHP

These proxy files support:

- Accessing cross domain resources
- Requests that exceed 2048 characters
- Accessing resources secured with token based authentication.
- [OAuth 2.0 app logins](#).
- Enabling logging
- Both resource and referer based rate limiting

<https://github.com/Esri/resource-proxy/tree/b701bd53500db2177d65fe822e7423c5b8053c35>

December 30, 2014

The website `github.com` includes a repository that provides examples of proxy server rate limiting in three different languages: DotNet, Java, and PHP, all common web server languages. I have highlighted the line that indicates the support for rate limiting. This is prior to the date that `oxylabs.io` became available (§8.1.2).

The functionality of rate limiting is standard in the computer networking field, including the field of proxy servers. It is also clearly present in product offerings that existed prior to the OX products. It is my opinion that the ability to provide rate limiting is neither novel nor interesting, within the context of evaluating the OX and LU products.

Svetainėje `github.com` yra saugykla, kurioje pateikiami keli įgaliootojo serverio greičio ribojimo pavyzdžiai trimis skirtingomis kalbomis: „DotNet“, „Java“ ir PHP, visomis įprastomis žiniatinklio serverio kalbomis. Aš paryškinau liniją, nurodančią normą riboti normą. Tai buvo galima įsigyti 2014 m. Gruodžio 25 d. Arba anksčiau, ty iki dienos, kurią `oxylabs.io` tapo prieinamas (8.1.2 skyrius).

Spartos ribojimo funkcija yra standartinė kompiuterių tinklo srityje, įskaitant tarpinius serverius. Tai taip pat aiškiai matyti gaminių pasiūloje, egzistavusioje prieš OX ir LU produktus. Mano nuomone, vertinant OX gaminius, galimybė riboti greitį nėra nei nauja, nei įdomi.

5.2.9 Unlimited Concurrent Sessions / Neribotas vienu metu vykstantis seansas

I agree with the analysis in Mr. Butenis'

Aš sutinku su p. Butenio pranešime pateikta

report that both OX and LU offer unlimited concurrent sessions. This is not unique to these two companies, as this is a common feature of other proxy service company offerings.

analize, kad tiek OX, tiek LU siūlo neribotas vienu metu vykstančias sesijas. Šioms dviem įmonėms tai nėra būdinga, nes tai yra bendras kitų tarpinių paslaugų įmonių pasiūlymų bruožas.

Questions about Proxy Connections

🕒 2014/07/23 16:55

Questions about Connections

- 1) I have set the "Maximum Connections" value as "10", why can I still connect to the 11th webpage?
- 2) What is the reason that when I am going to visit the webpage, it keeps connecting but never going to the content?

Introduction of Connections

- 1) Suppose that you have set the "Maximum Connections" value as "10" (Figure 1-1).



Figure 1-1

Note: Clients are only allowed to visit 10 proxy connections at the same time. If you open the 11th connection, there will be a connection prompt. It keeps connecting, but it will not send any proxy requests. However, as long as one of the previous 10 proxy connections is disconnected, the 11th connection will send the proxy request at once. If CCProxy does not work in this way, after you open the 11th proxy connection, you will find that it is not able to visit the webpage, which will affect the user experience.

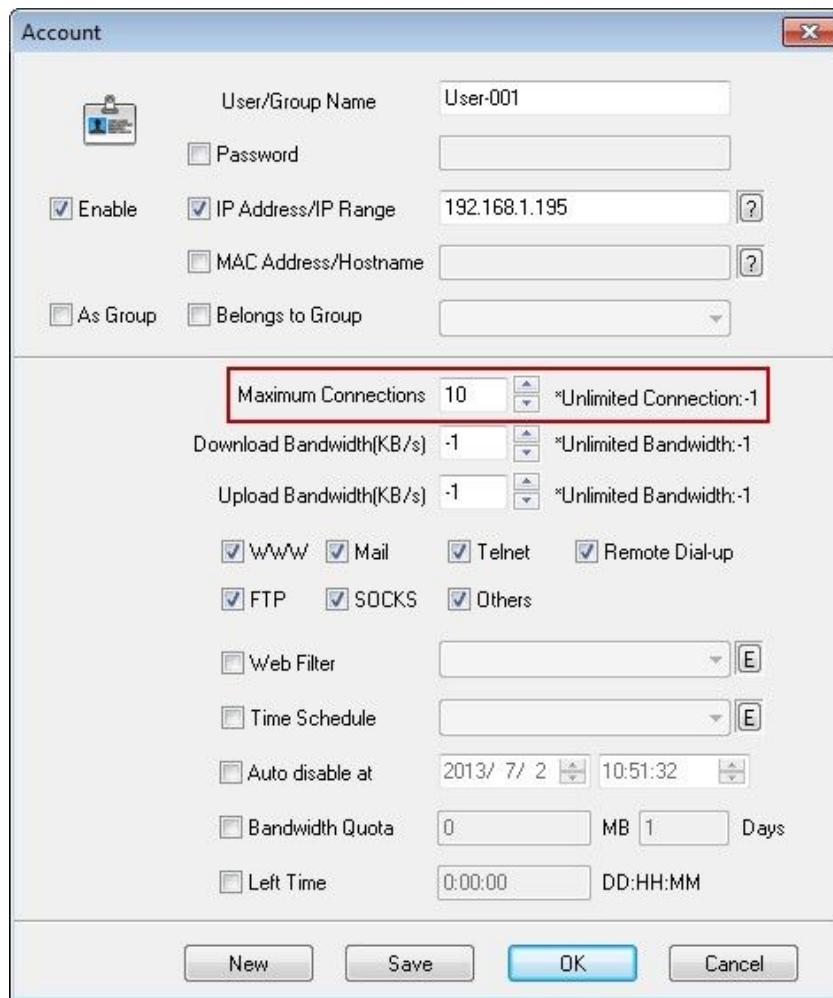
Related:

[How Many Users Can CCProxy Support](#)

[Why I Get a Dialog Box about Download Files and Fail to Open When I Visit Some Sites?](#)

<https://web.archive.org/web/20131121024334/https://www.youngzsoft.net/ccproxy/what-is-the-operating-principle-of-connections.htm>

November 21, 2013 (Page Dated July 2, 2013)



<http://www.youngzsoft.net/ccproxy/upload/what-is-the-operating-principle-of-connections-1.jpg>

Retrieved September 10, 2019

CCProxy server has been offered commercially since at least August 7, 2002 based upon the presence of their website as of that date (<https://bit.ly/2kdCeos>). Since at least November 21, 2013 they have documented their ability to configure connection limits. Their web page for this includes a screen capture that shows “unlimited” connections on their current website; this image is not captured by archive.org but appears to be the same image as on their current website (based

„CCProxy“ serveris buvo siūlomas komerciškai nuo mažiausiai 2002 m. Rugpjūčio 7 d., Atsižvelgiant į tai, kad tą dieną yra jų svetainė (<https://bit.ly/2kdCeos>). Nuo mažiausiai 2013 m. Lapkričio 21 d. Jie patvirtino savo sugebėjimą konfigūruoti ryšio ribas. Jų tinklalapyje yra ekrano fiksavimas, rodantis „neribotus“ ryšius jų dabartinėje svetainėje; šis vaizdas nėra užfiksuotas archive.org, bet atrodo, kad yra tas pats vaizdas, kaip ir dabartinėje jų svetainėje (atsižvelgiant į jo

upon its name).

pavadinimą).

25.3.2 To Configure a Resource Limits Policy

You can use the web interface Directory Service Control Center (DSCC) to perform this task.

1. View the properties of a resource limits policy.

```
$ dpconf get-resource-limits-policy-prop -h host -p port policy-name
```

The default properties of a resource limits policy are as follows:

connection-idle-timeout	: 1h
denied-presence-filter-attr	: all
denied-presence-filter-enabled	: false
description	: -
max-client-connections	: unlimited
max-connections	: unlimited
max-op-count-per-interval	: unlimited
max-simultaneous-operations-per-connection	: unlimited
max-total-operations-per-connection	: unlimited
minimum-search-filter-substring-length	: unlimited
op-count-per-interval-timeout	: 1s
referral-bind-policy	: default
referral-hop-limit	: default
referral-policy	: default
search-size-limit	: unlimited
search-time-limit	: unlimited
warning-op-count-per-interval	: unlimited



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[Legal Notices](#)

https://web.archive.org/web/20140422020014/https://docs.oracle.com/cd/E29127_01/doc.111170/e28972/client-dps-connections.htm

April 2, 2014

The Oracle® Fusion Middleware Administrator's Guide for Oracle Directory Server Enterprise Edition (<https://bit.ly/2kE37Ca>) documents how to configure connection limits between proxy clients and the Directory Proxy Server. The default is "unlimited". Thus, the concept of having unlimited proxy client connections to a proxy server existed no later than April 2, 2014.

„Oracle® Fusion“ tarpinės programinės įrangos administratoriaus vadovas, skirtas „Oracle Directory Server Server Enterprise Edition“ (<https://bit.ly/2kE37Ca>) dokumentuoja, kaip sukonfigūruoti tarpinių serverių klientų ir „Directory Proxy Server“ ryšio ribas. Numatytoji reikšmė yra „neribota“. Taigi neriboto tarpinio kliento ryšio su tarpiniu serveriu koncepcija egzistavo ne vėliau kaip 2014 m. Balandžio 2

§5.2.3 describes the process of load balancing. One reason to provide such load balancing is to enable “unlimited connections”. One name for this approach is *horizontal scaling* — the idea that we spread work out across multiple servers, to balance the load and permit an increase in the overall capacity of the system [7].

In §8.2.2 I describe my findings that Luminati was describing support for what, in my opinion, is a claim equivalent to “unlimited” concurrent sessions on their website, `luminati.io`, prior to the date that `oxylabs.io` became available (§8.1.2).

5.2.10 Choosing a Specific Peer / Konkretaus tinklo kolegos pasirinkimas

I agree with the analysis in Mr. Butenis’ report that both OX and LU offer the ability to choose a specific peer.

My analysis is that connecting to a single peer from a proxy server provider is the normal service provided. In §7.6 I provided a list of data center proxy providers that were active prior to the date that `oxylabs.io` became available (§8.1.2). For example, `anonymous-proxies.net` provides examples for using their proxy service, in which the specific proxy server to be used is specified.

In §5.2.3, I discuss the automatic assignment

d.

5.2.3 skyriuje aprašomas apkrovos balansavimo procesas. Tokio apkrovos balansavimo priežastis yra šios funkcijos įgalinimas. Vienas šio požiūrio pavadinimų yra horizontalus mastelio keitimas - idėja, kai mes paskirstome darbą keliuose serveriuose, kai subalansuotas apmokestinimas yra padidintas.

8.2.2 skyriuje aprašykite mano išvadą, kad Luminati apibūdino paramą tam, kas, mano manymu, yra „neribotos“ lygiagrečios sesijos jų tinklalapyje `luminati.io`, prieš datą, kai `oxylabs.io` tapo palaikoma. yra (8.1.2 skyrius).

Sutinku su p. Butenio pranešime pateikta analize, kad tiek OX, tiek LU siūlo galimybę pasirinkti konkretų bendraamžį.

Mano analizė yra tai, kad prisijungimas prie vieno tarpinio serverio teikėjo yra įprasta teikiama paslauga. 7.6 skyriuje pateikiau duomenų centro tarpinių paslaugų teikėjų, kurie buvo aktyvūs iki datos, kai atsirado `oxylabs.io`, sąrašą (8.1.2 skyrius). Pvz., `anonymous-proxies.net` pateikiami jų tarpinio serverio paslaugos naudojimo pavyzdžiai, kuriuose nurodomas konkretus naudojamas tarpinis serveris.

8.1.2 skyriuje aptariu automatinį skirtingų

of different proxy servers and agree with Mr. Butenis' analysis that both LU and OX provide this as part of their service offering. This is complementary to the specific peer model.

tarpinių serverių priskyrimą ir sutinku su p. Butenio analize, kad tiek LU, tiek OX tai teikia kaip dalį paslaugų. Tai papildo konkretų kolegų modelį.

This article discusses the various routing capabilities of the WebSphere® proxy server, which is a feature of IBM® WebSphere Application Server Network Deployment. Multiple configuration scenarios are presented, along with background information, setup instructions and tips to help you achieve success routing content using proxy server features in a secured environment.

https://www.ibm.com/developerworks/websphere/techjournal/1201_huebler/1201_huebler-pdf.pdf

January 25, 2012

The functionality for both single peer and multi-peer routing from a web client has existed as a feature in other products prior to January, 2015, the date that `oxylabs.io` was registered (§8.1.2). For example, the IBM WebSphere product offered proxy server routing. The configuration manual describes how to route proxy requests to both a specific server, as well as a group of servers. This mirrors the functionality of both “choosing a specific peer” as well as “load balancing/rotation” (§5.2.3).

Thus, it is my opinion that the fact OX and LU products permit choosing a specific proxy peer is neither novel nor interesting, within the context of evaluating the OX and LU products.

Pavienių ir daugialypių maršrutizatorių iš žiniatinklio kliento funkcionalumas jau egzistavo kituose produktuose iki 2015 m. Sausio mėn., Datos, kai `oxylabs.io` buvo įregistruotas (8.1.2 skyrius). Pavyzdžiui, „IBM WebSphere“ produktas pasiūlė tarpinio serverio maršruto parinkimą. Konfigūracijos vadove aprašoma, kaip nukreipti tarpinio serverio užklausas tiek į konkretų serverį, tiek į serverių grupę. Tai atspindi tiek pasirinkimo specifinį bendraamžį, tiek krovinių balansavimo / sukimosi (5.2.3 skyrius) funkcionalumą.

Taigi, mano nuomone, faktas, kad OX ir LU produktai leidžia pasirinkti konkretų tarpinį serverį, nėra nei naujas, nei įdomus vertinant OX ir LU produktus.

5.2.11 User's IP Address Anonymization / Vartotojo IP adreso anonimiškumas

I agree with the analysis in Mr. Butenis' report that both OX and LU support anonymizing the source computer's IP address. I did not find this surprising, since anonymization of the user's ("source") IP address is a fundamental reason for using a proxy service. Thus, I agree with Mr. Butenis' observation.

Sutinku su p. Butenio pranešime pateikta analize, kad tiek OX, tiek LU palaiko šaltinio kompiuterio IP adreso anonimizavimą. Manęs tai nestebino, nes vartotojo („šaltinio“) IP adreso anonimizavimas yra pagrindinė įgaliootojo serverio naudojimo priežastis. Taigi, aš sutinku su p. Butenio pastebėjimu.

- To hide the IP address of the client computer so that it can surf anonymous, this is mostly for security reasons. A proxy server can act as an intermediary between the user's computer and the Internet to prevent from attack and unexpected access.

<https://web.archive.org/web/20140823071552/https://www.youngzsoft.net/ccproxy/use-proxy-server.htm>

August 23, 2014.

The use of proxy servers to hide IP addresses is a common feature. Above I have shown a quotation from a website indicating this as a common use for proxy servers; the page is available on [archive.org](https://web.archive.org/web/20140823071552/https://www.youngzsoft.net/ccproxy/use-proxy-server.htm) as early as 2011 and continues to be found on the Internet as of the date of this report. See also §7.1.3. The CCProxy server has been offered commercially since at least August 7, 2002.

In §7.6 I have provided a list of data center proxy providers that were active prior to the date that oxylabs.io became available (§8.1.2). I have confirmed that all the listed providers (Your Private Proxy, Anonymous Proxies, Squid Proxies, Buy Proxies, My Private Proxy, SSL Private Proxy, Instant Proxies, Proxy-N-VPN, and Mexela)

Tarpinių serverių naudojimas norint paslėpti IP adresus yra bendra savybė. Aukščiau parodyčiau svetainės citatą, nurodančią, kad tai yra įprastas įgaliojimų serverių naudojimas; puslapis pasiekiamas [archive.org](https://web.archive.org/web/20140823071552/https://www.youngzsoft.net/ccproxy/use-proxy-server.htm) jau 2011 m. ir toliau yra internete internete nuo šios ataskaitos dienos. Taip pat žiūrėkite 7.1.3 skyrių. „CCProxy“ serveris komerciškai buvo siūlomas bent jau nuo 2002 m. Rugpjūčio 7 d.

7.6 skyriuje pateikiau duomenų centro tarpinių paslaugų teikėjų, kurie buvo aktyvūs iki datos, kai atsirado oxylabs.io, sąrašą (8.1.2 skyrius). Patvirtinau, kad visi išvardyti tiekėjai (jūsų privatus tarpinis serveris, anoniminiai tarpiniai serveriai, kalmarų tarpiniai serveriai, pirkimo tarpiniai serveriai, mano privatus tarpinis serveris,

supported source IP Address Anonymization prior to the existence of `oxylabs.io`.

In §8.2.2 I observed that Luminati claimed to have IP address anonymization on their website, `luminati.io`, prior to the date that `oxylabs.io` became available (§8.1.2).

The support of source (user's) IP address anonymization by the OX and LU products is not unique; indeed, if either of them did not support IP address anonymization, they would not have a viable web proxy product. Thus, it is my opinion that the support for IP address anonymization is neither novel nor interesting, within the context of evaluating the OX and LU products.

5.2.12 HTTPS Connection Support / HTTPS ryšio palaikymas

I agree with the analysis in Mr. Butenis' report that both OX and LU support the HTTPS protocol. I did not find this surprising, since using the HTTPS protocol has been a fundamental part of web proxy servers for almost as long as support for HTTP. See also §7.2.6.

In §7.6 I have provided a list of data center proxy providers that were active prior to the date that `oxylabs.io` became available

SSL privatus tarpinis serveris, momentiniai tarpiniai serveriai, tarpinis serveris-N-VPN ir „Mexela“) palaikė šaltinio IP adresų anonimiškumą anksčiau nei buvo `oxylabs.io`.

8.2.2 skyriuje pastebėjau, kad „Luminati“ teigė, kad jų interneto svetainėje `url luminati.io` jų IP adresas anonimizuojamas iki datos, kai `oxylabs.io` tapo prieinama (8.1.2 skyrius).

Šaltinio (vartotojo) IP adreso anonimizacijos palaikymas naudojant OX ir LU produktus nėra unikalus; Iš tikrųjų, jei nė vienas iš jų nepalaikytų IP adresų anonimiškumo, jie neturėtų perspektyvaus žiniatinklio tarpinio produkto. Taigi, mano nuomone, vertinant OX ir LU produktus, IP adresų anonimiškumo palaikymas nėra nei naujas, nei įdomus.

Aš sutinku su p. Butenio pranešime pateikta analize, kad tiek OX, tiek LU palaiko HTTPS protokolą. Manęs tai nestebino, nes HTTPS protokolo naudojimas buvo pagrindinė žiniatinklio tarpinių serverių dalis beveik tol, kol palaikė HTTP. Taip pat žiūrėkite 7.2.6 skyrių.

7.6 skyriuje pateikiau duomenų centro tarpinių paslaugų teikėjų, kurie buvo aktyvūs iki datos, kai atsirado `oxylabs.io`, sąrašą

(§8.1.2). I have confirmed that all the listed providers (Your Private Proxy, Anonymous Proxies, Squid Proxies, Buy Proxies, My Private Proxy, SSL Private Proxy, Instant Proxies, Proxy-N-VPN, and Mexela) supported HTTPS prior to the existence of `oxylabs.io`.

In §8.2.2 I describe my findings that Luminati was advertising HTTPS support on their website, `luminati.io`, prior to the date that `oxylabs.io` became available (§8.1.2).

The support of HTTPS by the OX and LU products is not unique; indeed, if either of them did not support the HTTPS protocol, they would not have a commercially viable web proxy product. Thus, it is my opinion that the support for HTTPS is neither novel nor interesting, within the context of evaluating the OX and LU products.

5.2.13 Resolution of Domain Name System in the Proxy Server / *Tarpinio serverio domenų vardų sistemos skiriamoji geba*

I agree with the analysis in Mr. Butenis' report that both OX and LU support DNS resolution in the proxy server.

(8.1.2 skyrius). Patvirtinau, kad visi išvardyti paslaugų teikėjai (jūsų privatus tarpinis serveris, anoniminiai tarpiniai serveriai, kalmarų tarpiniai serveriai, pirkimo tarpiniai serveriai, mano privatus tarpinis serveris, SSL privatus tarpinis serveris, momentiniai tarpiniai serveriai, tarpinis serveris-N-VPN ir „Mexela“) palaikė HTTPS, kol nebuvo `oxylabs.io`.

8.2.2 skyriuje aprašau savo išvadas, kad „Luminati“ reklamavo HTTPS palaikymą jų svetainėje `luminati.io` iki datos, kai `oxylabs.io` tapo prieinama (8.1.2 skyrius).

„HTTPS“ palaikymas OX ir LU gaminiais nėra unikalus; Iš tikrųjų, jei kuris nors iš jų nepalaikytų HTTPS protokolo, jie neturėtų komerciškai perspektyvaus žiniatinklio tarpinio produkto. Taigi, mano nuomone, vertinant OX ir LU produktus, HTTPS palaikymas nėra nei naujas, nei įdomus.

Aš sutinku su p. Butenio pranešime pateikta analize, kad tiek OX, tiek LU palaiko DNS skiriamąją galią įgaliotajame serveryje.

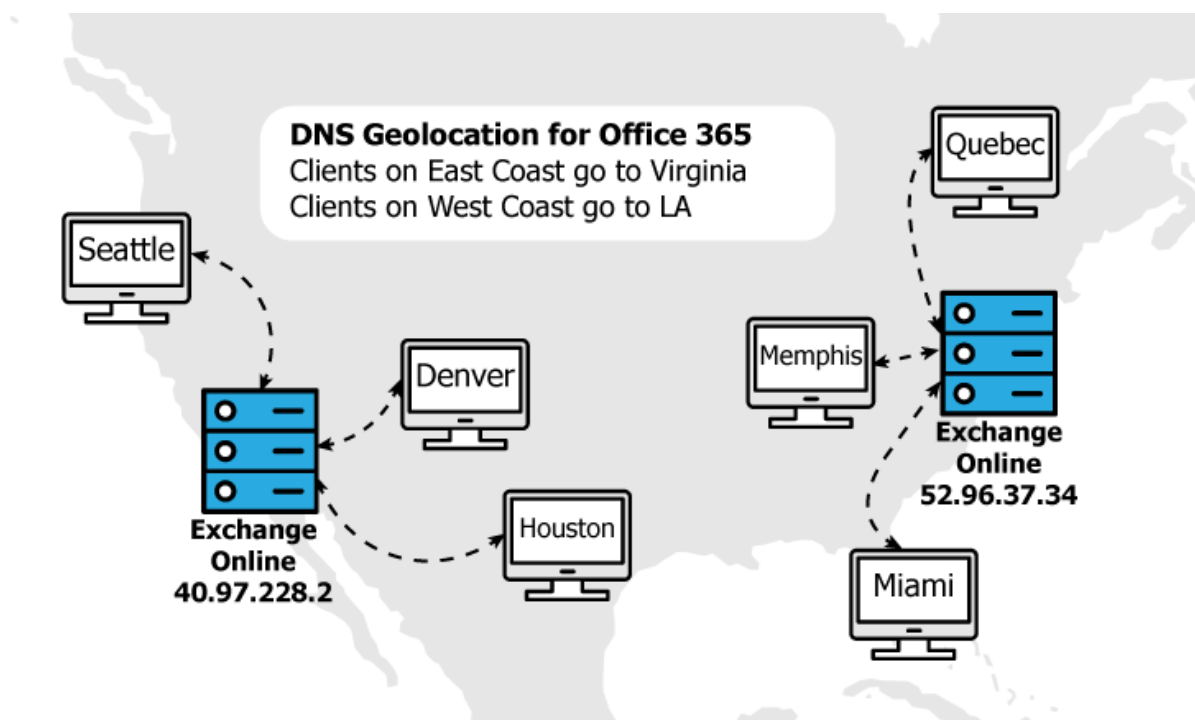
Controlling where DNS resolution is performed

- If you use `lum-customer-{your_customer_id}-zone-{your_zone}-dns-local` domain names will be resolved and cached by the Super Proxy.
- If you wish to perform the DNS resolution at the Proxy Peer, use `lum-customer-{your_customer_id}-zone-{your_zone}-dns-remote`
This is slower, but will give you the same IP as users in your chosen country receive.

<https://luminati.io/faq#control-dns-resolution>

Mr. Butenis' report combines two different DNS resolution mechanisms where he states "DNS resolution in the proxy server as a feature is indicated in the documentation available on the LU website and in the FAQ section" and "Website text: '... domain names will be resolved and cached by the Super Proxy'". The proxy server and the super proxy are two distinct options, not a single option. Thus, based upon Mr. Butenis' report, OX performs DNS resolution in the proxy server. I observe that LU provides the option of having the proxy server *or* the "super proxy" resolve the domain name. This distinction is important because it trades off performance of DNS lookup against geolocation correct lookup.

Pono Butenio ataskaita sujungia du skirtingus DNS skyrimo mechanizmus, kuriuose jis nurodo, kad „DNS skiriamoji geba tarpiniame serveryje kaip ypatybė yra nurodoma LU svetainėje esančiuose dokumentuose ir DUK skyriuose“ ir „Svetainės tekstas:“ .. Domenų vardus išspręs ir talpykloje kaups „Super Proxy“. Tarpinis serveris ir super tarpinis serveris yra dvi skirtingos parinktys, o ne viena. Taigi, remiantis p. Butenio ataskaita, OX vykdo DNS skiriamąją galią įgaliojajame serveryje. Pastebiu, kad LU suteikia galimybę įgaliojajį serverį arba atskirą vardų serverį („super proxy“) išspręsti domeno vardą. Šis skirtumas yra svarbus, nes jis pašalina DNS paieškos atlikimą nuo teisingos geografinės padėties nustatymo.



<https://www.exoprise.com/solutions/office-365-testing/>

This distinction between the proxy server and the super proxy for DNS resolution is because the proxy server may not be in the same location as the super proxy. Location sensitive domain name to IP address mapping is an important mechanism for permitting global services to scale, as well as to limit services to a specific geographical location. In the above example, the diagram explains how Microsoft uses location sensitive DNS resolution to direct geographically dispersed clients to nearby data centers. I describe the DNS process in §7.2.4.

Šis skirtumas tarp įgaliotojo serverio ir „super“ tarpinio serverio, skirtas DNS skiriamąją gebą, yra todėl, kad įgaliotasis serveris gali būti ne toje pačioje vietoje kaip ir pagrindinis tarpinis serveris. Domeno, nustatančio pagal vietovę jautraus domeno vardo IP adresams atvaizdavimas, yra svarbus mechanizmas, leidžiantis teikti globalias paslaugas mastui, taip pat apriboti paslaugas tam tikroje geografinėje vietoje. Aukščiau pateiktame pavyzdyje diagrama paaiškina, kaip „Microsoft“ naudoja vietoje jautrią DNS skiriamąją gebą nukreipdama geografiškai išsklaidytus klientus į netoliese esančius duomenų centrus. Aprašau DNS procesą 7.2.4 skyriuje .

```

tony@IAM-DESKTOP:~$ nslookup
> server 88.119.24.91
Default server: 88.119.24.91
Address: 88.119.24.91#53
> netflix.com
Server:      88.119.24.91
Address:     88.119.24.91#53

Non-authoritative answer:
Name:   netflix.com
Address: 52.209.224.161
Name:   netflix.com
Address: 54.229.68.77
Name:   netflix.com
Address: 52.49.6.246
Name:   netflix.com
Address: 52.48.104.170
Name:   netflix.com
Address: 54.171.21.76
Name:   netflix.com
Address: 34.249.125.167
Name:   netflix.com
Address: 52.51.246.114
Name:   netflix.com
Address: 52.49.120.6
Name:   netflix.com
Address: 2a01:578:3::34d1:eb8d
Name:   netflix.com
Address: 2a01:578:3::22fa:2993
Name:   netflix.com
Address: 2a01:578:3::341e:2dc6
Name:   netflix.com
Address: 2a01:578:3::34d2:745
Name:   netflix.com
Address: 2a01:578:3::341f:91b7
Name:   netflix.com
Address: 2a01:578:3::341f:b664
Name:   netflix.com
Address: 2a01:578:3::3648:d8f1
Name:   netflix.com
Address: 2a01:578:3::36ab:7445
> server 1.1.1.1
Default server: 1.1.1.1
Address: 1.1.1.1#53
> netflix.com
Server:      1.1.1.1
Address:     1.1.1.1#53

Non-authoritative answer:
Name:   netflix.com
Address: 54.201.91.38
Name:   netflix.com
Address: 54.213.182.74
Name:   netflix.com
Address: 52.37.69.124
Name:   netflix.com
Address: 52.37.77.112
Name:   netflix.com
Address: 52.37.219.6
Name:   netflix.com
Address: 52.41.20.47
Name:   netflix.com
Address: 52.41.193.16
Name:   netflix.com
Address: 54.149.101.155
Name:   netflix.com
Address: 2620:108:700f::3420:be97
Name:   netflix.com
Address: 2620:108:700f::3425:457c
Name:   netflix.com
Address: 2620:108:700f::3425:4d70
Name:   netflix.com
Address: 2620:108:700f::3425:db06
Name:   netflix.com
Address: 2620:108:700f::3429:142f
Name:   netflix.com
Address: 2620:108:700f::23a4:4198
Name:   netflix.com
Address: 2620:108:700f::23a4:93cf
Name:   netflix.com
Address: 2620:108:700f::341a:6f2b
>

```

The location of the name resolution is important for location sensitive services such as Netflix. In the figure above, I performed name resolution using two different name servers: one located in Vilnius, Lithuania (88.119.24.91), and another using the public Cloudflare DNS server (1.1.1.1). I note that the IP addresses returned for `netflix.com`

Vardo išsprendimo vieta yra svarbi tokioms vietinėms paslaugoms kaip „Netflix“. Aukščiau esančiame paveikslėlyje aš vykdžiau vardo skyrimą dviem skirtingais vardų serveriais: vieną, esantį Vilniuje, Lietuvoje (88.119.24.91), o kitą, naudodamas viešą „Cloudflare“ DNS serverį (1.1.1.1). Atkreipiu dėmesį, kad `netflix.com` grąžinti IP adresai

are different, depending upon the location of the DNS server used for resolving the query.

For the DNS server in Vilnius, Lithuania (88.119.24.191), the IP Version 4 addresses returned were, in sorted order:

34.249.125.167, 52.48.104.170, 52.49.6.246, 52.49.120.6, 52.51.246.114, 52.209.224.161, 54.171.21.76, and 54.229.68.77.

For the Cloudflare DNS server accessed from my location (Vancouver, Canada), the IP Version 4 addresses returned were, in sorted order: 52.37.69.124, 52.37.77.112, 52.37.219.6, 52.41.20.47, 52.41.193.16, 54.149.101.155, 54.201.91.38, and 54.213.182.74.

Thus, both the LU and OX products will work with geolocation sensitive DNS resolution. However, the LU product can offer DNS resolution inside their data center for customers that do not need proxy server specific geolocation sensitive DNS resolution. It is my opinion that this difference is significant and distinguishes the two products from one another.

The Internet Engineering Task Force (IETF) published a “Best Current Practices” document entitled “DNS Proxy Implementation Guidelines” as RFC 5625 in August 2009 [8]. Thus, by August 2009, DNS proxy implementations were common and well-understood.

skiriasi, atsižvelgiant į DNS serverio, naudojamo užklausai išspręsti, vietą.

DNS serverio Vilniuje, Lietuvoje (88.119.24.191), grąžinti IP versijos 4 adresai buvo surūšiuoti: 34.249.125.167, 52.48.104.170, 52.49.6.246, 52.49.120.6, 52.51.246.114, 52.209.224.161, 54.171.21.76, ir 54.229.68.77.

„Cloudflare“ DNS serveriui, prieinamam iš mano vietos (Vankuveris, Kanada), grąžinti 4 versijos IP adresai buvo surūšiuoti: 52.37.69.124, 52.37.77.112, 52.37.219.6, 52.41.20.47, 52.41.193.16, 54.149.101.155, 54.201.91.38, ir 54.213.182.74.

Taigi, ir LU, ir OX produktai veiks su geografine padėtimi jautria DNS skiriamąja geba. Tačiau LU produktas savo duomenų center gali pasiūlyti DNS skiriamąją gebą klientams, kuriems nereikia įgaliojoto serverio specifinės geografinės vietos nustatymo jautrios DNS rezoliucijos. Mano nuomone, šis skirtumas nėra akivaizdus ir išskiria abu produktus vienas nuo kito.

Interneto inžinerijos darbo grupė (IETF) 2009 m. Rugpjūčio mėn. Paskelbė „Geriausios dabartinės praktikos“ dokumentą pavadinimu „DNS tarpinio serverio diegimo gairės“ kaip RFC 5625. Taigi iki 2009 m [8]. Rugpjūčio mėn. DNS tarpinių serverių diegimas buvo įprastas ir suprantamas.

Sep
6
2014

Complete Guide on What is A Smart DNS Proxy Server and How Does it Work

Category: [Smart DNS Proxy](#)

What is a Smart DNS Proxy Server?



Smart DNS Proxy is sophisticated technology which allows Internet users all over the world to unblock Geo-restricted (or Geo-Blocked) websites like Netflix, Hulu, WWE Newtork, BelnSports.net, BBC and many more popular [websites which contain regional-restricted content](#).

<https://bit.ly/10PXR6p>

September 6, 2014 (Date of Article)

Smart DNS Proxy has an article on their website dated September 6, 2014 that describes the function of their DNS proxy service.

Išmanusis DNS tarpinis serveris jų svetainėje turi 2014 m. Rugsėjo 6 d. Straipsnį, kuriame aprašoma jų DNS tarpinio serverio paslaugos funkcija.



Why DNS Proxy?

A DNS Proxy allows you to unblock censorship filters and access websites, apps and services from anywhere in the world.

Benefits of SmartDNS

- ✓ Unblock Netflix, Hulu, and 80+ More Channels From Anywhere in the World
- ✓ Access Websites Such as Facebook, YouTube and Twitter
- ✓ SmartDNS Unblocks Geo-Restricted Sites, All Others Retain Your Local IP
- ✓ Quick and Easy Set Up, Access the Content You Want In Minutes
- ✓ Minimal To No Impact On Your Connection Speed

START NOW

Only \$4.16/Month

<https://web.archive.org/web/20140726235629/https://ironsocket.com/why-dns-proxy>

July 26, 2014

As of July 26, 2014, Ironsocket provided a DNS proxy service. This date is prior to the date that `oxylabs.io` became available (§8.1.2).

The support of DNS proxy services by the OX and LU products is consistent with broad understanding in the field of providing such services. Thus, it is my opinion that the support for DNS proxy services is neither novel nor interesting, within the context of evaluating the OX and LU products.

Nuo 2014 m. Liepos 26 d. „Ironsocket“ teikė DNS tarpinio serverio paslaugą. Ši data yra ankstesnė už datą, kai `oxylabs.io` tapo prieinama (8.1.2 skyrius).

DNS tarpinių serverių paslaugų palaikymas naudojant OX ir LU produktus atitinka platų supratimą tokių paslaugų teikimo srityje. Taigi, mano nuomone, vertinant OX ir LU produktus, DNS tarpinių serverių paslaugų palaikymas nėra nei naujas, nei įdomus.

5.3 Statement of Claims / Ieškinių pareiškimas

In this subsection I address paragraph 65 of the Statement of Claims, which sets forth six different “technical properties” that are common to both the Teso and Luminati data

Šiame poskirsnyje nagrinėju ieškinių pareiškimo 65 punktą, kuriame išdėstytos šešios skirtingos „techninės savybės“, būdingos ir „Teso“, ir „Luminati“ duomenų

center proxy products. It is my opinion that none of these properties are novel or interesting, within the context of evaluating the OX and LU products. These properties are found in many other products, including data center proxy products, as I explain in this subsection.

5.3.1 Sorting of IP Addresses / IP adresų rūšiavimas

The basic concept of sorting data is one that computer science has been exploring for decades. One of the most well-known books on the subject is *The Art of Computer Programming: Volume 3 Sorting and Searching* [9], which was published in 1973. One reason this is important in computer programs is efficiency: the time it takes to search an unordered set of data grows linearly with the size of the set. If the data is sorted, the time it takes to search the set is logarithmic. What that means is that if a set of data consists of 1,000 entries and it requires one second to process each entry, *on average* it would require 500 seconds to find the needed entry in the set, and worst-case 1,000 seconds. If the same set of data is sorted, the worst-case time is ten seconds.

IP addresses are fixed size numbers, exactly 32 bits in length. Sorting them makes sense, both for computer programs as well as presentation to humans, since both computers and humans can scan a sorted list

centro įgaliotiesiems gaminiams. Mano nuomone, vertinant OX ir LU produktus, nė viena iš šių savybių nėra nauja ar įdomi. Kaip aš paaiškinau šiame poskirsnyje, šios savybės yra daugelyje kitų produktų, įskaitant duomenų centro tarpinius produktus.

Pagrindinė susijusių duomenų rūšiavimo koncepcija yra ta, kurią informatika tyrinėjo dešimtmečius. Viena iš labiausiai žinomų knygų šia tema yra „Kompiuterių programavimo menas: 3 tomo rūšiavimas ir paieška“ [9], išleista 1973 m. Viena iš priežasčių, kodėl tai svarbu kompiuterinėms programoms, yra efektyvumas: laikas, kurio reikia paieškai netvarkytas duomenų rinkinys auga tiesiškai, atsižvelgiant į rinkinio dydį. Jei duomenys rūšiuojami, rinkinio paieškai reikia logaritminio laiko. Tai reiškia, kad jei duomenų rinkinį sudaro 1 000 įrašų ir kiekvienam įrašui apdoroti reikia vienos sekundės, vidutiniškai prireiks 500 sekundžių norint rasti reikiamą įrašą rinkinyje, o blogiausiu atveju - 1000 sekundžių. Jei rūšiuojami tie patys duomenys, blogiausias atvejis yra dešimt sekundžių.

IP adresai yra fiksuoto dydžio skaičiai, tiksliai 32 bitų ilgio. Rūšiuoti juos yra prasminga tiek kompiuterinėms programoms, tiek pateikimui žmonėms, nes tiek kompiuteriai, tiek žmonės gali rūšiuoti sąrašą

more quickly than an unsorted list. It also simplifies comparing two lists, which is why I sorted the IP addresses in §5.2.13.

Sort IP Addresses with GNU sort

Paul Heinlein <heinlein@madboa.com>

Initial publication: July 30, 2004

Most recent revision: August 3, 2004

How to sort IP addresses using the GNU **sort** utility.

The short answer. Here's the invocation that works. It's explained in the long answer that follows.

```
sort -n -t . -k 1,1 -k 2,2 -k 3,3 -k 4,4
```

<https://web.archive.org/web/20140717082534/https://www.madboa.com/geek/sort-addr/>

For example, one description of this, dating from 2004, and shown via this website screen shot dated July 1, 2014, explains how to sort IP addresses using standard UNIX/Linux command line tools — related to the tools that Mr. Butenis used in his report.

Pavyzdžiui, viename šio aprašymo, kilusio nuo 2004 m., Kuris buvo parodytas per 2014 m. Liepos 1 d. Šios svetainės ekrano kopiją, paaiškinta, kaip rūšiuoti IP adresus naudojant standartinius UNIX / Linux komandinės eilutės įrankius, susijusius su įrankiais, kuriuos p. Butenis naudojamas jo ataskaitoje.

The order of the proxy servers in the list is not relevant. You can sort the list by address, port and type (protocol).

<https://web.archive.org/web/20140702201856/http://www.proxifier.com/documentation/v3/Proxifier.pdf>

One example for sorting IP addresses in a web proxy server is shown above. In this case, Proxifier Version 3.15 provides an interface that permits the human user the ability to choose how information in a list is sorted, including sorting by IP address.

Thus, it is my opinion that the feature of sorting IP addresses is neither novel nor

Aukščiau pateiktas vienas pavyzdys, kaip rūšiuoti IP adresus žiniatinklio tarpiniame serveryje. Šiuo atveju „Proxifier“ 3.15 versijoje yra sąsaja, leidžianti vartotojui pasirinkti, kaip rūšiuoti sąrašo informaciją, įskaitant rūšiavimą pagal IP adresą.

Taigi, mano nuomone, vertinant „Teso“ ir „Luminati“ produktus, IP adresų rūšiavimo

interesting, within the context of evaluating the Teso and Luminati products.

ypatybė nėra nei nauja, nei įdomi.

5.3.2 Balancing (rotation) of queries / Užklausų balansavimas (pasukimas)

I addressed this point in §5.2.3. It is my opinion that the features of load balancing and IP address rotation are neither novel nor interesting when comparing the Teso and Luminati products.

Aš atkreipiau dėmesį į šį punktą 5.2.3 skyriuje. Mano nuomone, palyginus „Teso“ ir „Luminati“ produktus, apkrovos balansavimo ir IP adresų sukimo ypatybės nėra nei naujos, nei įdomios.

5.3.3 SOCKS5 Protocol Support / SOCKS5 protokolo palaikymas

I addressed this point in §5.2.4. It is my opinion that the feature of supporting the SOCKS5 protocol is neither novel nor interesting when comparing the Teso and Luminati products.

Į tai atkreipiau dėmesį 5.2.4 skyriuje. Manau, kad palyginus „Teso“ ir „Luminati“ gaminius, „SOCKS5“ protokolo palaikymo bruožas nėra nei naujas, nei įdomus.

5.3.4 HTTP Protocol Support / HTTP protokolo palaikymas

I addressed this point in §5.3.4. It is my opinion that LU does support the HTTP protocol because it is a proxy server for websites, which use the HTTP protocol. If they did not support this protocol, they would not have a web proxy server product offering — *all* proxy web servers support the HTTP protocol.

Aš atkreipiau dėmesį į šį punktą 5.3.4 skyriuje. Manau, kad „Luminati“ duomenų centro produktas palaiko HTTP protokolą, nes jis yra tarpinis serveris, skirtas svetainėms, kurios naudoja HTTP protokolą. Jei jie nepalaikytų šio protokolo, jie neturėtų interneto tarpinio serverio produkto, siūlančio — visi tarpiniai serveriai palaiko HTTP protokolą.

5.3.5 Authentication / Autentifikavimas

10.4.8 407 Proxy Authentication Required

This code is similar to 401 (Unauthorized), but indicates that the client must first authenticate itself with the proxy. The proxy MUST return a Proxy-Authenticate header field (section 14.33) containing a challenge applicable to the proxy for the requested resource. The client MAY repeat the request with a suitable Proxy-Authorization header field (section 14.34). HTTP access authentication is explained in "HTTP Authentication: Basic and Digest Access Authentication" [43].

<https://www.w3.org/Protocols/rfc2616/rfc2616-sec10.html#sec10.4.8>

The use of proxy authentication is a well-known feature. An error code indicating that authentication is required for a proxy server is present in RFC 2616, which was first published in 1999 [10]. The specific mechanisms to be used for authentication were also published in 1999 [11].

It is my opinion that the presence of authentication in a proxy server is a well understood and commonly found feature in many proxy server products. The presence of the ability to perform authentication has no value with respect to comparing the Teso and Luminati products.

Tarpinio serverio autentifikavimo naudojimas yra gerai žinoma funkcija. Klaidos kodas, nurodantis, kad reikia įgaliootojo serverio autentifikavimo, yra RFC 2616, pirmą kartą paskelbtame 1999 m [10]. Specifiniai autentifikavimo mechanizmai taip pat buvo paskelbti 1999 m [11].

Manau, kad autentifikavimas tarpiniame serveryje yra gerai suprantama ir dažniausiai sutinkama funkcija daugelyje tarpinių serverių. Galimybė atlikti autentifikavimą neturi jokios reikšmės lyginant „Teso“ ir „Luminati“ produktus.

5.3.6 Encrypted Communications Channel Capacity / Šifruoto ryšio kanalo talpa

Both the Teso and Luminati data center products use encryption for communications. This may be done using the HTTPS protocol, which is a standard protocol. Other protocols may have been used, such as those described in §7.1.5. It is my opinion that the presence of this feature is neither novel nor interesting, within the context of evaluating the OX and LU products, as it is an existing mechanism, which can be implemented using

Tiek „Teso“, tiek „Luminati“ duomenų centro produktai naudoja šifravimą ryšiams. Tai gali būti padaryta naudojant HTTPS protokolą, kuris yra standartinis protokolas. Gali būti naudojami kiti protokolai, tokie kaip aprašyti 7.1.5 skyriuje. Manau, kad vertinant OX ir LU produktus, ši savybė nėra nei nauja, nei įdomi, nes tai yra esamas mechanizmas, kuris gali būti įgyvendinamas naudojant visuotinai žinomus privataus

commonly known mechanisms for ensuring private (encrypted) communications.

(užšifruoto) ryšio užtikrinimo mechanizmus.

5.4 Server Locations / Serverio vietos

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Top locations

 United States 3,589,769 IPs	 Germany 1,198,423 IPs	 Russia 1,182,670 IPs	 United Kingdom 1,156,437 IPs
 France 973,264 IPs	 Canada 731,834 IPs	 Australia 347,348 IPs	 China 208,513 IPs

oxylabs.io/locations

IP addresses are assigned in *blocks*, or ranges of sequential values, from the total space, which consists of 2^{32} possible values in the commonly used IP version 4 address space; many of these values have been assigned in such a way that they are defined as being unused on the internet and thus, the actual

IP adresai yra priskiriami blokais arba nuosekliųjų verčių diapazonams nuo visos erdvės, kurią sudaro 2^{32} galimos vertės dažniausiai naudojamoje IP versijos 4 adresų erdvėje; daugelis šių verčių buvo priskirtos taip, kad jos būtų apibrėžtos kaip nenaudojamos internete, taigi tikrasis

number of valid public IP version 4 addresses is smaller than the maximum number of 4,294,967,296. The distribution of public IP addresses is managed by various internet authorities; the primary authority is the American Registry for Internet Numbers (ARIN). ARIN in turn assigns control over ranges of IP network addresses to other regional authorities.

Mr. Butenis used an IP address identification service called icanhazip.com. This service is managed by an individual named Major Hayden, as described on his website <https://major.io/icanhazip-com-faq>. It is a popular service to use because it only returns the specific information — the IP address corresponding to the sender of the request — and does not include unnecessary information, such as advertising, which is often the case for other services.

Mr. Butenis observes that the address returned by <http://icanhazip.com> was the address of the proxy server, not of the original client. I find this reasonable, since the purpose of using a proxy is to avoid identification of the original computer. In fact, Mr. Butenis used multiple different proxy servers in his testing. For the OX product, Mr. Butenis used servers with IP Addresses 66.234.192.245, 66.234.192.2, 64.140.133.239, 148.251.73.163, and 82.135.144.188. Mr. Butenis used servers

galiojančių viešosios IP versijos 4 adresų skaičius yra mažesnis nei maksimalus 4 294 967 296. Viešųjų IP adresų platinimą tvarko įvairios interneto institucijos; pagrindinė institucija yra Amerikos interneto numerių registras (ARIN). ARIN savo ruožtu IP tinklo adresų diapazonus kontroliuoja kitoms regionų valdžios institucijoms.

Ponas Butenis pasinaudojo IP adreso identifikavimo tarnyba, vadinama icanhazip.com. Šią paslaugą valdo asmuo, vardu Major Hayden, kaip aprašyta jo svetainėje <https://major.io/icanhazip-com-faq>. Tai populiaru naudoti paslauga, nes ji grąžina tik konkrečią informaciją — IP adresą, atitinkantį užklauskos siuntėją - ir neapima nereikalingos informacijos, tokios kaip reklama, kuri dažnai taikoma kitoms paslaugoms.

Ponas Butenis pastebi, kad [url http://icanhazip.com](http://icanhazip.com) grąžintas adresas buvo įgaliootojo serverio, o ne pradinio kliento adresas. Manau, kad tai yra pagrįsta, nes tarpinio serverio naudojimo tikslas yra išvengti originalaus kompiuterio identifikavimo. Tiesą sakant, p. Butenis bandymuose naudojo kelis skirtingus tarpinius serverius. OX produktui ponas Butenis naudojo serverius, kurių IP adresai buvo 66.234.192.245, 66.234.192.2, 64.140.133.239, 148.251.73.163 ir

with domain names of proxy.oxyllabs.io and stats.oxyllabs.io.

For the LU product, Mr. Butenis used at least one server with the domain name of luminati.io.

Mr. Butenis used the server with domain name of icanhazip.com.

82.135.144.188. Ponas Butenis naudojo serverius su proxy.oxyllabs.io ir stats.oxyllabs.io domenų vardais.

Savo produktui ponas Butenis naudojo bent vieną serverį, kurio domeno vardas buvo luminati.io.

Ponas Butenis naudojo serveriu, kurio domeno vardas buvo icanhazip.com.

5.4.1 IP Address 66.234.192.245 Location / IP adresas 66.234.192.245 Vieta

Geolocation data from IP2Location (Product: DB6, updated on 2019-9-1)

IP Address	Country	Region	City
66.234.192.245	United States 	California	Concord
ISP	Organization	Latitude	Longitude
Astound Broadband	Not Available	37.9780	-122.0311

Geolocation data from ipinfo.io (Product: API, real-time)

IP Address	Country	Region	City
66.234.192.245	United States 	California	Concord
ISP	Organization	Latitude	Longitude
vanoppen.biz LLC	Wave Broadband (wavebroadband.com)	37.9780	-122.0311

Geolocation data from DB-IP (Product: Full, 2019-9-1)

IP Address	Country	Region	City
66.234.192.245	United States 	California	San Jose
ISP	Organization	Latitude	Longitude
Astound Broadband	Wave Broadband	37.3394	-121.895

The following figure was obtained from <https://www.iplocation.net/> for the

Šis paveikslėlis buvo gautas iš serverio, kurio IP adresas 66.234.192.245, iš

server with IP address 66.234.192.245.

<https://www.iplocation.net/>.

Based upon this information, it is my opinion that this server is physically located in California, United States.

Remdamasis šia informacija, manau, kad šis serveris yra fiziškai Kalifornijoje, JAV.

5.4.2 IP Address 66.234.192.2 Location / IP adresas 66.234.192.2 Vieta

The following figure was obtained from <https://www.iplocation.net/> for the server with IP address 66.234.192.2.

Šis paveikslėlis buvo gautas iš <https://www.iplocation.net/> serveriui, kurio IP adresas 66.234.192.2.

Geolocation data from [IP2Location](#) (Product: DB6, updated on 2019-9-1)

IP Address	Country	Region	City
66.234.192.2	United States 	California	Concord
ISP	Organization	Latitude	Longitude
Astound Broadband	Not Available	37.9780	-122.0311

Geolocation data from [ipinfo.io](#) (Product: API, real-time)

IP Address	Country	Region	City
66.234.192.2	United States 	California	Concord
ISP	Organization	Latitude	Longitude
vanoppen.biz LLC	Wave Broadband (wavebroadband.com)	37.9780	-122.0311

Geolocation data from [DB-IP](#) (Product: Full, 2019-9-1)

IP Address	Country	Region	City
66.234.192.2	United States 	California	San Jose
ISP	Organization	Latitude	Longitude
Astound Broadband	Wave Broadband	37.3394	-121.895

Based upon this information, it is my opinion that this server is physically located in California, United States.

Remdamasis šia informacija, manau, kad šis serveris yra fiziškai Kalifornijoje, JAV.

5.4.3 IP Address 64.140.133.239 Location / IP adresas 64.140.133.239 Vieta

The following figure was obtained from <https://www.iplocation.net/> for the server with IP address 64.140.133.239.

Šis paveikslėlis gautas iš <https://www.iplocation.net/> serveriui, kurio IP adresas 64.140.133.239.

Geolocation data from [IP2Location](#) (Product: DB6, updated on 2019-9-1)

IP Address	Country	Region	City
64.140.133.239	United States 	Washington	Kenmore
ISP	Organization	Latitude	Longitude
CloudVPN Inc.	Not Available	47.7500	-122.2400

Geolocation data from [ipinfo.io](#) (Product: API, real-time)

IP Address	Country	Region	City
64.140.133.239	United States 	Florida	West Palm Beach
ISP	Organization	Latitude	Longitude
CLOUDVPN INC.	CLOUDVPN INC. (c3ntris.com)	26.7153	-80.0534

Geolocation data from [DB-IP](#) (Product: Full, 2019-9-1)

IP Address	Country	Region	City
64.140.133.239	United States 	Wyoming	Cheyenne
ISP	Organization	Latitude	Longitude
Sentris Network LLC	Cloudvpn Inc	41.1339	-104.814

Based upon this information, it is my opinion that this server is physically located in the United States.

Remdamasis šia informacija, manau, kad šis serveris yra fiziškai JAV.

5.4.4 IP Address 148.251.73.163 Location / IP adresas 148.251.73.163 Vieta

The following figure was obtained from <https://www.iplocation.net/> for the server with IP address 148.251.73.163.

Šis paveikslėlis gautas iš <https://www.iplocation.net/> url serveriui, kurio IP adresas 148.251.73.163.

Geolocation data from [IP2Location](#) (Product: DB6, updated on 2019-8-1)

IP Address	Country	Region	City
148.251.73.163	Germany 	Bayern	Nuremberg
ISP	Organization	Latitude	Longitude
Hetzner Online AG	Not Available	49.4478	11.0683

Geolocation data from [ipinfo.io](#) (Product: API, real-time)

IP Address	Country	Region	City
148.251.73.163	Germany 	Not Available	Not Available
ISP	Organization	Latitude	Longitude
Hetzner Online GmbH	H05-391652 (hetzner.de)	51.2993	9.4910

Geolocation data from [DB-IP](#) (Product: Full, 2019-8-2)

IP Address	Country	Region	City
148.251.73.163	Germany 	Saxony	Falkenstein
ISP	Organization	Latitude	Longitude
Hetzner Online GmbH	Not Available	50.475	12.365

Based upon this information, it is my opinion that this server is physically located in Germany.

Remdamasis šia informacija, manau, kad šis serveris yra Vokietijoje.

5.4.5 IP Address 82.135.144.188 Location / IP adresas 82.135.144.188 Vieta

The following figure was obtained from <https://www.iplocation.net/> for the server with IP address 82.135.144.188.

Šis paveikslėlis buvo gautas iš serverio, kurio IP adresas 82.135.144.188, iš <https://www.iplocation.net/>.

Geolocation data from [IP2Location](#) (Product: DB6, updated on 2019-9-1)

IP Address	Country	Region	City
82.135.144.188	Lithuania 	Vilniaus apskritis	Vilnius
ISP	Organization	Latitude	Longitude
UAB Tesonet	Not Available	54.6892	25.2798

Geolocation data from [ipinfo.io](#) (Product: API, real-time)

IP Address	Country	Region	City
82.135.144.188	Lithuania 	Vilnius	Vilnius
ISP	Organization	Latitude	Longitude
Telia Lietuva, AB	UAB TESONET (tesonet.com)	54.6892	25.2798

Geolocation data from [DB-IP](#) (Product: Full, 2019-9-1)

IP Address	Country	Region	City
82.135.144.188	Lithuania 	Vilnius	Vilnius
ISP	Organization	Latitude	Longitude
Telia Lietuva	Tesonet	54.6872	25.2797

Based upon this information, it is my opinion that this server is physically located in Lithuania.

Remdamasis šia informacija, manau, kad šis serveris yra Vokietijoje.

5.4.6 Domain name icanhazip.com Server Location / Domeno vardas icanhazip.com Serverio vieta

The following figure was obtained from <https://www.iplocation.net/> for the server with the name icanhazip.com.

Šis paveikslėlis buvo gautas iš serverio, kurio pavadinimas icanhazip.com, iš <https://www.iplocation.net/>.

Geolocation data from [IP2Location](#) (Product: DB6, updated on 2019-9-1)

Domain Name	Country	Region	City
icanhazip.com	United States 	California	San Francisco
ISP	Organization	Latitude	Longitude
CloudFlare Inc.	Not Available	37.7757	-122.3952

Geolocation data from [ipinfo.io](#) (Product: API, real-time)

Domain Name	Country	Region	City
icanhazip.com	United States 	Not Available	Not Available
ISP	Organization	Latitude	Longitude
Cloudflare, Inc.	Cloudflare, Inc. (cloudflare.com)	37.7510	-97.8220

Geolocation data from [DB-IP](#) (Product: Full, 2019-9-1)

Domain Name	Country	Region	City
icanhazip.com	United States 	Virginia	Ashburn
ISP	Organization	Latitude	Longitude
Cloudflare, Inc.	Cloudflare, Inc.	39.0438	-77.4874

While the meaning of IP addresses does not vary based upon location, the translation of a domain name to the corresponding IP address may be location sensitive. Thus, to ensure that this translation was likely the same for Mr. Butenis as for me, I used a DNS checker, which queries the domain name to IP mapping from DNS servers in multiple locations. That tool was <https://www.nexcess.net/web-tools/dns-checker/>. The following image was taken from that site for a location in the Russian Republic and thus relatively close to Lithuania. The returned IP addresses were the same for

Nors IP adresų reikšmė skiriasi ne pagal vietą, domeno vardo vertimas į atitinkamą IP adresą gali būti jautrus vietai. Taigi, norėdamas įsitikinti, kad p. Buteniui šis vertimas greičiausiai bus toks pats kaip ir man, aš naudočiau DNS tikrintuvą, kuris užklausia domeno vardo į IP žemėlapių sudarymą iš DNS serverių keliose vietose. Šis įrankis buvo <https://www.nexcess.net/web-tools/dns-checker/>. Iš šios vietos buvo paimtas vaizdas iš vietos Rusijos Respublikoje, taigi palyginti arti Lietuvos. Gražinti IP adresai buvo vienodi kiekvienam įrankio naudojamam DNS serveriui.

every DNS server used by the tool.

Mastertel	Moscow, Russian Federation	104.20.16.242 104.20.17.242
-----------	----------------------------	--------------------------------

Based upon this information, it is my opinion that the domain name icanhazip.com mapped to 104.20.16.24 or 104.20.17.242, both of which are located in the United States.

Remdamasis šia informacija, manau, kad domeno vardas icanhazip.com priskyrė 104.20.16.24 arba 104.20.17.242, kurie abu yra JAV.

5.4.7 Domain name luminati.io Server Location / Domeno vardas luminati.io Serverio vieta

You've entered a domain name. We've found an IP address from the domain name you've entered. Your translated IP address is **54.83.59.224**

Geolocation data from [IP2Location](#) (Product: DB6, updated on 2019-8-1)

Domain Name	Country	Region	City
luminati.io	United States 🇺🇸	Virginia	Ashburn
ISP	Organization	Latitude	Longitude
Amazon Technologies Inc.	Not Available	39.0437	-77.4875

Geolocation data from [ipinfo.io](#) (Product: API, real-time)

Domain Name	Country	Region	City
luminati.io	United States 🇺🇸	Virginia	Virginia Beach
ISP	Organization	Latitude	Longitude
Amazon.com, Inc.	Amazon Technologies Inc. (amazontechnologies.com)	36.8512	-76.1692

Geolocation data from [DB-IP](#) (Product: Full, 2019-8-2)

Domain Name	Country	Region	City
luminati.io	United States 🇺🇸	Virginia	Ashburn
ISP	Organization	Latitude	Longitude
Amazon.com, Inc.	Amazon Technologies Inc.	39.0438	-77.4874

The following figure was obtained from <https://www.iplocation.net/> for the server with the name luminati.io.

While the meaning of IP addresses does not vary based upon location, the translation of a domain name to the corresponding IP address may be location sensitive. Thus, to ensure that this translation was likely the same for Mr. Butenis as for me, I used a

Šis paveikslėlis buvo gautas iš serverio, kurio vardas luminati.io, iš URL <https://www.iplocation.net/>.

Nors IP adresų reikšmė skiriasi ne pagal vietą, domeno vardo vertimas į atitinkamą IP adresą gali būti jautrus vietai. Taigi, norėdamas įsitikinti, kad p. Buteniui šis vertimas greičiausiai bus toks pats kaip ir man, aš naudojau DNS tikrintuvą, kuris

DNS checker, which queries the domain name to IP mapping from DNS servers in multiple locations. That tool was <https://www.nexcess.net/web-tools/dns-checker/>. The following image was taken from that site for a location in the Russian Republic and thus relatively close to Lithuania. The returned IP addresses were the same for every DNS server, *other than China*, used by the tool.

Mastertel	Moscow, Russian Federation	184.73.155.216 54.83.59.224
-----------	----------------------------	--------------------------------

Based upon this information, it is my opinion that the domain name luminati.io mapped to 184.73.155.216 or 54.83.59.224, both of which are servers located in the United States.

užklausia domeno vardo į IP žemėlapių sudarymą iš DNS serverių keliose vietose. Šis įrankis buvo <https://www.nexcess.net/web-tools/dns-checker/>. Iš šios vietos buvo paimtas vaizdas iš vietos Rusijos Respublikoje, taigi palyginti arti Lietuvos. Gražinti IP adresai buvo vienodi kiekvienam įrankio naudojamam DNS serveriui, išskyrus Kiniją.

Remdamasis šia informacija, manau, kad domeno vardas luminati.io priskyrė 184.73.155.216 arba 54.83.59.224, kurie abu yra serveriai, esantys JAV.

6 Documents Reviewed / Dokumentai peržiūrėti

In preparing my report, I reviewed English language documents that were provided by Attorney Rapolas Kasparavičius. I have listed those documents in §6.1. In addition, in my own research to prepare this report, I relied upon the documents listed in §6.2.

Rengdamas savo pranešimą, aš peržiūrėjau dokumentus anglų kalba, kuriuos pateikė advokatas Rapolas Kasparavičius. Aš išvardijau tuos dokumentus 6.1 skyriuje. Be to, atlikdamas savo tyrimus, rengdamas šią ataskaitą, remiausi §6.2 skyriuje išvardytais dokumentais ir biblioteka.

6.1 Documents Provided by Counsel / Advokatės pateikti dokumentai

In preparing this report, I relied upon the following documents, which were provided to me by Luminati's legal counsel:

Rengdamas šią ataskaitą remiausi šiais dokumentais, kuriuos man pateikė „Luminati“ teisinis patarėjas:

- Statement of Claims

- Ieškinių pareiškimas

- Specialist Report

- Specialisto ataskaita

6.2 Other documents / Kiti dokumentai

A list of my peer reviewed publications, books, and patents is in the Bibliography. Additional articles I have written and disseminated are listed on my attached *Curriculum Vitae*. I have included references to third-party materials that I used in preparing my report, either in line with the quoted material, and/or in the Bibliography.

Mano recenzuotų leidinių, knygų ir patentų sąrašas pateiktas bibliografijoje. Kiti mano parašyti ir išplatinti straipsniai yra išvardyti pridedamame *textit Curriculum Vitae*. Įtraukiau nuorodas į trečiųjų šalių medžiagą, kurią panaudojau rengdamas savo ataskaitą, arba įtraukdamas į cituojamą medžiagą, ir (arba) į bibliografiją.

7 Background / Papildoma informacija

In this section, I provide background information on Internet technologies that I have evaluated as part of preparing this report. This information supports my opinions and describe specific detail useful in understanding the technologies evaluated within this report.

Šiame skyriuje pateikiu pagrindinę informaciją apie interneto technologijas, kurias įvertinau rengdamas šią ataskaitą. Ši informacija pagrindžia mano nuomones ir apibūdina konkrečią detalę, naudingą suprantant šioje ataskaitoje įvertintas technologijas.

7.1 Computer Network Concepts / Kompiuterių tinklo sąvokos

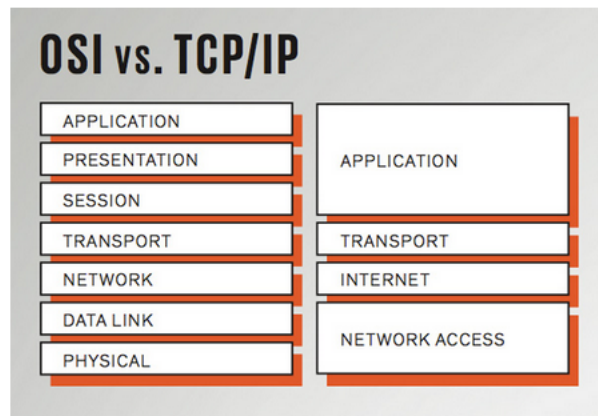
In this section, I provide a brief description of concepts useful in understanding computer networks. This includes computer network models (section A), tunneling (section B), proxy servers (section C), and encapsulation

Šiame skyriuje pateikiu trumpą sąvokų, naudingų suprantant kompiuterinius tinklus, aprašymą. Tai apima kompiuterių tinklo modelius (7.1.1 skyrius), tunelių nustatymą (7.1.2 skyrius), tarpinius serverius (7.1.3

(section D).

skyrius) ir kapsulę (7.1.4 skyrius).

7.1.1 Computer Network Models / *Kompiuterių tinklo modeliai*



A layered approach: The OSI reference model [left column] divides computer communications into seven distinct layers, from physical media in layer 1 to applications in layer 7. Though less rigid, the TCP/IP approach to networking can also be construed in layers, as shown on the right.

Source [12]

“A network protocol defines rules and conventions for communication between network devices. Network protocols include mechanisms for devices to identify and make connections with each other, as well as formatting rules that specify how data is packaged into sent and received messages. Some protocols also support message acknowledgment and data compression designed for reliable and/or high-performance network communication [13].”

Modern computer networks rely upon protocols for their normal function. Much

Nuoroda [12]

“Tinklo protokolas nustato ryšio ir tinklo įrenginių tarpusavio ryšio taisykles ir taisykles. Tinklo protokolus sudaro mechanizmai, skirti įrenginiams identifikuoti ir užmegzti ryšius vienas su kitu, taip pat formatavimo taisyklės, nurodančios, kaip duomenys sudedami į išsiųstus ir gautus pranešimus. Kai kurie protokolai taip pat palaiko pranešimų patvirtinimą ir duomenų glaudinimą, skirtus patikimam ir (arba) didelio našumo tinklo ryšiui palaikyti [13].”

Šiuolaikiniai kompiuterių tinklai normaliai funkcionuoja remdamiesi protokolais.

like *human* protocols, they are important for communications. While the following sections will discuss pertinent details of particular network protocols, this is important because these protocols form common agreements on how computer systems communicate with one another.

Panašiai kaip human protokolai, jie yra svarbūs bendraujant. Tolesniuose skyriuose bus aptarta svarbi tam tikrų tinklo protokolų informacija, tačiau tai yra svarbu, nes šie protokolai sudaro bendras sutartis dėl to, kaip kompiuterinės sistemos bendrauja viena su kita.

7.1.2 Tunneling / Tuneliai

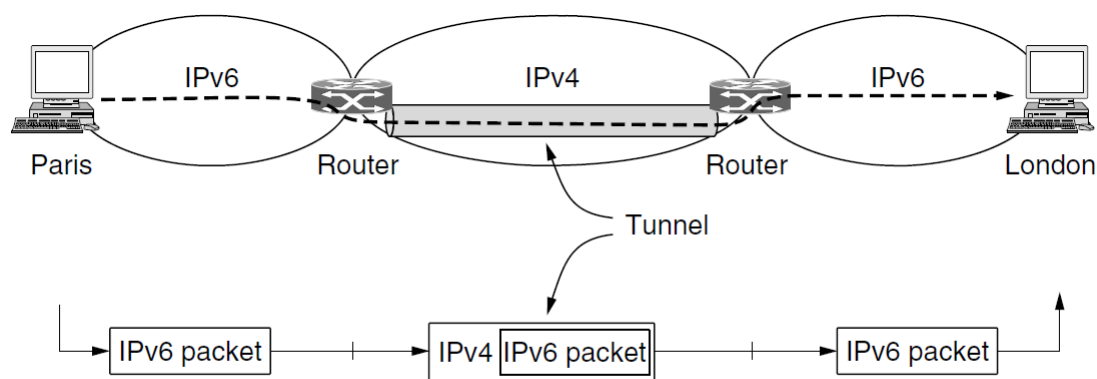


Figure 5-40. Tunneling a packet from Paris to London.

Source [14, p. 430].

Tunneling is a common technique used to create pathways for encapsulating and transmitting one protocol within another protocol. It is commonly used to take existing protocol messages and sending them via a different network protocol. For example in §7.1.2 I include a figure from Dr. Tanenbaum's book that describes how packets for the newer IPv6 protocol are tunneled by encapsulating them and carrying them over an IPv4 network, with ultimate delivery as an IPv6 protocol. The IP Version

Nuoroda [14, p. 430].

Tuneliavimas yra įprasta technika, naudojama sukurti vieno protokolo kapsuliavimo ir perdavimo kelius. Paprastai jis naudojamas pasiimti esamus protokolinius pranešimus ir siųsti juos per kitą tinklo protokolą. Pavyzdžiui, 7.1.2 skyriuje yra paveikslėlis iš Dr. Tanenbaumo knygos, kuriame aprašoma, kaip tuneluojami naujesnio IPv6 protokolo paketai, juos kapsuliuojant ir nešiojant per IPv4 tinklą, o galutinis pristatymas yra kaip IPv6 protokolas. IP versijos 4 protokolas (IPv4) ir

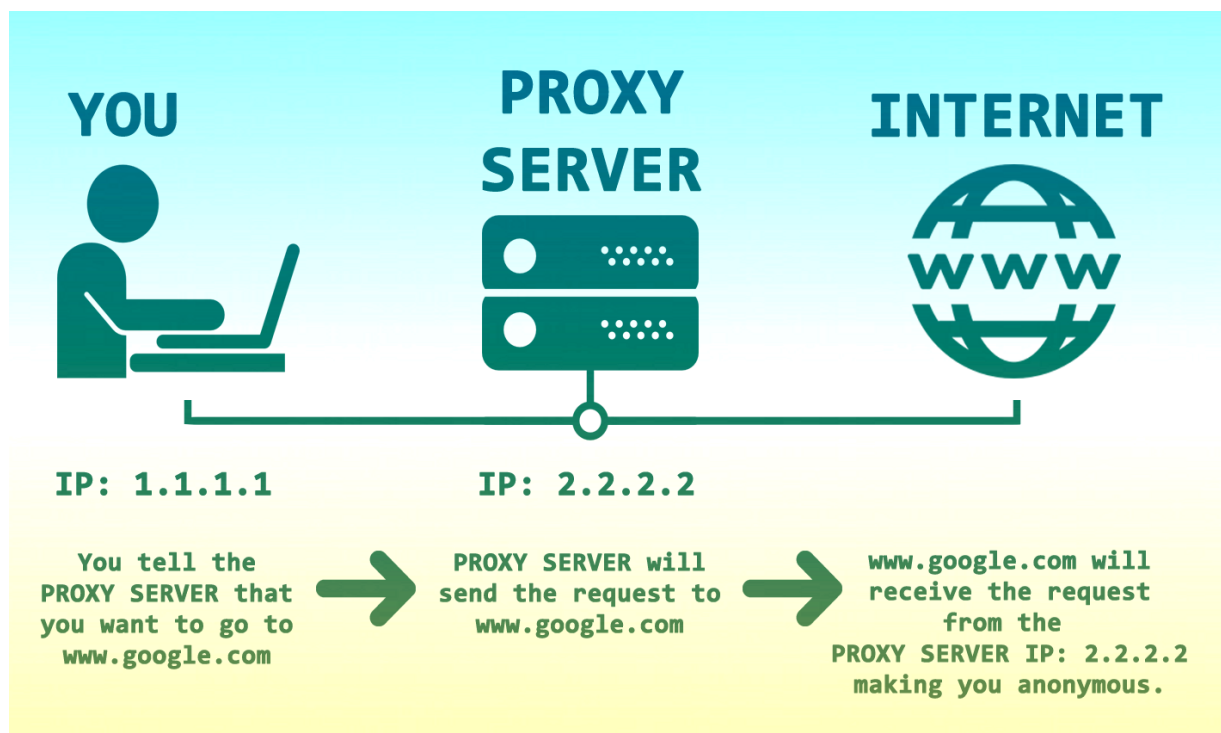
4 protocol (IPv4) and IP Version 6 protocol are explained in section §7.2.1.

This tunneling technique is a fundamental part of how the OX and LU products function: they provide this type of encapsulation for HTTP protocol messages so that they may be sent from the client, to a server. That server then sends a new message, encapsulating the original HTTP protocol message within the new message. In this way, the recipient of the HTTP protocol message perceives that the HTTP message was sent by the server. One specific technique for tunneling, the proxy server, is described in §7.1.3.

IP versijos 6 protokolas yra paaiškinti skyriuje 7.2.1.

Ši tuneliavimo technika yra pagrindinė OX ir LU produktų veikimo dalis: jie teikia tokio tipo HTTP protokolo pranešimų kapsulę, kad juos būtų galima siųsti iš kliento į serverį. Tada tas serveris išsiunčia naują pranešimą, į jį įtraukdamas originalų HTTP protokolo pranešimą. Tokiu būdu HTTP protokolo pranešimo gavėjas suvokia, kad HTTP pranešimą atsiuntė serveris. Viena specifinė tuneliavimo technika - įgaliotasis serveris, aprašyta 7.1.3 skyriuje.

7.1.3 Proxy Server / Tarpinis serveris



<https://www.stupidproxy.com/proxy-servers/>

The concept of a proxy server has existed for decades. One of the earlier proxy servers, Squid, was written and released in the 1990s. The source code for Squid is still available on Github.com (<https://github.com/squid-cache/squid/network>). While the project has been around since the mid-1990s, the source code was only moved to Github.com in 2012. The **Changelog** file in the Github.com repository includes a change history with the earliest date being October 29, 1997.

I note that the version from 2012 includes support for **DNS** (§7.2.4), **HTTP** (§7.2.5), and ssl — which is used for **HTTPS** support (§7.2.6).

Early use of a proxy server was to permit caching of web content to improve the performance of web pages. This is because web information that was stored close to the web client could be delivered to the client more quickly, much like a copy of a critical document on your desk takes less time to retrieve than the original copy in a filing cabinet in a different office.

The more general concept of a *proxy* is that it performs some action on behalf of another computer. For example, if the reader has internet service at home, it is quite likely that they have a physical device to which the computers inside the home connect, whether

Tarpinio serverio koncepcija egzistuoja dešimtmečius. Dešimtajame dešimtmetyje buvo parašytas ir išleistas vienas iš ankstesnių tarpinių serverių - „Squid“. Kalmarų šaltinio kodą vis dar galima rasti Github.com (<https://github.com/squid-cache/squid/network>). Nors projektas buvo vykdomas nuo 1990-ųjų vidurio, šaltinio kodas į Github.com buvo perkeltas tik 2012 m. Į failą **Changelog** į Github.com saugyklą įtraukta pakeitimų istorija, anksčiausia data buvo spalio 29 d., 1997.

Atkreipiu dėmesį, kad nuo 2012 m. Versijos yra palaikomas DNS (7.2.4 skyrius), HTTP (7.2.5 skyrius) ir ssl, kuris naudojamas HTTPS palaikymui (7.2.6 skyrius).

Ankstyvasis tarpinis serveris turėjo leisti talpinti žiniatinklio turinį, kad pagerėtų tinklalapių našumas. Taip yra todėl, kad žiniatinklio informaciją, kuri buvo saugoma šalia žiniatinklio kliento, klientui būtų galima pristatyti greičiau, panašiai kaip kritinio dokumento kopija ant jūsų darbo stalo užtrunka mažiau laiko, nei originalios kopijos, esančios kitos tarnybos biure. .

Bendresnė tarpinio serverio samprata yra ta, kad jis kai kuriuos veiksmus atlieka kito kompiuterio vardu. Pavyzdžiui, jei skaitytojas namuose naudojasi interneto paslaugomis, gana tikėtina, kad jie turi fizinį įrenginį, prie kurio prisijungia namų

by wired service (“ethernet”) or wireless (“Wi-Fi”). When a computer inside the home wishes to connect to a computer outside the home, it will first communicate with the physical device that separates “inside” from “outside”. Thus, a network connection is created from the home computer to the separation device and then from there to the destination. Frequently, these devices are called “firewalls”, “internet gateway”, or “routers” or sometimes even “modems”. Their function is to protect the computers inside the home. One way in which they do this is that only the “firewall” is visible to the outside work. This is accomplished by using the firewall as a “proxy” for the computers within the home network.

One important observation relevant to my analysis is that one of the effects of using a proxy is that it makes a group of computers *look like* a single computer. This is because only the proxy is visible. This has the important effect that it permits composition of proxies - so we may insert additional proxies between the source computer and the destination computer. This insertion is transparent to the source computer and the destination computer; it may be possible for them to detect the presence of the proxy, but normally the presence or absence of proxies along the link between the source and destination computer would not be observed.

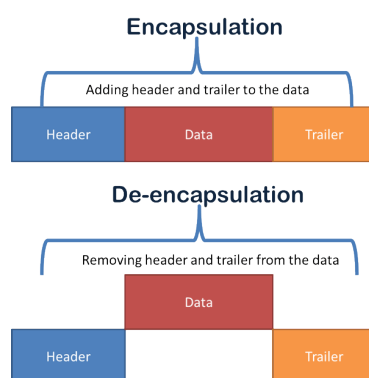
kompiuteriai, naudodamiesi laidiniu ryšiu („ethernet“) ar belaidžiu ryšiu („Wi-Fi“). Kai namie esantis kompiuteris nori prisijungti prie kompiuterio, esančio ne namuose, jis pirmiausia bendraus su fiziniu įrenginiu, kuris skiria „vidų“ nuo „lauko“. Tokiu būdu sukuriamas tinklo ryšys iš namų kompiuterio į atskyrimo įrenginį ir iš ten į paskirties vietą. Dažniausiai šie įtaisai yra vadinami „ugniasienėmis“, „interneto šliuzais“ arba „maršrutizatoriais“ arba kartais net „modemais“. Jų funkcija yra apsaugoti kompiuterius namuose. Vienas iš būdų, kaip jie tai daro, yra tas, kad išorinis darbas yra matomas tik kaip ugniasienė. Tai atliekama naudojant ugniasienę kaip „tarpinį serverį“ namų kompiuteriams.

Vienas svarbus pastebėjimas, susijęs su mano analize, yra tas, kad vienas iš įgaliotojo serverio naudojimo yra tas, kad jis priverčia kompiuterių grupę atrodyti kaip vienas kompiuteris. Taip yra todėl, kad matomas tik tarpinis serveris. Tai daro svarbų poveikį, nes leidžia sudaryti tarpinius serverius - todėl galime įterpti papildomus tarpinius serverius tarp šaltinio kompiuterio ir paskirties kompiuterio. Šis įterpimas yra skaidrus šaltinio ir paskirties kompiuteriams; jiems gali būti įmanoma nustatyti tarpinį serverį, tačiau paprastai tarpinių šaltinių buvimo ar nebuvimo ryšyje tarp šaltinio ir paskirties kompiuterio nebus stebima.

The OX and LU products provide proxy services that take messages from the source computer and send them to the destination computer using a collection of one or more intermediate proxies. This is conceptually similar to the single proxy model that I have just described, but some uses benefit from having multiple proxy agents. Both products implement numerous standard internet protocols (§7.2) to permit a computer, or network of computers. Some of these protocols are proxy oblivious — they are not aware of the proxy — while other protocols relate to specific mechanisms particular to proxies.

„OX“ ir „LU“ produktai teikia tarpinio serverio paslaugas, kurios paima pranešimus iš šaltinio kompiuterio ir siunčia juos į paskirties kompiuterį naudodamos vieno ar kelių tarpinių tarpinių duomenų rinkinį. Tai konceptualiai panaši į vieną tarpinio serverio modelį, kurį ką tik aprašiau, tačiau kai kuriems naudojimo būdams naudinga turėti kelis tarpinius serverius. Abu produktai diegia daugybę standartinių interneto protokolų (7.2 skyrius), kad būtų galima naudoti kompiuterį ar kompiuterių tinklą. Kai kurie iš šių protokolų yra pamiršti tarpinio serverio - jie nežino apie tarpinį serverį -, o kiti protokolai yra susiję su konkrečiais mechanizmais, ypač susijusiais su tarpiniais serveriais.

7.1.4 Encapsulation / Inkapsuliacija



Source [15]

An important aspect of computer networking is the idea of “encapsulating” information. This is similar to when someone places documents inside an envelope. The envelope

Nuoroda [15]

Svarbus kompiuterių tinklų kūrimo aspektas yra mintis „įterpti“ informaciją. Tai panašu į atvejus, kai kažkas įdeda dokumentus į voką. Vokas nepriklauso nuo jo viduje esančio

is independent of the contents placed inside it — provided that it fits — and the same kind of envelope can be used with different kinds of documents.

In the same way, a message to be sent from a source computer to a destination computer over a computer network assembles the message and then delivers that to a protocol layer. The protocol layer then encapsulates the message and includes information the destination computer requires to deliver that message to the proper agent; this is similar to the way in which an envelope might have the name of the intended recipient.

One benefit of this approach is that the message is not dependent upon the encapsulation, and the method of encapsulation is not dependent on the contents of the message. Thus, we can change the specifics of the envelope without affecting the message. For example, we can use different network protocols, as appropriate, much like we might use different color envelopes.

This approach is also composable. Much like we can place one envelope inside another envelope, we can encapsulate an already encapsulated computer network message. This allows us to take an HTML web page description and encapsulate it using the HTTP protocol, then encapsulate that using the TCP protocol, and then encapsulate the

turinio - su sąlyga, kad jis tinka -, ir tos pačios rūšies voką galima naudoti su įvairių rūšių dokumentais.

Tokiu pačiu būdu, pranešimas, kuris turi būti siunčiamas iš šaltinio kompiuterio į kompiuterį, į kurį naudojamas kompiuteris, kompiuterio tinkle, surenka pranešimą ir perduoda jį protokolo sluoksniui. Tada protokolo sluoksnis apklijuoja pranešimą ir apima informaciją, kurios reikia kompiuteriui, kuris perduoda tą pranešimą tinkamam agentui; tai yra panašu į tai, kaip voke gali būti numatytas gavėjo vardas.

Vienas šio požiūrio pranašumų yra tas, kad žinutė nepriklauso nuo kapsulės, o kapsuliavimo būdas nėra priklausomas nuo pranešimo turinio. Taigi, pakeisdami voko specifikaciją nepakenksime žinutei. Pvz., Mes galime naudoti įvairius tinklo protokolus, kaip tinkama, panašiai kaip mes galime naudoti skirtingų spalvų vokus.

Šis požiūris taip pat yra sudėtingas.

Panašiai, kaip galime įklijuoti vieną voką į kitą voką, galime įterpti jau įdėtą kompiuterinio tinklo pranešimą. Tai leidžia mums paimti HTML tinklalapio aprašą ir jį kapsuliuoti naudojant HTTP protokolą, tada įterpti jį naudojant TCP protokolą, o tada į TCP apklijuotą pranešimą kapsuliuoti

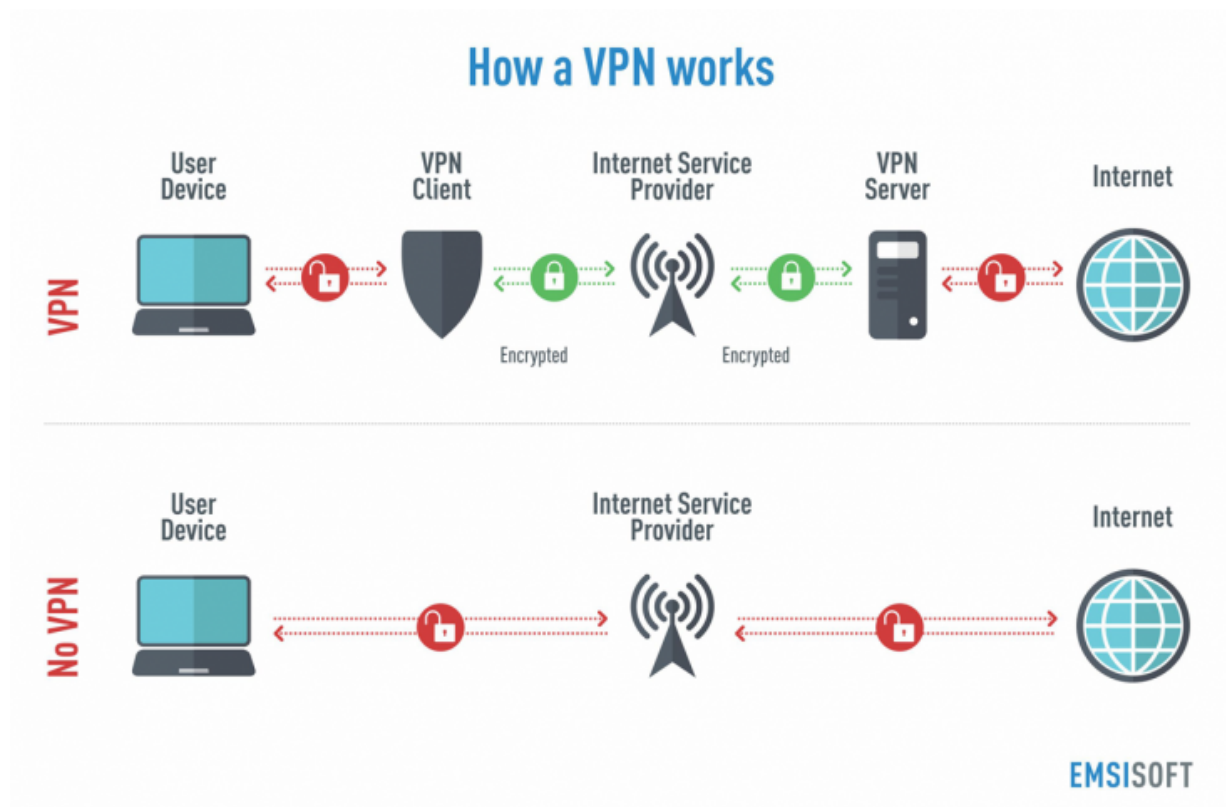
TCP wrapped message using the IP protocol, until we have sufficient information to send the message.

It is the ability to encapsulate arbitrary computer network messages inside additional protocols that allows us to construct proxies.

naudojant IP protokolą, kol turėsime pakankamai informacijos pranešimui siųsti.

Tai yra galimybė kapsuliuoti savavališkus kompiuterio tinklo pranešimus papildomų protokolų viduje, leidžiantį sukonstruoti tarpinius serverius.

7.1.5 Virtual Private Network (VPN) / Virtualus privatus tinklas (VPN)



<https://www.yellowstonecomputing.net/blog/vpns-your-personal-tunnel-to-privacy-part-2>

A virtual private network (VPN) server is similar to a proxy server (§7.1.3). One difference is that a virtual private network sends all messages to a single computer, the VPN server. Often a VPN is implemented to

Virtualiojo privataus tinklo (VPN) serveris yra panašus į tarpinį serverį (7.1.3 skyrius). Vienas skirtumas yra tas, kad virtualus privatus tinklas siunčia visas žinutes į vieną kompiuterį, VPN serverį. Dažnai VPN yra

route most or all network messages be sent through the VPN.

Another difference between a VPN and a proxy is that a proxy server usually changes the original message in such a way that the original source computer's identity is masked. A VPN normally does not modify the original message.

Another difference between a VPN and a proxy is that a VPN connection is normally encrypted independent of any encryption that might be used by the application, such as HTTPS (§7.2.6). Because a VPN involves encryption, it must address how keys are shared between the client computer and the VPN server. Examples for VPN key management protocols include PPTP, WireGuard, SSTP, L2TP/IPsec, IKEv2/IPsec, and SSL/TLS. SSL/TLS is also used by web browsers and web clients to encrypt information for the HTTPS protocol (§7.2.6).

Sometimes it can be difficult to distinguish the difference between a VPN and a proxy server because they provide similar functionality. Further, because they can be composed, it is possible to use both VPN and proxy server functionality together. For example, some network gateways (firewall or router) provide proxy services to the computers within the network, but also provide VPN functionality for specific

įgyvendinamas taip, kad dauguma tinklo pranešimų būtų siunčiami per VPN.

Kitas skirtumas tarp VPN ir tarpinio serverio yra tas, kad įgaliojasis serveris paprastai keičia pirminį pranešimą taip, kad pirminio šaltinio kompiuterio tapatybė būtų užmaskuota. Paprastai VPN nekeičia pirminio pranešimo.

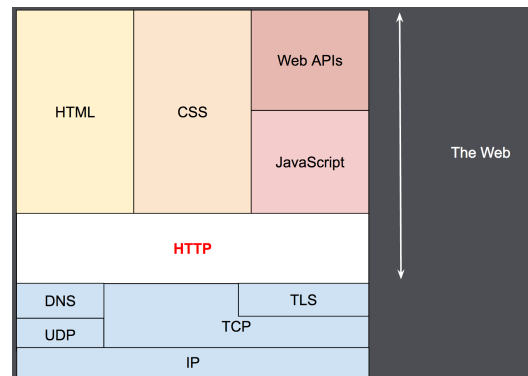
Kitas skirtumas tarp VPN ir tarpinio serverio yra tas, kad VPN ryšys paprastai užšifruojamas nepriklausomai nuo šifravimo, kurį gali naudoti programa, pavyzdžiui, HTTPS (7.2.6 skyrius). Kadangi VPN apima šifravimą, jis turi nurodyti, kaip raktai yra dalijami tarp kliento kompiuterio ir VPN serverio. VPN raktų valdymo protokolų pavyzdžiai yra PPTP, „WireGuard“, SSTP, L2TP / IPsec, IKEv2 / IPsec ir SSL / TLS. SSL / TLS taip pat naudoja interneto naršyklės ir interneto klientai, norėdami užšifruoti informaciją apie HTTPS protokolą (7.2.6 skyrius).

Kartais gali būti sunku atskirti skirtumą tarp VPN ir įgaliojo serverio, nes jie teikia panašias funkcijas. Be to, kadangi jie gali būti sudaryti, galima kartu naudoti ir VPN, ir tarpinio serverio funkcijas. Pvz., Kai kurie tinklo šliuzai (ugniasienė ar maršrutizatorius) teikia tarpinio serverio paslaugas tinklo kompiuteriams, bet taip pat teikia VPN funkcijas, skirtas konkrečiam paskirties srautui. Tokio tipo konfigūracija dažnai būna

destination traffic. This type of configuration is often found in corporate environments, but even residential grade gateways can provide this type of functionality (§5.2.1).

įmonių aplinkoje, tačiau tokio tipo funkcionalumą gali suteikti net gyvenamojo tipo vartai (5.2.1 skyrius).

7.2 Network Protocols / Tinklo protokolas



Source [16]

Much of the analysis that I have performed relates to the use of “internet protocols”. In the context of a computer network, a protocol is a mutually agreed upon process to facilitate communications. Protocols permit *interoperability* between multiple computers. Standardized protocols allow multiple vendors to construct products that can communicate. These protocols fit into a common layered model and the protocols fit within those models. I describe the two most common computer network models in §7.1.1. I describe the Internet Protocol (IP) in §7.2.1. I describe the Unreliable Datagram Protocol (UDP) in §7.2.2. I describe the Transmission Control Protocol (TCP) in §7.2.3. I describe the Domain Name System

Nuoroda [16]

Didžioji mano atliktos analizės dalis yra susijusi su interneto protokolų naudojimu. Kompiuterių tinklo kontekste protokolas yra abipusiai sutartas procesas, palengvinantis ryšį. Protokoliai leidžia suderinti daugelį kompiuterių. Standartizuoti protokoliai leidžia keliems pardavėjams kurti produktus, kurie gali bendrauti. Šie protokoliai tilptų į bendrą sluoksнинį modelį, o protokoliai tilptų į tuos modelius. Aš apibūdinau du dažniausiai pasitaikančius kompiuterinio tinklo modelius skyriuje N. Aš aprašau interneto protokolą (IP) 7.2.1 skyriuje. Aš apibūdinau nepatikimą duomenų schemas protokolą (UDP) 7.2.2 skyriuje. Aprašau perdavimo perdavimo protokolą (TCP) 7.2.3 skyriuje. Aš aprašau domenų vardų sistemą (DNS) 7.2.4 skyriuje.

(DNS) in §7.2.4. I describe the HyperText Transfer Protocol (HTTP) in §7.2.5. I describe the secure HTTP protocol in §7.2.6. I describe the SOCKS5 proxy protocol in §7.3. My goal is not to provide a complete description of each of these. Instead, I have tried to identify the important concepts that are central to the subject of my report.

Aš aprašau HyperText Transfer Protocol (HTTP) 7.2.5 skyriuje, aprašau saugų HTTP protokolą 7.2.6 skyriuje. Aprašau SOCKS5 tarpinį protokolą 7.3 skyriuje. Mano tikslas nėra pateikti išsamų kiekvieno iš jų aprašą. Vietoj to, aš bandžiau nustatyti svarbias sąvokas, kurios yra pagrindinės mano pranešimo temoje.

7.2.1 Internet Protocol (IP) / Interneto protokolas (IP)

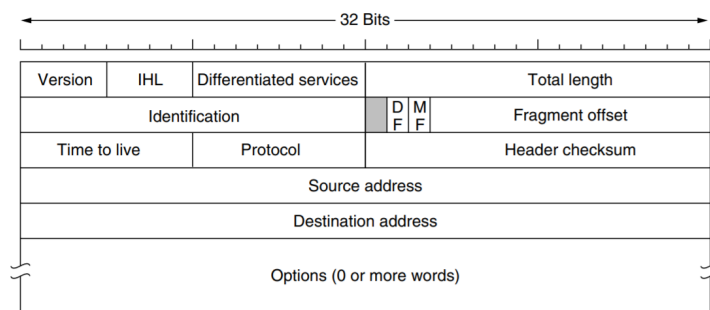


Figure 5-46. The IPv4 (Internet Protocol) header.

Source [14, p. 439]

The Internet Protocol (IP) was first defined in 1981 in RFC 791 [17]. It has been updated several times since 1981 but remains the primary network protocol used in computer networks. Other protocols, including TCP (§7.2.3) and UDP (§7.2.2) are used with IP to implement numerous application level protocols.

Nuoroda [14, p. 439]

Internetinis protokolas (IP) pirmą kartą apibrėžtas 1981 m. Kiti protokolai, įskaitant TCP (7.2.3 skyrius) ir UDP (7.2.2 skyrius), naudojami kartu su IP, siekiant įgyvendinti daugybę programų lygio protokolų.

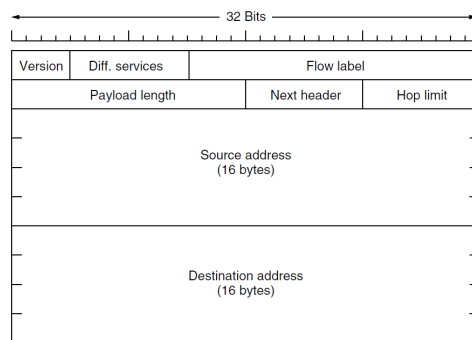


Figure 5-56. The IPv6 fixed header (required).

Source [14, p. 457]

In December 1998, a major revision to the IP protocol was standardized; it is referred to as IP Version 6. While it is widely supported, it has not become the primary protocol. This is because the primary problem this protocol was intended to address — the exhaustion of the IPv4 address space — has been mitigated through the use of two techniques that are core to the OX and LU technologies: the ability to use *proxies* between private and public networks, and the ability to *tunnel* existing protocols between two private networks. I describe proxy servers in §7.1.3 and the concept of protocol tunneling in §7.1.2.

Nuoroda [14, p. 457]

1998 m. Gruodžio mėn. Buvo suvienodinta esminė IP protokolo peržiūra; jis vadinamas IP 6 versija. Nors jis yra plačiai palaikomas, jis netapo pirminiu protokolu. Taip yra todėl, kad pagrindinė problema, kuria buvo siekiama išspręsti šį protokolą - IPv4 adresų erdvės išnaudojimas - buvo sušvelninta naudojant du metodus, kurie yra pagrindinės OX ir LU technologijos: galimybė naudoti tarpinius serverius tarp privatūs ir viešieji tinklai ir galimybė paisyti esamų protokolų tarp dviejų privačių tinklų. Aš aprašau tarpinius serverius 7.1.3 skyriuje ir protokolo tuneliavimo sąvoką 7.1.2 skyriuje.

7.2.2 Unreliable Datagram Protocol (UDP) / Nepatikimas duomenų diagramos protokolas (UDP)

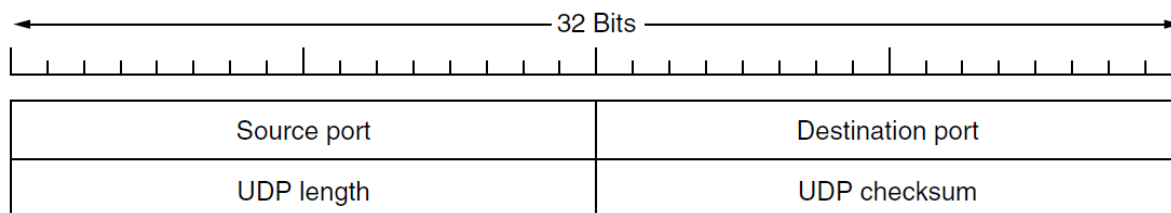


Figure 6-27. The UDP header.

Source [14, p. 542]

The Unreliable Datagram Protocol (UDP) is a simple protocol because it does not provide reliable delivery — hence the *unreliable* portion of its name. It does not provide segmentation and reassembly. It has no connection state, either. Despite this protocol's simplicity, it is used in a wide variety of computer network protocols. In some cases, a protocol may work over UDP and TCP (§7.2.3). For example, the HTTP protocol (§7.2.5) over UDP is an emerging standard known as HTTP/3 or QUIC [18]. Earlier HTTP protocols worked over TCP.

Nuoroda [14, p. 542]

Nepatikimas duomenų diagramos protokolas (UDP) yra paprastas protokolas, nes jis neužtikrina patikimo pristatymo - vadinasi, nepatikima jo pavadinimo dalis. Tai nepateikia segmentavimo ir pakartotinio surinkimo. Ji taip pat neturi ryšio būsenos. Nepaisant šio protokolo paprastumo, jis naudojamas daugelyje kompiuterių tinklo protokolų. Kai kuriais atvejais protokolas gali veikti per UDP ir TCP (7.2.3 skyrius). Pavyzdžiui, HTTP protokolas (7.2.5 skyrius) per UDP yra naujas standartas, žinomas kaip HTTP / 3 arba QUIC [18]. Ankstesni HTTP protokolai veikė per TCP.

7.2.3 Transmission Control Protocol (TCP) / Perdavimo perdavimo protokolas (TCP)

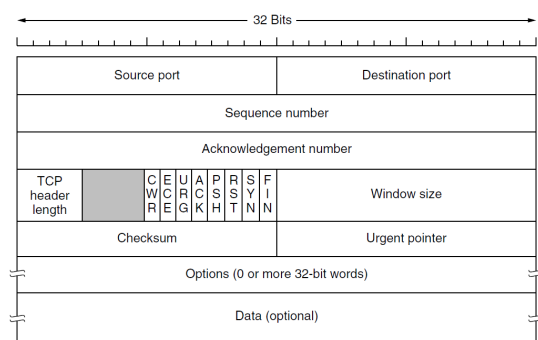


Figure 6-36. The TCP header.

Source [14, p. 557]

The Transmission Control Protocol (TCP) specification was published in 1981 [19]. While it has been updated a number of times with additional options, this remains the primary communications protocol of the internet. It provides ordered, reliable delivery over unreliable networks (usually using IP, see §7.2.1) using a connection abstraction. The HTTP (§7.2.5) and HTTPS (§7.2.6) protocols are designed to run over TCP. Note that TCP does not *require* the IP protocol; it happens to be the most commonly used.

Nuoroda [14, p. 557]

Transmisijos valdymo protokolo (TCP) specifikacija buvo paskelbta 1981 m. Nors ji keletą kartų buvo atnaujinta su papildomomis galimybėmis, ji išlieka pagrindiniu interneto ryšių protokolu. Tai teikia užsakytą, patikimą pristatymą per nepatikimus tinklus (paprastai naudojant IP, žr. Skyrių 7.2.1), naudojant ryšio abstrakciją. HTTP (7.2.5 sekcija) ir HTTPS (7.2.6 sekcija) protokolai skirti naudoti per TCP. Atminkite, kad TCP nereikia IP protokolo; tai būna dažniausiai naudojami.

7.2.4 Domain Name System (DNS) / Domenų vardų sistema (DNS)

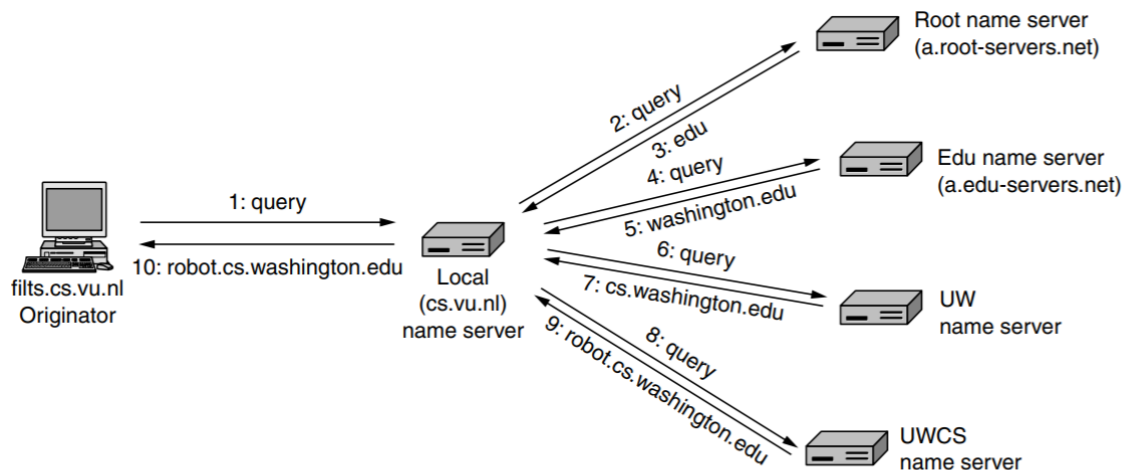


Figure 7-6. Example of a resolver looking up a remote name in 10 steps.

Source [14, p. 620]

The Domain Name System is a standard way to translate domain names, such as `netflix.com` to an IP Address. The IP Address is used by the IP protocol (§7.2.1) to route messages from the source computer to the target computer. The IP Protocol does not understand domain names directly.

The mapping between a domain name and relevant IP addresses is defined by the DNS server. Different DNS servers may implement different mappings. For example, when you use a public Wi-Fi service, it will redirect you to a service-specific page where you can log in, agree to “terms and conditions”, or perform any other steps the Wi-Fi network owner wishes you to perform before being given internet access.

Nuoroda [14, p. 620]

Domenų vardų sistema yra standartinis būdas išversti domenų vardus, pvz., `netflix.com`, į IP adresą. IP adresas naudojamas IP protokole (7.2.1 skyrius) pranešimams nukreipti iš šaltinio kompiuterio į tikslinį kompiuterį. IP protokolas tiesiogiai nesupranta domenų vardų.

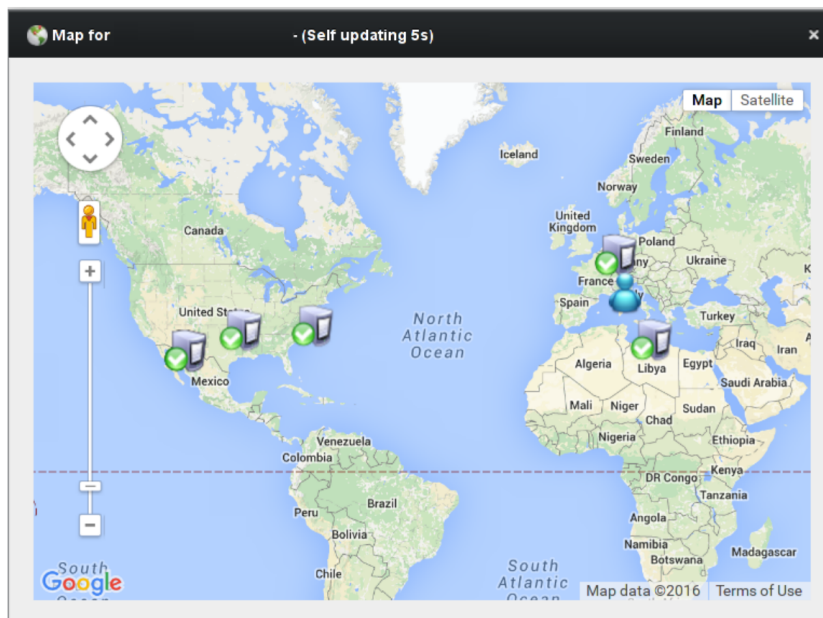
Domeno vardo ir atitinkamų IP adresų atvaizdą nustato DNS serveris. Skirtingi DNS serveriai gali įgyvendinti skirtingas atvaizdavimus. Pvz., Kai naudojate viešą „Wi-Fi“ paslaugą, ji nukreips jus į tam tikros paslaugos puslapį, kuriame galite prisijungti, sutikti su „sąlygomis ir sąlygomis“ ar atlikti kitus veiksmus, kuriuos jums nori atlikti „Wi-Fi“ tinklo savininkas. prieš suteikiant prieigą prie interneto.

A proximity balancing DNS dynamically routes your users and customers to the closest or most suitable available datacenter from a geographical perspective so that application load can be spread across your infrastructure and network response time is kept to a minimum.

GSLB.me flexible geographical balancing can determine the best location to serve content for your users based on the **distance between clients and servers**. **edns-client-subnet** DNS extension is used for granular and accurate geolocalization. This is crucial to achieve a high accuracy DNS balancing.

Geographical Proximity

Geographical proximity balancing handles DNS clients requests by returning the IP address(es) of geographically close datacenters and servers, this is particularly effective for latency-sensitive applications (webservices, online gaming, etc) and in all situations where users must be sent to different datacenters based on their physical position.



Real-time geographical map displays each target status

Datacenters and servers are automatically geo-positioned using **GSLB.me** geolocation database enabling fast rollout of your geographically-balanced services. For additional flexibility you can override the built-in geolocation database and enable **custom geopositioning** to manually define where your servers, datacenters and applications are based on criteria such as continent, region, country, city and street address.

Custom geopositioning makes it easy to manually set targets location

<http://www.gslb.me/solutions/geographical-balancing/>

Another common use of this — and a use case that is important to understand for this case — is that the DNS service may use the IP address of the source computer to determine the best target computer for the requested service. In a proxy environment,

Kitas paplitęs šio atvejo - ir naudojimo atvejis, kurį svarbu suprasti šiuo atveju - yra tai, kad DNS paslauga gali naudoti šaltinio kompiuterio IP adresą, kad nustatytų geriausią tikslinį kompiuterį, kuriam teikiama prašoma paslauga. Tarpinio

the domain name to IP address mapping may be controlled by carefully choosing the DNS server that is used to translate the domain name to the IP address.

For example, one common use of location sensitive domain name to IP address mapping is for “content distribution networks” (CDNs). Amazon Web Services provides CDN services to its clients (<https://aws.amazon.com/cloudfront/>). Netflix provides a popular streaming video service to customers in more than 100 countries, including Lithuania. In order for Netflix to provide optimal streaming video delivery to its clients, it uses source location aware routing, similar to the geolocation that I used in §5.4 to identify the locations of the servers used in Mr. Butenis’ report.

When a customer uses a proxy, they may choose to have the DNS name to IP address mapping done relative to the source computer’s IP address, or some other server, such as the proxy service or the proxy server. This allows the source computer’s user to control which DNS server is used.

The choice of which DNS server to use is an option of the software running on the source computer. When I use the **nslookup** utility (§5.2), I did so because it allows me to

serverio aplinkoje domeno vardo ir IP adresų atvaizdavimas gali būti kontroliuojamas kruopščiai pasirenkant DNS serverį, kuris naudojamas domeno vardo vertimui į IP adresą.

Pvz., Vienas įprastas vietos jautrumo domeno vardo naudojimas IP adresams atvaizduoti yra skirtas „turinio paskirstymo tinklams“ (CDN). „Amazon Web Services“ savo klientams teikia CDN paslaugas (<https://aws.amazon.com/cloudfront/>). „Netflix“ teikia populiarias vaizdo transliavimo paslaugas klientams daugiau nei 100 šalių, įskaitant Lietuvą. Kad „Netflix“ galėtų optimaliai pateikti srautinių vaizdo įrašų siuntimą savo klientams, jis naudoja šaltinio buvimo vietos nustatymo maršrutus, panašius į geografinę vietą, kurią aš naudojau 5.4 skyriuje, siekiant nustatyti p. Butenio ataskaitoje naudojamų serverių vietas.

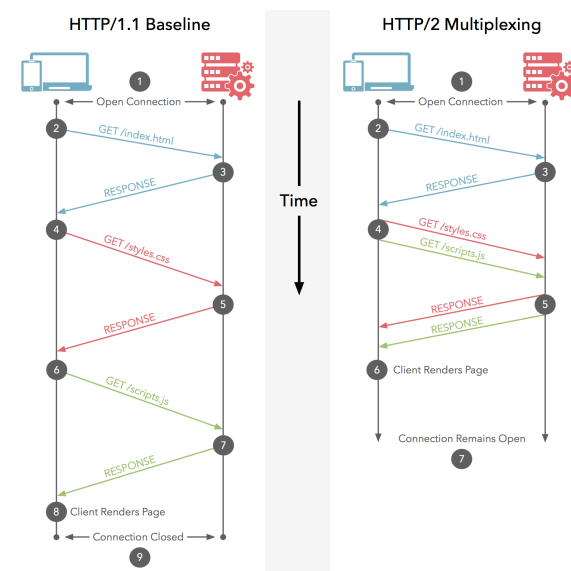
Kai klientas naudoja tarpinį serverį, jis gali nuspręsti, kad DNS vardo ir IP adresų atvaizdavimas būtų atliktas atsižvelgiant į šaltinio kompiuterio IP adresą arba į kokį nors kitą serverį, pavyzdžiui, įgalintojo serverio paslaugą ar įgalintojo serverio serverį. Tai leidžia pirminio kompiuterio vartotojui valdyti naudojamą DNS serverį.

Pasirinkite, kurį DNS serverį naudoti, pasirinkdami šaltinio kompiuteryje esančią programinę įrangą. Kai naudojuosi „nslookup“ programine įranga (5.2 skyrius),

specify the server to use, as well as the kind of DNS information to return.

The subject of geolocation sensitive DNS service is one that has been explored both commercially, such as with content distribution networks (CDNs) as well as within the academic literature [20], [21].

7.2.5 HyperText Transport Protocol (HTTP) / „HyperText“ transportavimo protokolas (HTTP)



Source [22]

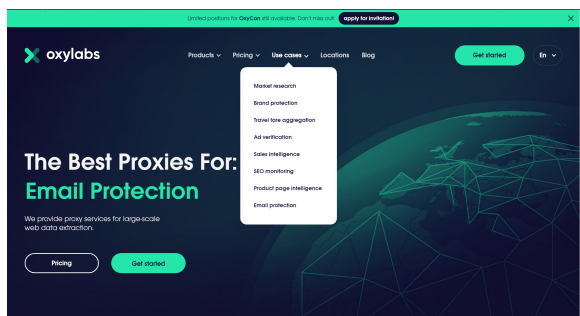
The HTTP protocol was first created in the early 1990s to permit the exchange of information between a web client and a web server. Note that this protocol does not define the contents of a web page — that is done by another protocol called HyperText Markup Language (HTML); the HTML protocol is out of scope for this report.

tai padariau todėl, kad tai leidžia man nurodyti, kokį serverį naudoti, taip pat DNS informacijos rūšį, kurią reikia grąžinti.

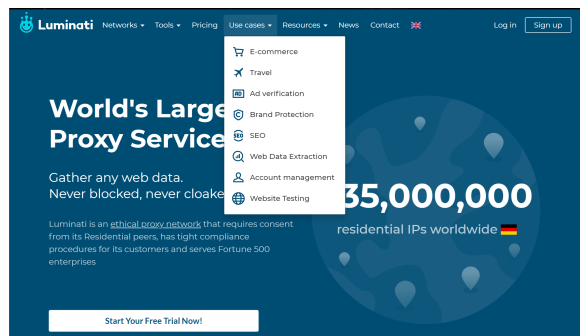
Geografinės padėties nustatymo požiūriu jautri DNS paslauga yra ta, kuri buvo išnagrinėta tiek komerciškai, pavyzdžiui, turinio paskirstymo tinkluose (CDN), tiek akademinėje literatūroje [21].

Nuoroda [22]

HTTP protokolas pirmą kartą buvo sukurtas 1990 m. Pradžioje, kad būtų galima keisti informaciją tarp žiniatinklio kliento ir tinklo serverio. Atminkite, kad šis protokolas neapibrėžia tinklalapio turinio - tai atliekama kitu protokolu, vadinamu „HyperText Markup Language“ (HTML); HTML protokolas netaikomas šiai ataskaitai.



oxylabs.io



luminati.io

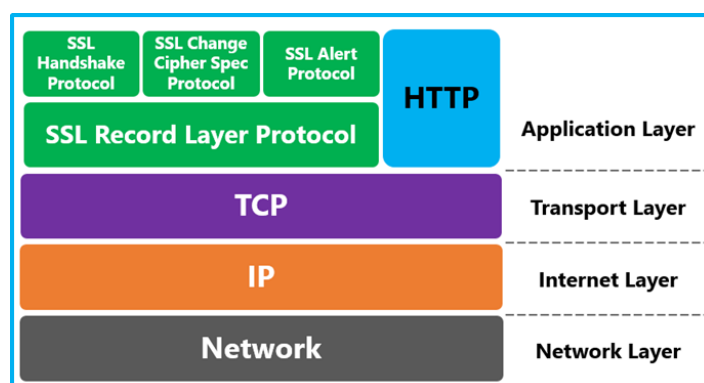
The HTTP protocol is central to the functionality of both the OX and LU products, because primary use cases for both products relate to analyzing the content of existing web pages. A proxy service is a well-known method for providing anonymity, content filtering, content filter avoidance, and performance [23].

Based upon my experience in developing systems software, the fact that there is substantial overlap in the business requirements of the OX and LU product offerings, it is natural that there would be an overlap in the protocols they would support and implement. By itself, the presence of identical features is not indicative of misappropriation of intellectual property.

HTTP protokolas yra labai svarbus tiek OX, tiek LU produktų funkcionalumui, nes pirminiai abiejų produktų naudojimo atvejai yra susiję su esamų tinklalapių turinio analize. Tarpinio serverio paslauga yra gerai žinomas būdas anonimiškumui užtikrinti, turiniui filtruoti, turinio filtrui išvengti ir našumui užtikrinti [23].

Remiantis mano patirtimi kuriant sistemų programinę įrangą ir tuo, kad labai skiriasi OX ir LU produktų verslo verslo reikalavimai, natūralu, kad protokolai, kuriuos jie palaikys ir įgyvendins, sutaps. Pats identiškų bruožų buvimas dar nereiškia neteisėto intelektinės nuosavybės pasisavinimo.

7.2.6 Secure HTTP (HTTPS) / Saugus HTTP (HTTPS)



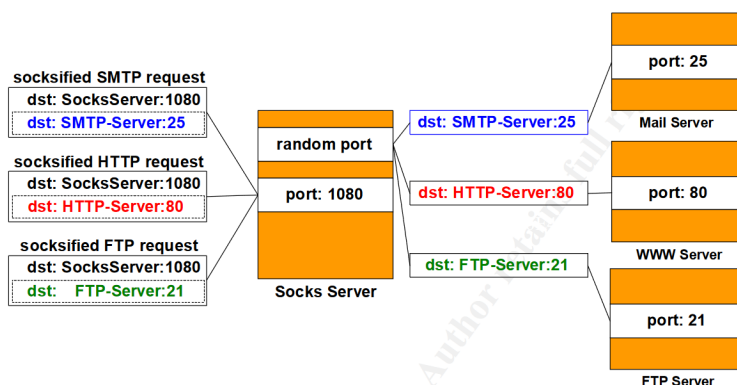
Source [24]

The Secure HyperText Transfer protocol (HTTPS) is a mechanism for establishing a cryptographically secure communications channel for HTTP. The HTTP protocol itself does not change, but the HTTP messages are sent over an encrypted communications channel between the client and the server. The techniques for handling HTTPS over a proxy server have been well-known for more than a decade [25].

Nuoroda [24]

Saugus HyperText perdavimo protokolas (HTTPS) yra kriptografiškai saugaus ryšio kanalo HTTP nustatymo mechanizmas. Pats HTTP protokolas nesikeičia, tačiau HTTP pranešimai siunčiami užšifruotu ryšio kanalu tarp kliento ir serverio. HTTPS tvarkymo tarpiniame serveryje būdai yra gerai žinomi daugiau nei dešimtmetį [25].

7.3 SOCKS5

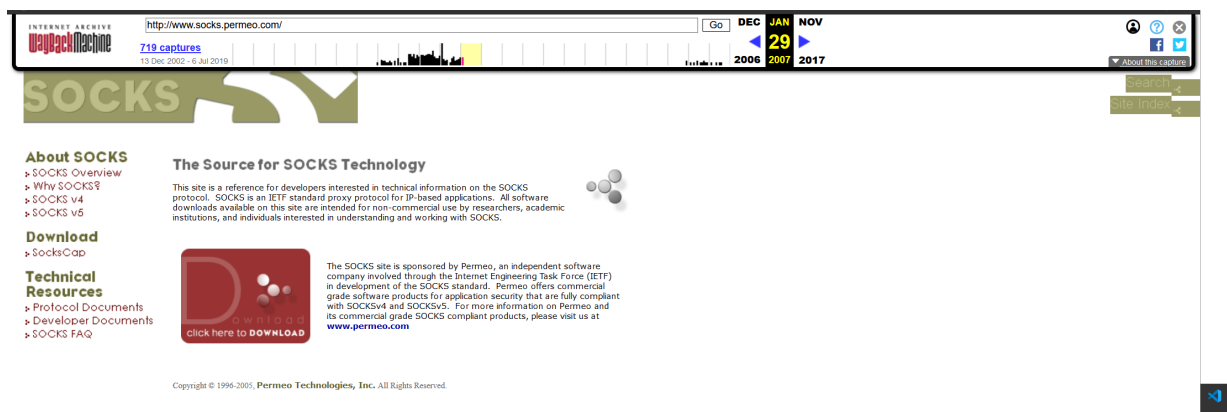


Source [26, p. 6]

Nuoroda [26, p. 6]

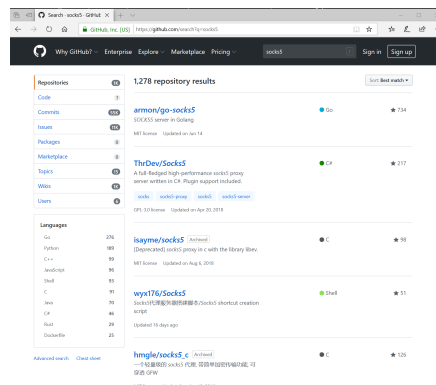
The SOCKS protocol was first published in 1992 and the most recent version of the protocol (Version 5) was published in 1999 [27], [28]. The purpose of the SOCKS protocols are to enable a program running on the source computer (such as a web browser) to securely communicate with a program running on the destination computer (such as a web server), despite the presence of an intermediate system, such as a firewall (which is one type of proxy, see §7.1.3).

SOCKS protokolas pirmą kartą buvo paskelbtas 1992 m., O naujausia protokolo versija (5 versija) buvo išleista 1999 m. [27], [28]. SOCKS protokolų tikslas yra suteikti galimybę šaltinio kompiuteryje (pvz., Žiniatinklio naršyklėje) veikiančiai programai, nepaisant tarpinės sistemos buvimo, saugiai susisiekti su tikslo kompiuteryje (pvz., Žiniatinklio serveryje) veikiančia programa, pvz., užkardą (kuri yra viena tarpinių serverių rūšių, žr. 7.1.3 skyrių).



A SOCKS5 reference implementation was published by Permio Technologies, Inc. no later than December 13, 2002, and was available through at least January 29, 2007, according to information archived by <https://web.archive.org> from the web page originally at <http://www.socks5.permio.com>.

„Permio Technologies, Inc.“ paskelbė „SOCKS5“ nuorodos įgyvendinimą ne vėliau kaip 2002 m. Gruodžio 13 d., Ir ji buvo prieinama bent iki 2007 m. Sausio 29 d. adresu <http://www.socks5.permio.com>.



A search of the open source software website <https://github.com> indicates there are over 1,000 repositories that contain the search term “socks5” in their name. Many of these appear to be implementations of the SOCKS5 protocol in a variety of computer programming languages.

Atvirojo kodo programinės įrangos svetainės <https://github.com> paieška rodo, kad yra daugiau nei 1 000 saugyklų, kurių pavadinime yra paieškos terminas „socks5“. Atrodo, kad daugelis iš jų yra SOCKS5 protokolo įgyvendinimas įvairiomis kompiuterio programavimo kalbomis.

WHOIS search results

Domain Name: LUMINATI.IO

Registry Domain ID: D503300000040504182-LRMS

Registrar WHOIS Server: whois.marcaria.com

Registrar URL: www.marcaria.com

Updated Date: 2017-11-24T12:53:15Z

Creation Date: 2014-09-09T14:49:37Z

Registry Expiry Date: 2023-09-09T14:49:37Z

Registrar Registration Expiration Date:

Registrar: Marcaria.com International, Inc.

Registrar IANA ID: 1086

Registrar Abuse Contact Email: abuse@marcaria.com

Registrar Abuse Contact Phone: +1.3478970218

Reseller:

Domain Status: ok <https://icann.org/epp#ok>

Registrant Organization: Luminati

Registrant State/Province: Israel

Registrant Country: IL

Name Server: NS-1031.AWSDNS-00.ORG

Name Server: NS-293.AWSDNS-36.COM

Name Server: NS-748.AWSDNS-29.NET

Name Server: NS-1965.AWSDNS-53.CO.UK

DNSSEC: unsigned

>>> Last update of WHOIS database: 2019-09-09T01:04:36Z <<<

whois.marcaria.com

The luminati.io domain was registered on
September 9, 2014.

Domenas luminati.io buvo įregistruotas
2014 m. Rugsėjo 9 d.

7.5 hola.org

WHOIS search results

Domain Name: HOLA.ORG
Registry Domain ID: D85808702-LROR
Registrar WHOIS Server: whois.godaddy.com
Registrar URL: http://www.godaddy.com
Updated Date: 2017-09-07T08:30:05Z
Creation Date: 2002-04-21T03:45:09Z
Registrar Registration Expiration Date: 2023-04-21T03:45:09Z
Registrar: GoDaddy.com, LLC
Registrar IANA ID: 146
Registrar Abuse Contact Email: abuse@godaddy.com
Registrar Abuse Contact Phone: +14806242505
Domain Status: clientTransferProhibited
http://www.icann.org/epp#clientTransferProhibited
Domain Status: clientUpdateProhibited
http://www.icann.org/epp#clientUpdateProhibited
Domain Status: clientRenewProhibited http://www.icann.org/epp#clientRenewProhibited
Domain Status: clientDeleteProhibited http://www.icann.org/epp#clientDeleteProhibited
Registry Registrant ID: CR28459335
Registrant Name: Derry Shribman
Registrant Organization: Hola Networks Ltd.
Registrant Street: 3 Hamachshev st
Registrant City: Netanya
Registrant State/Province:
Registrant Postal Code: 42935
Registrant Country: IL
Registrant Phone: +972.542271300
Registrant Phone Ext:
Registrant Fax:
Registrant Fax Ext:
Registrant Email: sysadmin@hola.org
Registry Admin ID: CR28459339
Admin Name: Derry Shribman
Admin Organization: Hola Networks Ltd.
Admin Street: 3 Hamachshev st
Admin City: Netanya
Admin State/Province:
Admin Postal Code: 42935
Admin Country: IL
Admin Phone: +972.542271300
Admin Phone Ext:
Admin Fax:
Admin Fax Ext:
Admin Email: xderry@yahoo.com
Registry Tech ID: CR28459337
Tech Name: Derry Shribman
Tech Organization: Hola Networks Ltd.
Tech Street: 3 Hamachshev st
Tech City: Netanya
Tech State/Province:
Tech Postal Code: 42935
Tech Country: IL
Tech Phone: +972.542271300
Tech Phone Ext:
Tech Fax:
Tech Fax Ext:
Tech Email: sysadmin@hola.org
Name Server: NS-404.AWSDNS-50.COM
Name Server: NS-1131.AWSDNS-13.ORG
Name Server: NS-1718.AWSDNS-22.CO.UK
Name Server: NS-813.AWSDNS-37.NET
DNSSEC: unsigned
URL of the ICANN WHOIS Data Problem Reporting System: http://wdprs.internic.net/
>>> Last update of WHOIS database: 2019-09-12T22:00:00Z <<<

godaddy.com/whois

The hola.org domain was registered on April 21, 2002.

The hola.org website describes Ofer Vilenski and Derry Shribman, while the luminati.io website lists Ofer Vilenski as President and Derry Shribman as

hola.org domenas buvo įregistruotas 2002 m. Balandžio 21 d.

Advokatas man patarė, kad „Luminati“ technologija pagrįsta bendra technologija su „Hola“. Tinklalapyje hola.org aprašomi Oferis Vilenski ir Derry Shribmanas, o

CTO/CSO.

svetainėje `luminati.io` Oferis Vilenskis nurodomas kaip prezidentas, o Derry Shribmanas nurodomas kaip CTO / CSO.

I have been advised by counsel that until September 2017, Luminati and Hola were the same company. Thus, for the purposes of my analysis, I have considered the dates of Luminati's technology prior to September 2017 to be the same as Hola's technology.

Man patarė advokatas, kad iki 2017 m. Rugsėjo mėn. „Luminati“ ir „Hola“ buvo ta pati įmonė. Taigi savo analizės tikslais aš laikiau, kad „Luminati“ technologijos datos iki 2017 m. Rugsėjo mėn. Yra tokios pačios kaip „Hola“ technologijos.

7.6 Data Center Proxy Providers / Duomenų centro tarpinių paslaugų teikėjai

Your Private Proxy	January 21, 2009	https://bit.ly/2kmpjRf
Anonymous Proxies	February 23, 2010	https://bit.ly/2lCD4LV
Squid Proxies	June 11, 2010	https://bit.ly/2m5BKkP
Buy Proxies	March 24, 2011	https://bit.ly/2lIIqFu
My Private Proxy	September 13, 2011	https://bit.ly/2kB9UfL
SSL Private Proxy	August 27, 2012	https://bit.ly/2kBM08X
Instant Proxies	February 28, 2013	https://bit.ly/2m6zVUN
Proxy-N-VPN	July 22, 2013	https://bit.ly/2lBAcN
Mexela	September 30, 2014	https://bit.ly/2lBAkhH

This list shows data center proxy services offered dedicated IP addresses prior to January 28, 2015, the first date of registration for `oxylabs.io`. In each row I provide a common name for the service, a date prior to January 2015 on which there is evidence that the service was being advertised to the public, the URL of the web page that demonstrates this use prior to January 2015. Each of those services has been determined to still be offering proxy

Šiame sąrašė pateikiamos duomenų centrų tarpinių serverių paslaugos, kurias iki 2015 m. Sausio 28 d., Pirmąją `oxylabs.io` registracijos datą, buvo siūlomos specialiaisiais IP adresais. Kiekvienoje eilutėje pateikiu bendrą paslaugos pavadinimą, datą iki 2015 m. Sausio mėn., Kai yra įrodymų, kad paslauga buvo reklamuojama visuomenei, tinklalapio URL, kuris parodo tokį naudojimą iki 2015 m. Sausio mėn. buvo nuspręsta, kad nuo 2019

servers commercially as of September 8, 2019, based upon the continuing operation of a responsive website.

m. rugsėjo 8 d. šios paslaugos vis dar siūlys tarpinius serverius, atsižvelgiant į nuolatinį interaktyvios svetainės veikimą.

8 Company Background / Bendrovės informacija

In this section I provide background for both Teso and Luminati.

Šiame skyriuje pateikiu „Test“ ir „Illuminati“ pagrindus.

8.1 Teso / Teso

In this section I provide background for Teso's websites `shader.io` and `oxylab.io`.

Šiame skyriuje pateikiu „Teso“ svetainių `shader.io` ir `oxylab.io` pagrindus.

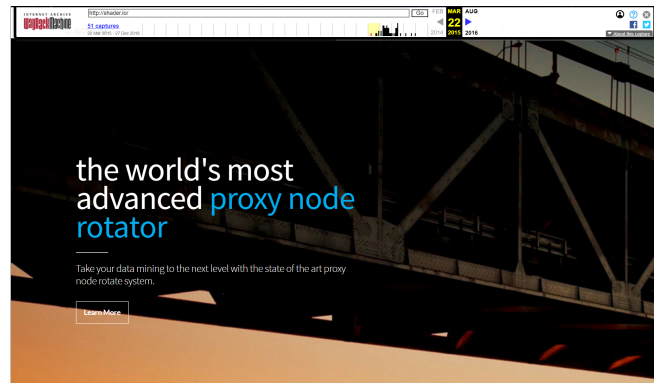
8.1.1 `shader.io`

```
Domain Name: SHADER.IO
Registrar WHOIS Server: whois.101domain.com
Registrar URL: https://www.101domain.com
Updated Date: 2018-11-05T13:00:35Z
Creation Date: 2015-01-28T07:30:56Z
Registrar Registration Expiration Date: 2020-01-28T07:30:56Z
Registrar: 101Domain GRS Ltd
Registrar IANA ID: 1011
Registrar Abuse Contact Email: abuse@101domain.com
Registrar Abuse Contact Phone: +1.7604448674
Name Server: JANET.NS.CLOUDFLARE.COM
Name Server: DOM.NS.CLOUDFLARE.COM
URL of the ICANN WHOIS Data Problem Reporting System: http://wdprs.internic.net
>>> Last update of WHOIS database: 2019-09-08T23:16:13Z <<<
```

`nic.io/whois-search.htm`

The `shader.io` domain was registered on January 28, 2015.

Domenas `shader.io` „buvo įregistruotas 2015 m. Sausio 28 d.



According to [archive.org](#) on March 22, 2015, the website was providing specific content. Thus, I consider this website was active no earlier than January 28, 2015.

When I attempted to access [shader.io](#) on September 14, 2019, my web browser was forwarded to [oxylabs.io](#). Thus, I have considered them together for the purposes of historical analysis.

[Archive.org](#) duomenimis, 2015 m. Kovo 22 d., Svetainė teikė konkretų turinį. Taigi aš manau, kad ši svetainė buvo aktyvi ne anksčiau kaip 2015 m. Sausio 28 d.

Kai 2019 m. Rugsėjo 14 d. Bandžiau pasiekti [shader.io](#), mano žiniatinklio naršyklė buvo persiųsta į [oxylabs.io](#). Taigi istorinės analizės tikslais aš juos laikiau kartu.

8.1.2 oxylabs.io

```
Domain Name: OXYLAB.IO
Registrar WHOIS Server: whois.101domain.com
Registrar URL: https://www.101domain.com
Updated Date: 2019-08-05T13:35:42Z
Creation Date: 2015-11-30T14:44:18Z
Registrar Registration Expiration Date: 2019-11-30T14:44:18Z
Registrar: 101Domain GRS Ltd
Registrar IANA ID: 1011
Registrar Abuse Contact Email: abuse@101domain.com
Registrar Abuse Contact Phone: +1.7604448674
Name Server: BRIT.NS.CLOUDFLARE.COM
Name Server: HANK.NS.CLOUDFLARE.COM
URL of the ICANN WHOIS Data Problem Reporting System: http://wdprs.internic.net
>>> Last update of WHOIS database: 2019-09-09T01:01:17Z <<<
```

nic.io/whois-search.htm

The [oxylabs.io](#) domain was registered on November 30, 2015.

According to [archive.org](#) on November 13, 2016, the website was returning redirection

Domenas [oxylabs.io](#) buvo įregistruotas 2015 m. Lapkričio 30 d.

Remiantis [archive.org](#), 2016 m. Lapkričio 13 d., Svetainė pateikė peradresavimo

requests, which means it was not directly serving content. On May 13, 2017, the website was serving content directly. Thus, I consider this website was active no earlier than November 14, 2016.

Thus, when I refer to the “date that oxylabs.io became available” I have used this to mean January 28, 2015, which is the date that `shader.io` became available.

8.2 Luminati / Luminati

In this section I describe background information about Luminati, including a description of their open source program “Luminati Proxy Manager” and their website `luminati.io`.

užklaudas, tai reiškia, kad ji tiesiogiai neteikia turinio. 2017 m. Gegužės 13 d. Svetainė tiesiogiai teikė turinį. Taigi aš manau, kad ši svetainė buvo aktyvi ne anksčiau kaip 2016 m. Lapkričio 14 d.

Man advokatas patarė, kad `shader.io` ir `oxylabs.io` “priklauso”, Teso “. Taigi, analizuodamas datas, analizuodamas „Teso“ gaminių pasiūlą, naudoju kuo ankstesnę datą - 2015 m. Sausio 28 d.

Šiame skyriuje aprašoma pagrindinė informacija apie „Luminati“, įskaitant jų atvirojo kodo programas „Luminati Proxy Manager“ aprašymą ir jų svetainę `luminati.io`.

8.2.1 Luminati Proxy Manager / „Luminati“ tarpinio serverio valdytojas

 **Luminati Proxy Manager**

An open-source software for seamlessly managing multiple proxies via API and admin UI

- One entry point
- Concurrent connections
- Auto retry rules
- Real time statistics

How to install

Latest version: 1.152.345
Previous versions

 Windows installer [Download exe file v.1.152.345](#)

 Bash install script (recommended for Mac OS and Linux)
`curl -L https://luminati.io/static/lpm/luminati-proxy-latest-setup.sh | bash` [Copy](#)

 NPM package [Link](#)
`sudo npm install -g @luminati-io/luminati-proxy` [Copy](#)

 Docker image [Link](#)
`docker pull luminati/luminati-proxy` [Copy](#)

 Github source code [Link](#)

To send a request

 Run the request code in console [Generate request code](#)

To manage proxies via console

 Use Proxy Manager commands [LPM commands](#)

Use raw API without Proxy Manager

- ✔ No client needed
- ✘ No advanced features

[Generate request code](#)

`https://luminati.io/cp/lpm`

The Luminati Proxy Manager is an optional package that Luminati makes freely available for managing proxy servers, including those provided by the Luminati data center proxy

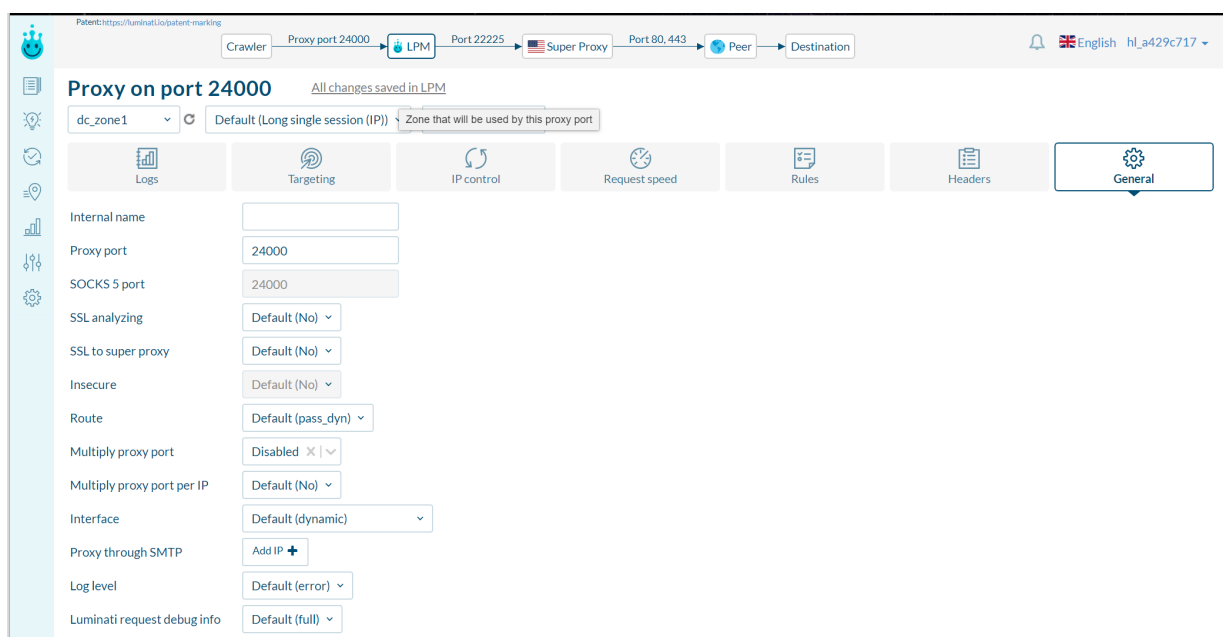
„Luminati Proxy Manager“ yra pasirenkamas paketas, kurį „Luminati“ suteikia laisvai prieinamiems tarpiniams serveriams valdyti, įskaitant tuos, kuriuos

product. Note that the page on the `luminati.io` site says it is not required to use the Luminati data center proxy product.

“Luminati Proxy Manager (LPM) is an advanced open-source proxy manager for web scraping, Lots of features and functions of LPM, You can Limit or auto-retry your requests, Create rules for IP rotation, enable the SOCKS5 Proxy via LMP and more...It’s an advanced proxy tool to handle your proxies request...” (<https://www.stupidproxy.com/luminati-guide/>).

teikia „Luminati“ duomenų centro tarpinis produktas. Atminkite, kad `luminati.io` svetainės puslapyje sakoma, kad nebūtina naudoti „Luminati“ duomenų centro įgalintojo produkto.

„Luminati Proxy Manager“ (LPM) yra patobulintas atvirojo kodo tarpinis serveris, skirtas žiniatinklio įrašui, daug LPM funkcijų ir funkcijų, galite apriboti arba automatiškai pakartoti savo užklausas, sukurti IP sukimosi taisykles, įgalinti SOCKS5 tarpinį serverį per LMP ir dar daugiau ... Tai patobulintas tarpinio serverio įrankis, skirtas jūsų įgaliojimų asmenų užklausoms tvarkyti ... ”(<https://www.stupidproxy.com/luminati-guide/>).



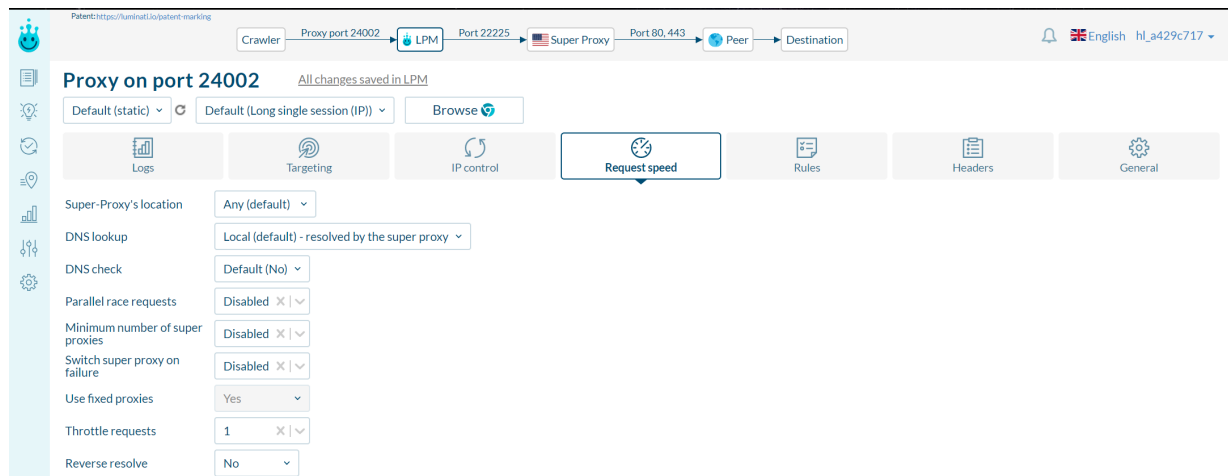
<http://127.0.0.1:22999/proxy/24002/speed>

The Luminati Proxy Manager permits configuration of a SOCKS5 gateway service. However, this runs on the local client

„Luminati Proxy Manager“ leidžia konfigūruoti „SOCKS5“ šliuzo paslaugą. Tačiau tai veikia vietiniame kliento

computer. It is not limited to use with Luminati products.

kompiuteryje. Naudojimas ne tik „Luminati“ gaminiais.



<http://127.0.0.1:22999/proxy/24002/speed>

The Luminati Proxy Manager permits throttling the number of operations. As noted in §5.2.8, the Luminati data center product does not implement rate limiting itself. The Luminati Proxy Manager can provide client-side rate limiting if needed.

„Luminati Proxy Manager“ leidžia nustatyti operacijų skaičių. Kaip pažymėta 5.2.8 skyriuje, „Luminati“ duomenų centro produktas neįgyvendina greičio ribojimo. Jei reikia, „Luminati Proxy Manager“ gali apriboti kliento dažnį.

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SOCKS protocol version 5 server and client implementations for node.js
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luminati_client
Luminati client API
JavaScript MIT 2 stars 3 forks 0 issues 1 watcher Updated on Jul 26, 2018

api
luminati.io API
Java 4 stars 8 forks 0 issues 1 watcher Updated on Apr 4, 2018

vim-jsx-improve
Forked from neoclides/vim-jsx-improve
Syntax and indent plugin for React jsx.
Vim script 19 stars 0 forks 0 issues 0 watchers Updated on Oct 10, 2017

ruby-Luminati
Forked from Sebastian/Luminati
Ruby-wrapper for communicating with the Luminati.io-network.
Ruby MIT 9 stars 5 forks 0 issues 0 watchers Updated on Mar 20, 2015

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<https://github.com/luminati-io/luminati-proxy>

The Luminati Proxy Manager is an open source software package; the source code for this is freely available on [Github.com](https://github.com). The initial distribution of source code (a

„Luminati Proxy Manager“ yra atvirojo kodo programinės įrangos paketas; šaltinio kodas yra laisvai prieinamas [url Github.com](https://github.com). Pradinis pirminio kodo („įsipareigoti“)

“commit”) was on April 13, 2016. It an optional proxy manager that may be run on a computer controlled by the user. It is not a part of LU.

Neither Mr. Butenis’ report nor the Statement of Claims describe comparing this publicly available Luminati code with any Teso code, public or private.

8.2.2 luminati.io / luminati.io

This section provides historical records from the `luminati.io` website. I have provided the date of the information obtained from `archive.org`.

The `luminati.io` website as of the dates discussed in this section are descriptions of their residential proxy service, as they did not have a data center proxy service offering (the “LU” product described in Mr. Butenis’ report) at that time. While there are common elements to both products, they are not identical. However, Luminati certainly used their existing infrastructure, experience, knowledge, and code in the development of their data center product.

platinimas buvo 2016 m. Balandžio 13 d. Tai yra neprivalomas įgaliotasis serverio tvarkytuvas vartotojo kompiuterinei sistemai. Tai nėra „Luminati“ duomenų centro įgaliooto serverio paslaugų dalis.

Nei p. Butenio ataskaitoje, nei ieškinių pareiškime nėra aprašyta, kaip palyginti šį viešai prieinamą „Luminati“ kodą su koku nors valstybiniu ar privačiu „Teso“ kodu.

Šiame skyriuje pateikiami istoriniai įrašai iš `luminati.io` svetainės. Aš pateikiau informacijos, gautos iš `archive.org`, datą.

Svetainė `luminati.io` nuo šiame skyriuje aptariamų datų yra jų gyvenamosios vietos tarpinio serverio paslaugos aprašymai, nes jie neturėjo duomenų centro įgaliooto serverio paslaugų pasiūlymo (p. Butenio aprašytas produktas „LU“). ataskaita) tuo metu. Nors abiemis gaminiais būdingi bendri elementai, jie nėra tapatūs. Tačiau kurdami duomenų centro produktą, „Luminati“ neabejotinai panaudojo turimą infrastruktūrą, patirtį, žinias ir kodą.

Luminati is the world's largest anonymity network. Use it to route your data through any of our millions of IPs that are located in every city around the world.

<https://web.archive.org/web/20141016080718/http://luminati.io/>

October 16, 2014

The `luminati.io` website, as of October 16, 2014 indicates that they provide anonymous proxies.

`luminati.io` svetainėje nuo 2014 m. Spalio 16 d. Nurodoma, kad jie teikia anoniminius įgaliotinius.

Luminati Benefits

How is Luminati better than alternative solutions?

- **All countries** – Luminati is the only network that provides you with IPs in ALL countries in the world! (except N. Korea)
- **Real anonymity** – the proxies in the Luminati network are regular PCs, laptops and phones, and thus are not identified as proxies or as Tor network nodes
- **Super scalable** – Scale up to millions of IPs and back down to a few without costs or re-configuration
- **Super speed** – Much faster than Tor (similar in speed to regular Internet)
- **Highly secure** – VPN connection from your Super-Proxy to any of millions of nodes 'in the wild' reduce the risk of man-in-the-middle attacks, snooping, or control of end nodes (as may be the case in Tor)
- **Highly accurate** – Sites that identify you as a proxy may provide you with misleading information. Luminati's Network is comprised of real devices, not proxies and not Tor end points
- **No overhead** – No need to employ staff for managing proxies
- **Simple** – Set the Super-Proxy as your crawler's proxy, and go!
- **Proxies in every city** – Reroute through any city in the world (coming soon)!
- **Supported protocols** – Supports HTTP, HTTPS and TLS

<https://web.archive.org/web/20141016080718/http://luminati.io/>

October 16, 2014

The `luminati.io` website, as of October 16, 2014 indicates that they support HTTP (§7.2.5) and HTTPS (§7.2.6).

`luminati.io` svetainėje nuo 2014 m. Spalio 16 d. Nurodoma, kad jos palaiko HTTP (7.2.5 skyrius) ir HTTPS (7.2.6 skyrius).

How to Guide

Using the system – Overview

- **Setup** – Set the Super Proxy as your proxy, and it will relay your requests through a Peer on the Luminati network
- **Choose** – You can specify to the Super Proxy which country the Peer should be at, when to change IP within the country and set other parameters
- **Control** – The control of these parameters is done either by inserting special HTTP headers to the HTTP requests you send to the Super Proxy, or by routing through ports on the Super Proxy that are mapped to various countries
- **Try before you buy** – Luminati provides you with a free 7 day trial of the system. Sign up now to start
- **Billing** – You pay only for the traffic you route through Luminati, regardless of how many different IPs you use

<https://web.archive.org/web/20141016080718/http://luminati.io/>

October 16, 2014

The `luminati.io` website, as of October 16, 2014 indicates that they provide IP address rotation. In addition, it describes their implementation of using HTTP (§7.2.5) headers to control the behavior of the Luminati proxy service. This is an example of an Application Programming Interface (API) in internet (web) parlance.

`luminati.io` svetainėje nuo 2014 m. Spalio 16 d. Nurodoma, kad jos teikia IP adresų kaitaliojimą. Be to, aprašomas jų įgyvendinimas naudojant HTTP (7.2.5 skyrius) antraštes, kad būtų galima kontroliuoti „Luminati“ duomenų centro tarpinio serverio paslaugą. Tai yra internetinio (žiniatinklio) kalbų programinės įrangos sąsajos (API) pavyzdys.



<https://web.archive.org/web/20141016080718/http://luminati.io/>

October 16, 2014

The `luminati.io` website, as of October 16, 2014 indicates that they provide the ability to control IP address selection when using

`luminati.io` svetainė nuo 2014 m. Spalio 16 d. Nurodo, kad jos suteikia galimybę kontroliuoti IP adresų pasirinkimą, kai

the Luminati proxy service. It also graphically describes the process of making the original sending site anonymous. The description of the HTTP request changing the country and/or IP address of the proxy server using HTTP demonstrates that they support the HTTP protocol as well as provide a web service API for managing the Luminati proxy service's processing of the request.

naudojamasi „Luminati“ duomenų centro serveriu. Taip pat grafiškai aprašomas originalios siuntimo svetainės anonimiškumo procesas. HTTP užklauskos, pakeičiančios tarpinio serverio, naudojančio šalį, ir (arba) IP adresą, naudojant HTTP, aprašymas rodo, kad jie palaiko HTTP protokolą, taip pat pateikia interneto paslaugų API, skirtą „Luminati“ duomenų centro tarpinio serverio paslaugos užklauskos apdorojimui tvarkyti.

9 Declaration / Deklaracija

The opinions provided in this report are for the use of Rapolas Kasparavičius of the firm of Ellex Valiunas in the matter of UAB Teso LT versus Lumnati Network LTD.

In this capacity:

- I am aware that as an expert witness, I have the duty to assist the Court and I am not an advocate for any party;
- I have made my report in conformity with this duty; and
- I will, if called upon, give oral or written testimony, and give that testimony in conformity with that duty.

Pateiktos nuomonės pateiktos dėl „Ellex Valiunas“ firmos Rapolo Kasparavičiaus naudojimosi UAB „Teso LT“ ir „Lumnati Network LTD“.

Atlikdamas šias pareigas:

- Aš žinau, kad kaip ekspertas liudytojas turiu pareigą padėti Teismui ir nesu advokatas jokioms šalims;
- Aš parengiau savo pranešimą laikydamasis šios pareigos; ir
- Aš būsiu paprašytas duoti žodinius ar rašytinius parodymus ir duosiu tuos parodymus laikydamasis tos pareigos.


William Anthony Mason

September 16, 2019

Date

Bibliography / Bibliografija

Bibliography

- [1] C. Sar, *What is a shared proxy?*, <https://bit.ly/2k49y0w>, (accessed September 8, 2019), Aug. 2019.
- [2] ———, *What is a dedicated proxy?*, <https://bit.ly/2kKwnap>, (accessed September 8, 2019), Aug. 2019.
- [3] R. T. Fielding and R. N. Taylor, *Architectural styles and the design of network-based software architectures*. University of California, Irvine Doctoral dissertation, 2000, vol. 7.
- [4] D. Erickson, M. Casado, and N. McKeown, “The effectiveness of whitelisting: A user-study.”, in *CEAS*, 2008.
- [5] J. Wu, P. Teregowda, J. P. F. Ramírez, P. Mitra, S. Zheng, and C. L. Giles, “The evolution of a crawling strategy for an academic document search engine: Whitelists and blacklists”, in *Proceedings of the 4th Annual ACM Web Science Conference*, ACM, 2012, pp. 340–343.
- [6] C. J. Dietrich and C. Rossow, “Empirical research on ip blacklisting.”, in *CEAS*, 2008.
- [7] D. Beaumont, *How to explain vertical and horizontal scaling in the cloud*, <https://ibm.co/2LI6SPg>, (accessed September 14, 2019), Apr. 2014.
- [8] R. Bellis, “Dns proxy implementation guidelines”, 2009, (accessed September 13, 2019).
- [9] D. E. Knuth, “The art of computer programming. vol. 3: Sorting and searching”, 1973.
- [10] R. Fielding, J. Gettys, J. Mogul, H. Frystyk, L. Masinter, P. Leach, and T. Berners-Lee, *Rfc 2616: Hypertext transfer protocol-http/1.1*, <https://bit.ly/1D0Czhq>, 1999.
- [11] J. Franks, P. Hallam-Baker, J. Hostetler, S. Lawrence, P. Leach, A. Luotonen, and L. Stewart, *Http authentication: Basic and digest access authentication*, <https://bit.ly/2ktn3rs>, 1999.
- [12] A. L. Russel, *Osi: The internet that wasn't*, <https://spectrum.ieee.org/tech-history/cyberspace/osi-the-internet-that-wasnt>, (accessed September 6, 2019), Jul. 2013.
- [13] B. Mitchell, *What are network protocols?*, <https://www.lifewire.com/definition-of-protocol-network-817949>, (accessed September 4, 2019), Jun. 2019.

- [14] A. S. Tanenbaum and D. J. Wetherall, “Computer networks”, 2011.
- [15] C. N. Notes and S. Guide, *Data encapsulation and de-encapsulation explained*, <https://bit.ly/2kpGKjV>, (accessed September 7, 2019), Jan. 2019.
- [16] Mozilla, *An overview of http*, <https://mzl.1a/2k5zWHP>, (accessed September 7, 2019), Aug. 2019.
- [17] J. Postel *et al.*, “Rfc 791: Internet protocol”, 1981.
- [18] M. Bishop, “Quic internet draft”, Sep. 2019, (accessed September 7, 2019).
- [19] J. Postel, “Transmission control protocol”, 1981.
- [20] O. Dan, V. Parikh, and B. D. Davison, “Ip geolocation through reverse dns”, *arXiv preprint arXiv:1811.04288*, 2018.
- [21] C. Huang, N. Holt, A. Wang, A. G. Greenberg, J. Li, and K. W. Ross, “A dns reflection method for global traffic management.”, in *USENIX Annual Technical Conference*, 2010.
- [22] B. Patch, *Http/2 for a faster web*, <https://bit.ly/1MVtsN1>, (accessed September 7, 2019), Mar. 2015.
- [23] T. T. Reviews, *When and why should i use a proxy server*, <https://bit.ly/2lCm0p1>, (accessed September 8, 2019), Apr. 2015.
- [24] K. K. Panday, *Ssl handshake and https bindings on iis*, <https://bit.ly/2lYC1px>, (accessed September 7, 2019), Aug. 2013.
- [25] B. O’Donovan, *Secure communication with stunnel*, <https://bit.ly/2lIny16>, (accessed September 8, 2019), Oct. 2004.
- [26] S. Institute, *Global information assurance certification paper*, <https://bit.ly/2knXRCG>, (accessed September 6, 2019), 2002.
- [27] D. Koblas and M. R. Koblas, “Socks”, in *USENIX Security Symposium*, (accessed September 8, 2019), 1992.
- [28] M. Leech, “Socks protocol version 5”, Mar. 1996, (accessed September 8, 2019).
- [29] N. Finn and T. Mason, “Atm lan emulation”, *IEEE Communications Magazine*, vol. 34, no. 6, pp. 96–100, 1996.
- [30] R. Nagar and T. Mason, *Windows NT file system internals (OSR classic reprints)*. Osr Press, 2006.
- [31] W. A. Mason, P. G. Viscarola, M. J. Cariddi, and S. J. Noone, *Managing access to a resource*, US Patent 8,539,228, Sep. 2013.

- [32] P. Viscarola and W. A. Mason, *Windows NT device driver development*. New Riders Publishing, 1998.
- [33] W. A. Mason, P. G. Viscarola, M. J. Cariddi, and S. J. Noone, *Managing lock rankings*, US Patent 7,809,897, Oct. 2010.
- [34] —, *Managing application resources*, US Patent 8,024,433, Sep. 2011.
- [35] —, *Managing lock rankings*, US Patent 7,512,748, Mar. 2009.
- [36] W. A. Mason and P. G. Viscarola, *Log-structured host data storage*, US Patent 7,949,693, May 2011.
- [37] W. A. Mason and R. D. W. Widdowson, *Methods and systems for data storage*, US Patent App. 12/658,209, Aug. 2010.
- [38] W. A. Mason, P. G. Viscarola, M. J. Cariddi, and S. J. Noone, *Systems and methods for arbitrary data transformations*, US Patent 8,990,228, Mar. 2015.
- [39] W. A. Mason, *Methods and systems for data storage*, US Patent 9,830,329, Nov. 2017.
- [40] W. A. Mason and R. D. W. Widdowson, *Methods and systems for remote data storage utilizing content addresses*, US Patent App. 13/608,852, Feb. 2013.
- [41] W. A. Mason, *Work queue thread balancing*, US Patent 9,535,759, Jan. 2017.
- [42] M. L. Kazar, B. W. Leverett, O. T. Anderson, V. Apostolides, B. A. Bottos, S. Chutani, C. Everhart, W. A. Mason, S.-T. Tu, and E. R. Zayas, “Decorum file system architectural overview.”, in *USENIX Summer*, vol. 90, 1990, pp. 151–163.
- [43] S. Chutani, O. T. Anderson, M. L. Kazar, B. W. Leverett, W. A. Mason, R. N. Sidebotham, *et al.*, “The episode file system”, in *Proceedings of the USENIX Winter 1992 Technical Conference*, San Fransisco, CA, USA, 1992, pp. 43–60.
- [44] W. A. Mason, S. J. Noone, and R. D. W. Widdowson, *File system directory attribute correction*, US Patent 8,903,874, Dec. 2014.
- [45] W. A. Mason, P. G. Viscarola, M. J. Cariddi, and S. J. Noone, *Systems and methods for arbitrary data transformations*, US Patent 8,521,752, Aug. 2013.
- [46] T. Mason, A. Gavrilovska, and D. A. Joyner, “Collaboration versus cheating: Reducing code plagiarism in an online ms computer science program”, in *Proceedings of the 50th ACM Technical Symposium on Computer Science Education*, ACM, 2019, pp. 1004–1010.
- [47] M. Sysel and O. Doležal, “An educational http proxy server”, *Procedia Engineering*, vol. 69, pp. 128–132, 2014.

- [48] J. R. Levine, “Dns blacklists and whitelists”, 2010, (accessed September 8, 2019).

Curriculum Vitae / Gyvenimo aprašymas

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Curriculum Vitae



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DOCTORAL RESEARCH

“Percipience: Enabling Personal Data Mining”

My research examines mechanisms for extending traditional file systems functionality to enable the capture, storage, retrieval, and search of the relationships between data elements. Because of the vast body of existing applications, my work explores how to support existing applications at the same time enabling incremental evolution of applications to exploit this functionality to provide a richer set of tools to find data relationships amongst the user’s personal data.

WORK EXPERIENCE

wamason.com CURRENT, FROM JULY 2017 (PT)

Expert

Litigation support for clients requiring expert services. This work primarily focuses on assisting in the preparation of materials necessary for patent litigation. I provide research, prepare reports, explain technical terms clearly and lucidly, engage in depositions, etc. The focus is on clearly explaining potentially complex technical concept and terms to non-technical people.

Georgia Institute of Technology CURRENT, FROM AUGUST 2015 (PT)

Instructional Assistant

Provide support for the Online Master of Science in Computer Science programs course. Responsibilities included providing formative and summative analysis to students, develop automatic grading tools, explain concepts to students, provide direction to other team members, work with instructors regarding course execution and improvements. Responsible for detecting and reporting student plagiarism.

BitRaider CURRENT, FROM 2011 (PT)

Technical Adviser

Provide technical advice to BitRaider, a company specializing in streaming game delivery.

OSR Open Systems Resources, Inc. NOVEMBER 1994 – NOVEMBER 2016 (FT)

Consulting Partner and Treasurer/Vice-President

In this role I was responsible for providing services to commercial customers, by designing and implementing key file systems technologies for file systems and file system filter drivers. I was involved in critical aspects of our educational offerings, teaching classes in Driver Development, File Systems Development, File System mini-filter development and Kernel Debugging. Focus was on the Windows OS platform. Recent projects include: Isolation Filter Kit that provides multiple simultaneous views of

EDUCATION

2017 – PRESENT	Doctor of Philosophy IN PROGRESS Computer Science <i>University of British Columbia</i>
2015 – 2017	Master of Science Computer Science <i>Georgia Institute of Technology</i>
1983 – 1987	Bachelor of Science Department of Mathematics <i>University of Chicago</i>

EXPERT WITNESS EXPERIENCE

GENERAL INTERNET TECHNOLOGY -
STORAGE TECHNOLOGIES - US DISTRICT COURT DE
OS KERNEL, DEVICE DRIVERS - USPTO BOARD
OS KERNEL, DEVICE DRIVERS - US DISTRICT COURT DE
COMPUTER BACKUP - US DISTRICT COURT NORTHERN CA

COMPUTER SKILLS

LANGUAGES C, C++, Python, L^AT_EX, etc.
OPERATING SYSTEMS Windows, Linux, UNIX, etc.

COMMUNICATION SKILLS

CONFERENCES	Invited Speaker at WDC Invited Speaker at IDC Invited Speaker at AFSUG Retrospective
POSTERS	Percipience - UBC CS-50 – 2018
COURSES	Developing Windows File Systems Developing Windows FS Mini-Filters Advanced Windows Driver Development Windows Internals (Source Code) Porting Windows Drivers to 64-bit Windows Kernel Debugging

SKILLS

Goal Oriented

I believe in action over long-winded discussions. I listen to everyone’s viewpoints and use my judgement to act based on consensus to achieve goals quickly and efficiently.

Fail Fast

When working on projects it is important to work towards

a single file on either local or network storage without modifying the underlying file system. This abstraction was useful for transparent per-file encryption in that different applications could be given different views (raw or encrypted) of the file, with coherency, specialized file system mini-filter for transparently backing up SQL databases to Azure, another for doing block level change tracking for SQL databases.

SEPTEMBER 1993 TO NOVEMBER 1994 (FT)

FORE Systems *Technical Manager*

In this role I was responsible commercial ATM adapter device driver development for PC platforms (Windows, Novell, and OS/2). Actively participated in the ATM Forum (SA&A and LAN Emulation SWG). Worked with both Microsoft and Novell to define and establish architectural support for ATM.

JULY 1989 TO SEPTEMBER 1993 (FT)

Transarc Corporation *Area Manager (previously Member of Technical Staff)*

Technical management for DCE LFS physical media file systems work: coordinate activities of team members, managed DCE/DFS integration including working with key external partners: OSF, HP, IBM and Digital Equipment Corporation. Transarc AFS development on various UNIX platforms, DCE/DFS development. As a member of the Episode team I was responsible for designing and implementing the transactional system, still novel for its use of undo/redo model journaling. Part of the DCE/DFS development team, participated in architecture and design.

JUNE 1987 TO JUNE 1989 (FT)

Leland Stanford Jr. University *Research Staff*

Working in **Professor David Cheriton's** Distributed Systems Group. Responsible for systems administration for the group. Implemented the UNIX version of VMTP (Versatile Message Transport Protocol), a reliable transport protocol for use over IP, including IP multicast. Wrote kernel mode device drivers for Ethernet NICs for the V Operating System, as well as a primitive kernel debugger.

success. One way I use to achieve this is to establish mechanisms to tell *when* something is not going to work — to *fail fast* in order to maximize the useful work and minimize the cost of investigating paths that will not lead to success.

Education

One of the most rewarding aspects of my work is in helping other people learn and grow. It is deeply rewarding when former students, employees, and colleagues tell me how I helped them in improving their own work. To achieve this, I strive to break things down and make them understandable based upon the listener's current level of understanding.

Passionate

I have been involved in computer systems from an early age. My education, work, and research have fueled my passion for these topics. I enjoy learning new technologies and finding ways to re-examine old technologies in light of greater understanding. I greatly enjoy finding new ways to make systems better, not only in terms of performance but also as tools for creativity and productivity for *all* users.

PATENTS

US/9830329 METHODS AND SYSTEMS FOR DATA STORAGE
US/9600486 FILE SYSTEM DIRECTORY ATTRIBUTE CORRECTION
US/9535759 WORK QUEUE THREAD BALANCING
US/8990228 ARBITRARY DATA TRANSFORMATION
US/8903874 FILE SYSTEM DIRECTORY ATTRIBUTE CORRECTION
US/8539228 MANAGING ACCESS TO A RESOURCE
US/8521752 ARBITRARY DATA TRANSFORMATIONS
US/8024433 MANAGING APPLICATION RESOURCES
US/7809897 MANAGING LOCK RANKINGS
US/7512748 MANAGING LOCK RANKINGS
US/7949693 LOG-STRUCTURED HOST DATA STORAGE
US 15/791,486 METHODS AND SYSTEMS FOR DATA STORAGE

BOOKS

Windows NT Device Driver Development (with Peter Viscarola), New Riders, 1998

lex & yacc, O'Reilly & Associates, June 1990 (1st Edition), November 1992 (2nd Edition)

PEER REVIEWED PAPERS

Collaboration versus Cheating: Reducing Code Plagiarism in an Online MS Computer Science Program, SIGCSE February 27-March 2, 2019 (Primary author)

DEcorum File Systems Architecture, USENIX Technical Conference Summer 1990. (Co-author)

The Episode File System, USENIX Technical Conference Winter 1992. (Co-author)

ARTICLES

CHANGES IN WINDOWS 10 REDSTONE, THE NT INSIDER, JANUARY/FEBRUARY 2017

LOGICAL AND PHYSICAL FILE SIZES IN WINDOWS, THE NT INSIDER, MARCH/APRIL 2015

MINIFILTER LOAD AND UNLOAD ORDERING, THE NT INSIDER, MAY/JUNE 2014

DRIVE LETTER ALTERNATIVES, THE NT INSIDER, MAY/JUNE 2014

WINDOWS POOL MANAGER, THE NT INSIDER, JAN/FEB 2014

THE ISOLATION DRIVER (PART II), THE NT INSIDER, VOL. 18, ISS. 1, JANUARY 2011

THE ISOLATION DRIVER (PART I), THE NT INSIDER, VOL. 17, ISS. 2, JULY 2010

UNDOCUMENTED DFS & RDR INTERACTIONS, THE NT INSIDER, VOL. 17, ISS. 2, JULY 2010

FILE SYSTEMS, FILE SYSTEM FILTER DRIVERS AND REMOVABLE STORAGE DEVICES, THE NT INSIDER, VOL. 16, ISS. 2, MAY 2009

FILTERING FILE SYSTEMS – TEN THINGS YOU SHOULD KNOW, THE NT INSIDER, VOL. 16, ISS. 1, JANUARY 2009

DEBUGGING IO3: WHERE TO GO WITH A SYSTEM CRASH, THE NT INSIDER, VOL. 15, ISS. 2, JULY 2008

WINDOWS VISTA AND FILE SYSTEMS, THE NT INSIDER, VOL. 14, ISS. 4, NOVEMBER 2007

AN INTRODUCTION TO TRANSACTIONS, THE NT INSIDER, VOL. 14, ISS. 1, JANUARY 2007

MUP CHANGES IN WINDOWS VISTA, THE NT INSIDER, VOL. 14, ISS. 1, JANUARY 2007

OBTAINING A USEFUL NAME FOR THE EXECUTABLE IMAGE IN A PROCESS, THE NT INSIDER, VOL. 13, ISS. 4

AN INTRODUCTION TO FILE SYSTEM STREAMS, THE NT INSIDER, VOL. 13, ISS. 2, MARCH 2006

HASHING TECHNIQUES, THE NT INSIDER, VOL. 13, ISS. 1, JANUARY 2006

THE TRANSACTIONAL FILE SYSTEM (TXFS) IN WINDOWS, THE NT INSIDER, VOL. 12, ISS. 3 MAY 2005

FILE SYSTEM FILTER CONTEXT – OBSERVATIONS & COMMENTS, THE NT INSIDER, VOL. 11, ISS. 5, NOVEMBER 2004

TESTING FILE SYSTEMS, THE NT INSIDER, VOL. 11, ISS. 3, MAY 2004

BLOCKING SPECIAL KERNEL APCs AT IRQL PASSIVE_LEVEL, THE NT INSIDER, VOL. 11, ISS. 2, MARCH 2004

CLEVER WAYS TO SAVE STACK SPACE, THE NT INSIDER, VOL. 11, ISS. 2, MARCH 2004

FINDING FILE CONTENTS IN MEMORY, THE NT INSIDER, VOL. 11, ISS. 1, JANUARY 2004

DEBUGGING A SOUND DRIVER, THE NT INSIDER, VOL. 11, ISS. 1, JANUARY 2004

CACHING IN THE PENTIUM 4 PROCESSOR, THE NT INSIDER, VOL. 11, ISS. 1, JANUARY 2004

FINDING YOUR WAY THROUGH THE STACK, THE NT INSIDER, VOL. 9, ISS. 6, NOVEMBER 2003

EMERGING ISSUES IN IOCANCELFILEOPEN, THE NT INSIDER, VOL. 10, ISS. 4, SEPTEMBER 2003

DEBUGGING ANOTHER CRASH DUMP, THE NT INSIDER, VOL. 10, ISS. 2, MARCH 2003

CALLING CONVENTIONS FOR THE X86, THE NT INSIDER, VOL. 10, ISS. 1, JANUARY 2003

REPARSE POINTS IN WINDOWS, THE NT INSIDER, VOL. 10, ISS. 1, JANUARY 2003

MANAGING ADDRESS SPACE INCREASES FOR IA64, THE NT INSIDER, VOL. 9, ISS. 4, JULY 2002

DRIVE LETTER ASSIGNMENT AND THE MOUNT MANAGER, THE NT INSIDER, VOL. 9, ISS. 4, JULY 2002

BYTE RANGE LOCKING, THE NT INSIDER, VOL. 9, ISS. 3, MAY 2002

REFERENCE COUNTING FOR FILE SYSTEM FILTER DRIVERS, THE NT INSIDER, VOL. 9, ISS. 2, MARCH 2002

USING THE INVERTED CALL MODEL, THE NT INSIDER, VOL. 9, ISS. 1, JANUARY 2002

A REVIEW OF SYNCHRONIZATION PRIMITIVES, THE NT INSIDER, VOL. 9, ISS. 1, JANUARY 2002

NEW FILE SYSTEMS MATERIAL IN WINDOWS XP, THE NT INSIDER, VOL. 8, ISS. 4, JULY 2001

NAME TUNNELING IN WINDOWS 2000 FILE SYSTEMS, THE NT INSIDER, VOL. 8, ISS. 3, MAY 2001

WINDOWS NT SECURITY PART III, THE NT INSIDER, VOL. 8, ISS. 3, MAY 2001

OBSERVATIONS ON FILE SYSTEM FILTER DRIVERS, THE NT INSIDER, VOL. 7, ISS. 6, NOVEMBER 2000

ANALYZING A CRASH DUMP, THE NT INSIDER, VOL. 7, ISS. 2, MARCH 2000

WINDOWS NT SECURITY PART II, THE NT INSIDER, VOL. 6, ISS. 5, SEPTEMBER 1999

WINDOWS NT SECURITY PART I, THE NT INSIDER, VOL. 6, ISS. 3, MAY 1999

STRUCTURED EXCEPTION HANDLING, THE NT INSIDER, VOL. 6, ISS. 2, MARCH 1999

WINDOWS NT VIRTUAL MEMORY PART II, THE NT INSIDER, VOL. 6, ISS. 1, JANUARY 1999

WINDOWS NT VIRTUAL MEMORY PART I, THE NT INSIDER, VOL. 5, ISS. 2, MARCH 1998

ASYNCHRONOUS PROCEDURE CALLS, THE NT INSIDER, VOL. 5, ISS. 1, JANUARY 1998

HOW NT HANDLES I/O COMPLETION, THE NT INSIDER, VOL. 4, ISS. 3, MAY 1997

BUILDING IRPS TO PERFORM I/O, THE NT INSIDER, VOL. 4, ISS. 1, JANUARY 1997

THE LANMANAGER FILE SERVER ON NT, THE NT INSIDER, VOL. 3, ISS. 4, OCTOBER 1996

OPLOCKS ON WINDOWS NT, THE NT INSIDER, VOL. 3, ISS. 3, JULY 1996

USING THE NT CACHE MANAGER, THE NT INSIDER, VOL. 3, ISS. 2, APRIL 1996

LIFE IN THE FAST I/O LANE, THE NT INSIDER, VOL. 3, ISS. 1, JANUARY 1996

LAN EMULATION, IEEE COMMUNICATIONS, FALL 1994

DISTRIBUTED COMPUTING, ENCYCLOPEDIA OF COMPUTER SCIENCE & TECHNOLOGY, VOL. 30, 1994

OSF'S DISTRIBUTED COMPUTING ENVIRONMENT, UNIX REVIEW, JANUARY 1993

THE V OPERATING SYSTEM, BYTE MAGAZINE, NOVEMBER 1987